

RAJASTHAN RAJYA VIDYUT UTPADAN NIGAM LTD.
(Govt of Rajasthan Undertaking)



RVUNL SOLAR POWER PARK
at
Tehsil- Pugal, Bikaner, Rajasthan



Detailed Project Report of 1310 MW Phase-I
RVUNL Solar Park (2000 MW)
at Village- Ramsar Chhota, Bandarewala,
Distt-Bikaner, Rajasthan.

Submitted to:-
Rajasthan Rajya Vidyut Utpadan Nigam Limited

Submitted by:-
TÜV SÜD South Asia Private Limited
Mumbai, M.H.





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Responsible for	Job Title	Name	Date	Signature
Content	Manager Asst Manager	RS KK	14-11-2022	
Reviewed	Sr. Manager	SS	15-11-2022	
Approval	DGM	AI	15-11-2022	
Copyright:	TSSA ©	Doc. Ref.	Report No.: 2022/GES-000861/Ra/DPR/SP/41777	
Remarks	---	----	Report version: V04	

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List of Abbreviations

ABT	Availability Based Tariff
AC	Alternating Current
ACCC	Aluminium conductor composite core
ACSR	Aluminium Conductor Steel Reinforced
Amps	Amperes
APDCL	Assam Power Distribution Company Limited
APEDA	Arunachal Pradesh Energy Development Agency
APEPRL	Adani Renewable Energy Park Rajasthan Limited
APGCL	Assam Power Generation Corp. Limited
APGENCO	Andhra Pradesh Power Generation Corporation
APSPCL	Andhra Pradesh Solar Power Corporation Pvt. Limited
B.T. Road	Bituminous Road
BH	Bore holes
BM	Build Margin
BSR	Basic Schedule of Rates
CDM	Clean Development Mechanism
CdTe	Cadmium Tellerudide
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Commission
CIGS	Copper Indium Gallium Solenoid
CM	Combined Margin
CPV	Concentrated Photovoltaic
CREDA	Chhattisgarh State Renewable Energy Development Agency
C-Si	Crystalline silicon
CSP	Concentrated Solar Power
CSR	Corporate Social Responsibility
CTU	Central Transmission Utility
CUF	Capacity Utilization Factor
cum	Cubic meter
dB	Decibels
DC	Direct Current
DGPS	Differential Global Positioning System
DLC	District Level Committee
DNI	Direct Normal Irradiance
DPR	Detailed Project Report
EIA	Environment Impact Assessment
ELP	Electrical Lineworker Program
EPC	Engineering, Procurement & Construction
GEDCOL	Green Energy Development Corporation of Odisha
GHI	Global Horizontal Irradiance
GoI	Government of India
GPCL	Gujarat Power Corporation Limited
GPS	Global Positioning System



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GSS	Grid Sub Station
GW	Giga Watt
HPGCL	Haryana Power Generation Corporation Limited
HSIIDC	Haryana State Industrial and Infrastructure Development Corporation
HTF	Heat Transfer Fluid
IDC	Interest During Construction
IEGC	Indian Electricity Grid Code
IGNP	Indira Gandhi Nahar Pariyojana
IL&FS	Infrastructure Leasing and Financial Services
IMD	Indian Meteorological Department
IRR	Internal Rate of Return
ISRO	Indian Space Research Organisation
JAKEDA	J&K Energy Dev. Agency
Km	Kilo Meter
KREDL	Karnataka Renewable Energy Development Limited
KSEB	Kerala State Electricity Board
KSPDCL	Karnataka Solar Power Development Corporation Limited
kV	kilo Volt
LILLO	Loop-In Loop-Out
LSPDCL	Lucknow Solar Power Development Corporation Limited
MAHAGENCO	Maharashtra State Power Generation Company Limited
MDR	Major District Road
MePGCL	Meghalaya Power Generation Corporation Limited
MJ	Multi- Junction
MNRE	Ministry of New & Renewable Energy
MoEF	Ministry of Environment and Forest
Mono c-Si	Mono-crystalline silicon
MPUVNL	Madhya Pradesh Urja Vikas Nigam Limited
MSL	Mean Sea Level
multi c-Si	Multi-crystalline Silicon
MUs	Million Units
MW	Mega Watt
MWh	Mega Watt Hour
NASA	National Aeronautics and Space Administration
NLDC	National Load Dispatch Centre
NOC	No Objection Certificate
NREDCAP	New & Renewable Energy Development Corporation of A.P. Limited
NREL	National Renewable Energy Laboratory
NRLDC	Northern Regional Load Dispatch Centre
NTPC	National Thermal Power Corporation
NVVNL	NTPC Vidyut Vyapar Nigam Limited
O&M	Operation and Maintenance
OD	Oxygen Demand
OHSR	Over Head Storage Reservoir



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OM	Operating Margin
PGCIL	Power Grid Corporation of India Limited
PPE	Personal Protective Equipment
PPS	Park Pooling Substation
QCA	Qualified Coordinating Agency
R&R	Resettlement and Rehabilitation
RE	Renewable Energy
REGC	Rajasthan Electricity Grid Code
REMC	Renewable Energy Management Centre
RERC	Rajasthan Electricity Regulatory Commission
RES	Renewable Energy Sources
RLDC	Regional Load Dispatch Centres
ROW	Right of Way
RPO	Renewable Purchase Obligation
RRECL	Rajasthan Renewable Energy Corporation Limited
RREDF	Rajasthan Renewable Energy Development Fund
RSDCL	Rajasthan Solarpark Development Company Limited
RVPN	Rajasthan Vidyut Prasaran Nigam
RVUNL/RRVUNL	Rajasthan Rajya Viduyut Utpadan Nigam Limited
RWR	Raw Water Reservoir
SCADA	Supervisory Control And Data Acquisition
SECI	Solar Energy Corporation of India
SEZ	Special Economic Zones
SIDCUL	State Industrial Development Corporation Uttarakhand Limited
SLDC	State Load Dispatch Centres
SPD	Solar Project Developer
SPM	Suspended Particulate Matter
SPO	Solar Purchase Obligation
SPPD	Solar Power Park Developer
SPT	Standard Penetration Test
SPV	Solar Photo Voltaic
STE	Solar thermal electricity
STU	State Transmission Utility
SUCRL	Saurya Urja Company of Rajasthan Limited
SUN Haryana	Saur Urja Nigam Haryana Limited
TACSR	Thermal resistant Aluminium conductor Steel reinforced
TBM	Temporary Benchmark Points
TNREDC	Telangana New & Renewable Energy Development Corporation Limited
UPNEDA	Uttar Pradesh New and Renewable Energy Development Agency
W.B.M. Road	Water Bound Macadam Road or Gravel Road
WBSEDCL	West Bengal State Electricity Distribution Company



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1. Overview & Summary

1.1 Overview of RVUNL

Rajasthan Rajya Vidyut Utpadan Nigam limited (RVUNL) is the Solar Power Park Developer (SPPD) at Ramsar Chhota village which has been entrusted with the development of power projects under State Sector, in the state along with O&M of State-Owned Power Stations.

The proposed Solar Park is a part of development plans of RVUNL, which has been entrusted with the job of development of power projects under state owned power stations. Government of Rajasthan constituted RVUNL under Companies Act 1956 on 19th July 2000. The Nigam is since playing lead role in giving highest priority to the power generation for manifold and rapid development of the state. The generation station of RVUN have acquired a distinctive reputation in the country for their efficient and economic power generation.

RVUNL is undertaking of Govt. of Rajasthan holding CIN No- U40102RJ2000SGC016484 has authorized and has 8 Board of directors.

Registered office of RVUNL is at Vidyut Bhawan, Jyoti Nagar, Janpath, Jaipur -302 005, Rajasthan.

The multiple operational projects of RVUNL with capacity are as below

Table 1: Power Projects under RVUNL

Sl.No	Name of the Power Plant	Capacity (MW)
I	Kota Super Thermal Power Station (KSTPS) Kata	1240
li	Suratgarh Super Thermal Power Station (SSTPS), Suratgarh	1500
lii	Suratgarh Super critical Thermal Power Project (SScTPP), Suratgarh	1320
lv	Chhabra Thermal Power Project (CTPP), Chhabra	1000
v	Chhabra Super critical Thermal Power Project (CScTiPP), Chhabra	1320
vi	Kailsindh Thermal Power Project (KaTPP), Jharawar	1200
vii	Dholpur Combined Cycle Power Project (DCCPP), Dholpur	330
viii	Ramgar Gas Thermal Power Project (RGTPP), Jaisalmer	273.5
ix	Girai Lignite Power Limited (GLPL), Barmer	250
x	Mahi Hydrel Power Station, Banswara	140



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1.2 Executive Summary

Rajasthan Rajya Vidyut Utpadan Nigam Ltd (“RRVUNL”) intends to develop 2000 MW Solar power park under MNRE’s UMREPP scheme (mode 8) in phased manner under two phases (1310 MW – Phase I & 690 MW – Phase II) and installation of 810 MW solar power project in the above solar park. Land acquired is about 2.42 Ha (5.98 Acres) per MW. However, based on the high efficiency modules available at the market, the land requirement per MW DC capacity is estimated around 2.31Ha (5.70 Acres).

For the implementation of the above solar projects RVUNL has appointed TÜV SÜD South Asia Private Ltd (hereafter “Technical Consultant or TA”) for providing consultancy services wide LOI Ref RVUN/CE(TD)/SE(TD-III)/F/D 283 dated 20th May 2022 for development of 2000 MW capacity solar park and installation of 810 MW solar power project at village Ramsar Chhota, Bandarewala and Barala at Pugal Tehsil, Bikaner District of Rajasthan.

The solar park concept is similar to an economic zone dedicated to the generation of power through solar energy and also to the manufacturing of solar energy components. Solar energy is the most readily available source of non-polluting renewable energy resource. It could be utilized in two ways viz. direct conversion into electricity through solar photovoltaic (PV) cells and indirect conversion through generating high temperatures by concentrating collectors and thereby run the steam turbine in line with a conventional thermal power plant. The uniqueness of the solar technologies is that it offers a wide range of applications in solar PV as well as solar thermal technology in which case, the generated heat could be used for domestic as well as industrial applications and power generation.

India being a tropical country is blessed with good sunshine over most parts, and the number of clear sunny days in a year also being quite high. India is in the sunny belt of the world. As per Ministry of New and Renewable Energy (MNRE), Government of India (GoI), the country receives solar energy equivalent to more than 5,000 trillion kWh per year with a daily average solar energy incident over India which varies from 4.0 to 7.0kWh/m² depending upon the location. India’s equivalent solar energy potential is about 6,000 million GWh of energy per year. Hot and Dry climatic regions have the best solar radiations in the country with around 300 sunny days, which makes these region very appropriate locations for harnessing solar energy. India has a vast solar potential of 749 GW for power generation. Jawaharlal Nehru National Solar Mission had targeted installing up of 20,000 MW capacity Solar Power by the end of year 2022. In year 2015, India revised the target of renewable energy (RE) capacity from approx. 34 gigawatts (GW) in 2015 to 175GW by 2022. The revised target of installation of solar power generation is 100 GW capacity. Out of which, Rajasthan state has target of 8% capacity. The Western desert region of the State of Rajasthan have sparse population, dry climate, very low average rainfall, maximum sunshine hours, vast un-irrigated and non-agricultural land which is most appropriate for solar power generation.

The Power system of the country has been integrated to operate as all India Grid. Grid operation is monitored and controlled by National Load Dispatch Centre (NLDC), Regional Load Dispatch



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Centres (RLDCs) and State Load Dispatch Centres (SLDCs). Section 28(3) and 32(2) of the Electricity Act, specify their functions. For NRLDC, SLDC monitors operation of solar power plants in Rajasthan. Scheduling, despatch, monitoring, and real time operation of grid will be performed by Rajasthan SLDC in accordance with Indian Electricity Grid Code (IEGC), Rajasthan Electricity Grid Code (REGC) and applicable open access regulations. For transmission of Power from Renewable Energy Sources (RES), concentrated mostly in few solar and wind rich States, to the load centres in other states Green Energy Corridors have been conceived.

MNRE has come up with scheme for development of solar /Renewable Energy Ultra -Mega Parks. A park is a connected zone for development of Renewable Power generation projects and provides developers an area that is well characterized with proper infrastructure, access to amenities and where the risk of project can be minimized.

The solar park concept is similar to an economic zone dedicated to the generation of power through solar energy and also to the manufacturing of solar energy components. The first solar park in the country was established at Charanka in Gujarat. This was closely followed by the Bhadla solar park in Rajasthan. To facilitate development of solar park Ministry of New & Renewable Energy (MNRE) had issued scheme for setting-up Solar Parks. MNRE has been approved total 42 numbers of solar parks to be developed in 21 States and union territories with an aggregate capacity of 23403 MW.

Rajasthan Solar Energy Policy 2014 also facilitates Development of solar Park in the state with the recommendations of RRECL, Renewable Energy Corporation Limited (RRECL) is the State Nodal Agency of Government of Rajasthan for the development of renewable energy sources. Developer can get allocation of govt land/ acquisition of agricultural land & access of ceiling as per agricultural holding act. RVUN Solar Park is being developed considering this policy. Rajasthan Rajya Vidyut Utpadan Nigam Limited (RRVUNL) as a Solar Power Park Developer (SPPD) has initiated for development of infrastructure and management of 2000 MW Solar Park in phased manner.

Proposed RVUN Solar Park is being developed under MNRE Scheme of setting up solar power park. Currently this park is developed under scheme 8 of MNRE with a assistance CFA upto 30% of project cost or 20lakh/MW whichever is lower.

The selected land for rvunl solar park is in Ramsar Chhota, Bandarewala & Barala villages in Pugal tehsil in Bikaner district. The development of proposed entire solar park is in phased manner with required expert civil work to address the essential requirements of Solar Park. It has been noticed that the selected location satisfies most of the essential requirements towards setting up a Solar Park in the vicinity. The project is planned with Mono/ multi-crystalline based Solar PV technologies (which are the well mature and proven technology across the globe) along with central/String type inverter. The present Detailed Project Report (DPR) contains all key aspects of the Solar Park of the capacity of 1310 MW (Phase I) in the Pugal Tehsil, of the state of Rajasthan, India. This Solar Park Phase -I Contains three no's of plot. Plot- 1 comprises of 660 MW with 1537.56 Ha, Plot -2 comprises of 335



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MW with Land size of 756.85 Ha. And plot-3 comprises of 732.58 Ha with a capacity of 315 MW making cumulative installation capacity of 1310 MW for phase-I of RRVUNL solar park.

RVUNL solar park is having 1310 MW capacity conceived in Western Rajasthan in approximate 3026.99 hectares of waste land/ non-agricultural land surrounding the village Ramsar Chhota (28.3001N 72.8877E) & Bandarewala (28.1607N, 73.003E) in Pugal tehsil, Bikaner district in Rajasthan. The closest Railway station is Bikaner which is situated on Jaisalmer-Attarabad Road through National Highway No. 62 (NH-62). Both Mentioned Villages are at a distance of 45 kms approx. from Main Canal of Indira Gandhi Nahar Pari yojana (IGNP). Solar PV plants in Rajasthan are mainly with the fixed tilt module mounting. Currently about more than 500-Watt modules are now commercially available in market. For RVUNL Solar Park fixed tilt of 18 degree mounting with 540-Watt module has been considered.

Proposed solar park land is of flat type with undulations and pure silty/ loose sand. Average ground level has gradual slope from highest level in North-West and lowest level in Northeast.

Power from each plot will be evacuated at 33 kV cables and will be laid by solar project developer (SPD) up to 400 kV Park Pooling Sub-station (PSS) / nearest Interfacing connection point available at block of solar power park developer (SPPD). Further it will be evacuated by RVUNL PSS through 400 kV transmission line to RVPNL Grid Sub-Station (GSS) by SPPD's. All the Transmission losses upto RVPN GSS will be in the part of SPD's.

Based on the site visit and the details of the nearest GSS, it is evident that the nearby GSS doesn't have adequate redundancy to accommodate the power generated from the solar park. Hence, a dedicated GSS with a voltage level not less than 400 kV shall be constructed near the proposed project location for power evacuation of entire 1310 MW Solar Park Phase-I. There is a proposal of 400/765 kV substation from RVPN, which is dedicated for the power evacuation from the proposed solar park. The Land for proposed grid substation of RVPN is approx. 10Km near RVUNL park and the Grid Substation shall be developed by RVPN. The detail of proposed substation is shared in subsequent section in this report. New GSS to be developed by RVPNL near by their existing 132 KV GSS at Surasar.

Setting-up Control infrastructure is in the scope of Park Developer which will include optical fibre communication between 400 kV PSS and 400/765 kV GSS of RVPNL for voice communication, requisite data transmission and differential protection on this network. In addition to the data acquisition through SCADA for energy accounting and deviation settlement, Data telemetry adopted at the inverter or plant level, data relating to power system output & other parameters and weather-related data etc. shall be mandatorily provided by such generator or QCA to the SLDC in real time.

SPD will install instruments and test module, for measurement of GHI and DNI, anemometers for wind speed and direction and cloud cover (okta) measuring instruments and transmission of their data at 15 minute's intervals as above. Above provisions will be applicable to SPD, who shall also



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put-up SCADA at their generating stations for sending data from generating station through pooling station of Park to RVPNL GSS to SLDC to integrate it with SLDC SCADA, in line with SLDC requirement.

SPPD will develop & maintain boundary wall fencing, internal peripheral WBM road of 7.5mtr width at each plot of solar park and 10mtr wide bituminous road in the internal village road and road coming between land packets.

The report of study conducted by MNRE has expressed that RE technologies have relatively (compared to other energy options) lower ecological footprint and that as established by life-cycle assessment, RE projects have comparatively low environmental externalities that too are limited to project development phase. The RE projects don't generate solid or liquid effluents during operations and thereby pollution of land, surface water or ground water resources is not anticipated from such projects.

The infrastructure is planned to keep in view that if extension of the Solar Park is planned and/or nearby a Solar Park comes-up in future the same infrastructure can be utilised for the extension/new Solar Park/Solar Projects. RVUNL has proposed to build 810 MW solar power plant as SPD & for balance 500 MW developers will be selected by RVUNL through bid. The SPPD will develop green belt area in the park. Main objectives of the green belt plantation are to reduce the impact of erosion, to stabilise dunes and to provide aesthetic enhancement of the project. The selection of plant types and planting arrangements will be decided based on the discussions with the Forest Department considering no / little shadow falling on solar modules, adaptability, and suitability to agro-climatic reasons/ zones. Presently plantation of Acacia tortillas (Babul), at 5m x 5m spacing, along main roads and on periphery boundary of solar park and along the interior roads by keeping 1 – 2 meters away from boundaries of solar park.

Village Ramsar Chhota & Bandarewala is situated in desert belt and as such water is a scarce commodity. Water table is quite deep (more than 40 meters below ground level) and ground water is available in limited quantity. Ground Water is saline which is neither potable nor useful in module washing. The nearest water source is Indira Gandhi Nahar Pari yojana (IGNP) Canal. Nearest off-take point is at Pana Lal Pumping Station near by RD710 which is about 18 km South-west of Ramsar Chhota Village. Water supply for RVUNL Solar Park is conceived for Drinking water, washing of the PV modules, Green belt development, etc.

The quantum of water required to wash a module is considered to be 2.5 litres per Module surface per washing with total 10 washings in a year. For drinking water, sanitations etc, the quantum of water required are considered at the standard norms of 70 litres per person per day. Water will be pumped from IGPN Main Canal Panalal Pumping Station to Solar Park's raw water reservoirs of capacity to store 30 days water for supply during canal closure. Capacity of pumping station and pipeline for raw water supply considers requirement of filling reservoirs emptied during canal



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closure, evaporation loss, reservoir leakage, drinking water supply to adjoining village (if required) in future.

A total of 450 cusec water has been allocated to power sector by the Indira Gandhi Nahar Board, out of which 58 cusec water is reserved for solar power plants. Water requirement of the proposed RVUNL Solar Park will be 0.2 cusecs as per the estimates. The water available is saline which is neither potable nor useful in module washing. Its extraction is not advisable since the levels of the aquifers will be affected resulting in drying of the soil. The nearest water resource is Indira Gandhi Canal at Panna Lal pumping station which is at 20 km.

During the operations phase the impacts on the surrounding environment are negligible and are reversible in nature and can be mitigated by proper Environmental Management plan. The proposed infrastructure facilities for RVUNL solar park do not attract any environmental clearance as per EIA notification. RVUNL has kept a provision of Corporate Social Responsibility (CSR) specific for RVUNL Solar Park Project for both village Ramsar Chhota & Bandarewala which is equivalent to 1% of total cost of the Solar Park. The fund will be utilised for various programme and activities to be undertaken in villages situated nearby the Proposed Solar Park for development and upliftment of the local community.

Solar Park cost estimate is based on its indicative layout and preliminary design of infrastructure developments. Cost of the Govt. land has been estimated as per the DLC rates of both villages. Cost estimate for electrical works is based on annual plan of RVPN. Cost of 400 kV transmission line with AL-59 (ACSR) conductor and tower design for ACSR Zebra has been extrapolated based on lesser span and Aluminium equivalent of conductor.

The estimated cost of solar park including land cost, civil, electrical and water infrastructure cost etc. is Rs. 850.18 Crores excluding interest during constructions (IDC) and Rs. 897.24 Crores including IDC with 29% equity (considered mnre grant as equity) and 71% debt/loan at interest rate of 9.25% per annum.

Equity, Debt, and IDC takes into account the receipts from MNRE and also receipt of one-time lease premium for land from the Developer. The debt will be repaid with receipt of development charges. MNRE provides the grant of 30% of the total project cost or Rs.20 lakhs/MW, whichever is lower for development of internal infrastructure including cost of transmission to CTU/STU point. In case of RVUNL Solar Park, the lower applicable MNRE grant will be at the rate of Rs 20Lakh per MW which outcomes to Rs 262.00 Crore. Accordingly, one time upfront charges will be incurred by SPD Rs 51.54 Lakh/MW & Annual maintenance (O&M) charges for first year come around at Rs. 2.555 lakh/MW which will increase @5.85% per year cumulatively as per RERC regulation. The water consumption charges will be as per the PHED tariff for Industrial use.

Internal Rate of Return (IRR) for Project has been worked out as 12.94 % based on the assumptions detailed in respective sections. Sensitivity analysis of IRR has been worked out with changes in Development charges, Annual maintenance charges, operational expenses, return on equity etc.



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Project IRR is highly sensitive to development charges rate, operational expenses.

For a solar park, yield cannot be exactly determined as PV modules utilized by different developers may differ. Further the D.C. to A.C ratio may differ based on alignment of modules, inverter, cabling, transformers, Module installation (fixed tilt, seasonal tilt, angle of tilt), modules cleaning and maintenance practice and outage of transmission lines. Considering Average value of 20% CUF, annual yield will be 1.76 Million kWh/MW/ annum and for Solar Park of 1310 MW it will be 2358 Million unit / year.

The CO₂ emission reduction will be positive environmental impact of solar parks. 1310 MW solar park with 20% CUF will generate 2358000 MWh per annum and will affect at 2428237 tonnes of CO₂ emission per year.

The Environmental and social Impact from the proposed project is discussed in section 7 of this report. The proposed infrastructure facilities for Solar Park at Pugal do not attract any environmental clearance as per EIA notification 2006 and 2009 issued by Ministry of Environment and Forest (MoEF). However, provisions of other Act, rules and regulations will have to be complied. Accordingly, the project developer shall require obtaining consent from competent authorities. There is marginal or no impact is expected due to the development of the proposed project in land, Air quality noise level, or any other impacts.

Accordingly, for RVUNL Solar Park of 1310 MW phase-I capacity, the estimated job potential has been worked out as mentioned below:

- Total 460 number of Skilled and 10585 number of Unskilled Manpower during Construction & Commissioning works for both SPD & SPPD which includes construction of Roads, street lighting, building, water infrastructure, transmission lines, substations, installation of solar power plants, greenery development etc.
- Total 433 number of Skilled and 786 number of Unskilled Manpower during Operations & Maintenance (O&M) of the above referred works.

As setting of Solar Park does not impact the environment significantly therefore mitigations measures are minimal. However, creation of green belt & roadside plantation will bring positive impact to air quality and minimize the noise level. To mitigate social impact, there shall be a provision of Community Development Fund under CSR Program in which certain activities shall be proposed in the surround villages in 20 km. radius.

The completion of the solar park will take place in about 24 months from the Start date to match with the scheduled connectivity at 400/765 kV RVPNL sub-station.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

2. Purpose & Scope

2.1 Introduction Concept of Solar Park

Solar Park is a connected zone for development of Renewable Power generation projects and provides developers an area that is well characterized with proper infrastructure, access to amenities and where the risk of project can be minimized.

As per Ministry of New and Renewable Energy (MNRE), Government of India (GoI) has provided scheme for development of Renewable Energy (Solar) Ultra -Mega Parks.

India is one of the best recipients of solar energy due to its location in the solar belt and has vast solar potential of 749 GW for power generation. To harness this potential, Jawahar Lal Nehru National Solar Mission was announced by Union Minister for New and Renewable Energy in a statement in the Parliament on 23.11.09. The Solar Mission had targeted setting up of 20,000 MW installed capacity in 3 stages by the end of 2022.

Mission document recommended SEZ like incentives to be provided to the manufacturing parks. State of Gujarat took a lead and set-up Country's first Solar park at Charanka for large scale solar power generation by providing land, power evacuation and other infrastructure facility. The concept of concentrated zones of development like Industrial zones, business parks, technology parks, Special Economic Zones ("SEZ"), and other similar models have been very successful at facilitating and accelerating new industries when structured and implemented effectively.

The solar park concept is similar to an economic zone dedicated to the generation of power through solar energy and also to the manufacturing of solar energy components.

MNRE, vide notification No. 30/26/2014-15/NSM dated 12.12.2014, had notified the administrative approval for implementation of a Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects in the country commencing from 2014-15 to 2018-19. Draft of Implementation Agreement to be executed between Solar Power Park Developer (SPPD) and Solar Project Developer (SPD), inter alia provides that SPPD will collect one time Solar Power Park Development Expenses towards capital cost for Land development and other Infrastructure such as External Transmission Lines Network, Roads, Water Supply, Drainage System, Boundary fence, Security etc. in the Solar Park from the SPDs, including all applicable taxes, duties, cess and other Government levies.

In addition, SPPD will charge Annual O&M Charges / Annual Lease Rent including all applicable taxes, duties, cess, and other Government levies from SPDs.

In February 2016, the Ministry of New & Renewable Energy issued the guidelines for setting up of the solar park. Besides vast solar potential of 749 GW for power generation, India has also the wind power potential of 103 GW due to its long coastline, biomass power generation of around 25 GW



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and Small hydro (of capacity up to 25 MW) of around 20 GW. In 2015, India announced an increased target of the renewable energy (RE) capacity from nearly 34 Giga Watts (GW) to 175GW by 2022. The revised target has total installed capacity of 100,000 MW from solar power, 60,000 MW from wind energy, 10,000 MW from biomass energy and 5,000 MW from small hydro power projects. The targets set by Ministry of New & Renewable Energy (MNRE) to accomplish the scaled-up target of 1,75,000 MW installed capacity for Renewable power generation by 2022 consists of 40.737GW for ground mounted solar projects. The tentative target for Rajasthan is 8% (Ref: draft National Electricity plan Vol-I (Generation) – Dec 2016].

The Government of India had chalked out the target of developing 40737 MW of ground mounted solar power plants by 2022, mainly by Solar Parks. This target has been revised to 60000 MW. Out of these, decentralised solar power plants,

Solar parks are basically demarcated development zones with proper infrastructure like land, evacuation system, roads and other amenities. State Governments or EPC developers acquire the land, put up transmission lines, get the required government permissions and approvals, park developers offer the facility to the companies who can put up their solar projects on the land and offer the operators a fee. Thus, the risks associated with the projects are substantially minimized.

The first solar park in the country was established at Charanka in Gujarat. This was closely followed by the Bhadla Solar Park in Rajasthan. The concept of solar parks has given an impetus for rapid development of solar power projects in the country. Total 42 no of solar parks in 21 States and union territories with an aggregate capacity of 23404 MW have been approved under MNRE scheme for development of Solar Parks Commencing till date.

In Rajasthan, the Rajasthan Renewable Energy Corporation Limited (RRECL) is the State Nodal Agency of the Government of Rajasthan. RVUNL has Proposed for the development of infrastructure and management of Solar Parks under the implementation of mode-8 Scheme of MNRE.

2.2 Description of Solar Park

RRVUNL is planning to develop Solar Park at village Ramsar Chhota, Bandarewala, Barala at Pugal tehsil which comes under district Bikaner to accommodate 1310 MW installed capacity in Phase-I & 690 MW in Phase-II.

The capacity of Solar Park has been determined by considering 540-watt peak modules with fixed tilt modules mounting arrangement requiring estimated land as 5.7 Acre/MW or 2.31 Hectare/MW. This is considered looking to the facts that currently 500-600 Wp modules are commercially marketed and secondly, tracking system for Ramsar Chhota & Bandarewala (28.30 N & 72.88E) would not be beneficial. In fact, almost all solar PV plants in Rajasthan have the fixed tilt module mounting.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

Land Area of Solar Park is spread on both sides of Bikaner – Pugal major village road. Ramsar Chhota & Bandarewala village (GPS 28.300 N 72.887 E) is situated in Pugal tehsil, Bikaner district in Rajasthan. The nearest Village of the Proposed Solar Park is Ramsar Chhota & Bandarewala is situated at approx. 45 kms from headquarter Bikaner. Nearest railway station is Bikaner at 45 kms. While nearest Airport is Bikaner at approx. 45 kms. Nearest source of water from the park is IGNP Main Canal which is passing through village Ramsar Chhota (district Bikaner) at approx. 18Km. Nearest head of IGNP Main Canal is Pana Lal Pumping Station near RD 710 (district Bikaner) which is at a distance of approx. 20 kms.

2.3 Purpose of DPR

Purpose of the DPR is to provide details about the Proposed RVUNL Solar Park under MNRE's Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects. This DPR contains following details about the project:

- Land details
- Geo-technical Investigation (Soil type and Electrical Resistivity of the Site)
- Khasra Plan Layout
- Solar Plots Layout
- Bifurcation of Plots for allocation to Solar Project Developers
- Electrical Line Route Layout
- Water Storage & Distribution Layout
- Broad Cost Estimates
- One Time Lease Premium for Land
- Road Infrastructure (Approach + Internal) along with Road Lights
- Water Infrastructure
- Office Building and Dormitories at PPS
- Green Area Development
- Shifting of Existing Transmission Lines from the Proposed Park
- Grid Substations
- Transmission Lines
- Contingency, Administration Charges, Community Development Fund
- Project Financing



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

- Interest During Constructions (IDC) Charges
- Debt: Equity Ratio
- Project Internal Rate of Return (IRR)
- Sensitivity Analysis based on IRR

The MNRE under CFA provision will provide the grant of upto 30% of the Total Project Cost or Rs. 20 Lakh / MW whichever is lower in the milestones as prescribed by MNRE.

For RVUNL Solar Park the lower amount of CFA grant eligibility has been derived at the rate of Rs 20 Lakhs per MW which outcomes to Rs 262Crores for Phase-1 (1310MW).

3. Solar Sector Overview

The desert region of the Rajasthan state has sparse population, dry climate, very low average rainfall, maximum sunshine hours, vast un-irrigated and non- agricultural land. It receives the maximum solar irradiation intensity in India with availability of low-cost land. Thus, Rajasthan has a huge potential for setting up of Solar Power Projects.

The Sun rays have ultra-violet as well as infrared radiations. Solar thermal projects utilise infra-red component of solar energy and Solar PV cells utilise ultra-violet component of solar rays. Both technologies have been used for power generation but with falling prices of PV modules, Solar PV generation has become viable and have achieved grid price parity.

The Sun rays have ultra-violet as well as infra-red radiations. Solar thermal projects utilise infra-red component of solar energy and Solar PV cells utilise ultra-violet component of solar rays. Both technologies have been used for power generation but with falling prices of PV modules, Solar PV generation has become viable and have achieved grid price parity.

Solar infrared radiation is a dilute form of heat energy and must be concentrated to generate adequate temperature to heat either high temperature heat transfer fluid (HTF) for Rankine cycle / steam based solar thermal plant or inert helium gas for Stirling cycle engines. The tracking of sun is required to have concentration of solar energy. Solar thermal electricity (STE) is therefore also known as concentrated solar power (CSP). It has been a proven technology with almost 30 years of experience. Its strengths lie on inherent thermal storage giving firm availability of electric power during sunshine hours and additional thermal storage to generate electric power during non-sunshine hours. There are four major types of CSP technologies for electricity generation. These are parabolic trough system, Linear Fresnel Reflector system, parabolic dish system and Power tower system. Parabolic trough system uses curved reflectors in parabolic shape while a Linear Fresnel Reflector system employs plain reflectors. They are single axis tracking system concentrating solar energy on cylindrical tube containing HTF or water and surrounded by



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concentric evacuated glass tube to prevent heat losses by conduction and convection. HTF is usually synthetic oil with high boiling point or the molten salts. Hot HTF is circulated through a heat exchanger where water is transformed to steam through heat exchange. In CSP technology employing water is directly converted to superheated steam in cylindrical tubes. Power tower system and Parabolic dish system have two axis tracking concentrating solar energy on an object. Power tower system has number of plain reflectors, also called heliostats, focusing on small window of receiver on a stationary tower. The receivers are made of several materials – being ceramics the most common - and the HTF has been water, air and other gases or molten salts. Parabolic dish system has reflectors arranged in parabolic shape focusing solar rays on spherical surface of Stirling engine placed at focus of the paraboloid. Stirling engines development has not attained commercial viability. Efficiency of solar thermal power stations depends on peak temperature achieved.

3.1 Solar PV Technology

In the present scenario, renewable resources emerge as the best alternative. At present, renewable energy accounts for about 20 % of India's installed generation capacity Target of 344 GW (as on 31.03.2018). Much of this capacity is wind-based (about 34GW, as on 30.04.2018), with the share of solar power being only about 59.3GW. India is blessed with an abundance of non-depleting and environmentally friendly renewable resources, such as solar, wind, biomass, hydro, and cogeneration and geothermal.

Solar PV technologies, which utilize the 'light' aspect of sunlight. Photovoltaic technologies use a device called a solar cell to directly convert sunlight into electricity. It is decided to venture into PV technologies and hence, only these technologies will be discussed in this report.

PV technologies can be classified as under:-

3.1.1 Bulk Silicon Technology

Bulk silicon is most common and mature technology and covers about 80% of the global market share. In the bulk silicon photovoltaic technology, the solar cells are derived from pure silicon ingots or blocks by slicing them into wafers. The type of solar cell produced depends upon the quality of silicon material used. Two main types of solar cells based on this principle are available in the market today: Crystalline silicon (c-Si), which is the same as single or mono-crystalline silicon solar cell, and multi-crystalline silicon (mc-Si), which is the same as poly-crystalline silicon solar cell.

Solar cells are the heart of the photovoltaic power plant and convert sunlight directly into electricity. The industry standard for solar cell size today is 156 mm x 156 mm (or often called a 6-inch cell), and a thickness in the range of 160-180 micrometres (μm). Typical efficiencies of such cells vary from 14% to 22%, and give power outputs in the range of 3.4 watts (W) to 5.3 W. Standard crystalline



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and poly-crystalline silicon modules comprise of 60 or 72 solar cells, and are rated for power between 200 and 300W

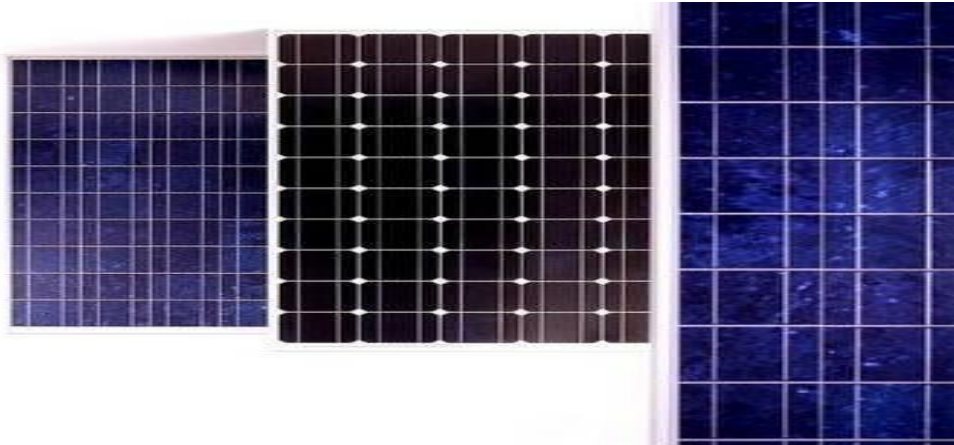


Figure 1: PV Modules

3.1.2 Thin Film Technology

Thin film photovoltaic modules are constructed by depositing extremely thin layers of photosensitive material on to a substrate like glass, stainless steel, metal foils or plastic. The photosensitive material can be silicon, other semiconductors or even combination of different metals.

Thin film solar cells use absorber photosensitive material thickness around 1 μm compared to their bulk silicon counterparts with thicknesses around 160-180 μm . Thus, the thin film solar technology offers tremendous cost savings though economical use of expensive semiconductors. However, it is generally being observed that the crystalline quality of semiconductors deposited as thin films are not as high as those in bulk silicon solar cells. The inherent defects in the thin film material result generally into a lower efficiency. However, a substantial amount of research and development is carried out around the world to enhance thin film solar cell efficiencies, and positive results are gradually observed.

3.1.3 Third Generation PV Technology

Third-generation PV systems include technologies, such as concentrating PV (CPV) and Organic PV cells that are still under demonstration or have not yet been widely commercialized. Third-generation PV technologies are at the pre-commercial stage and vary from technologies under demonstration (e.g. multi-junction concentrating PV) to novel concepts still in need of basic R&D (e.g. quantum-structured PV cells). Some third-generation PV technologies are beginning to be commercialized, but it remains to be seen how successful they will be in taking market share from existing technologies.



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3.2 Installed Solar Capacity in India:

Solar Power in India is a fast-developing industry. India's Solar installed capacity was 59.302 GWAC as of 31 August 2022. Solar power generation in India, ranks fourth globally in 2021. At present, renewable energy accounts for about 20% of India's installed generation capacity of 344 GW (as on 31.03.2018). India is blessed with an abundance of non-depleting and environmentally friendly renewable resources, such as solar, wind, biomass, hydro and cogeneration and geothermal. Wind energy sector, which has shown tremendous growth in the recent year, dominates the renewable energy sector in India. India has an abundance of solar radiation, with the peninsula receiving more than 300 sunny days in a year.

Indian rooftop and utility targets

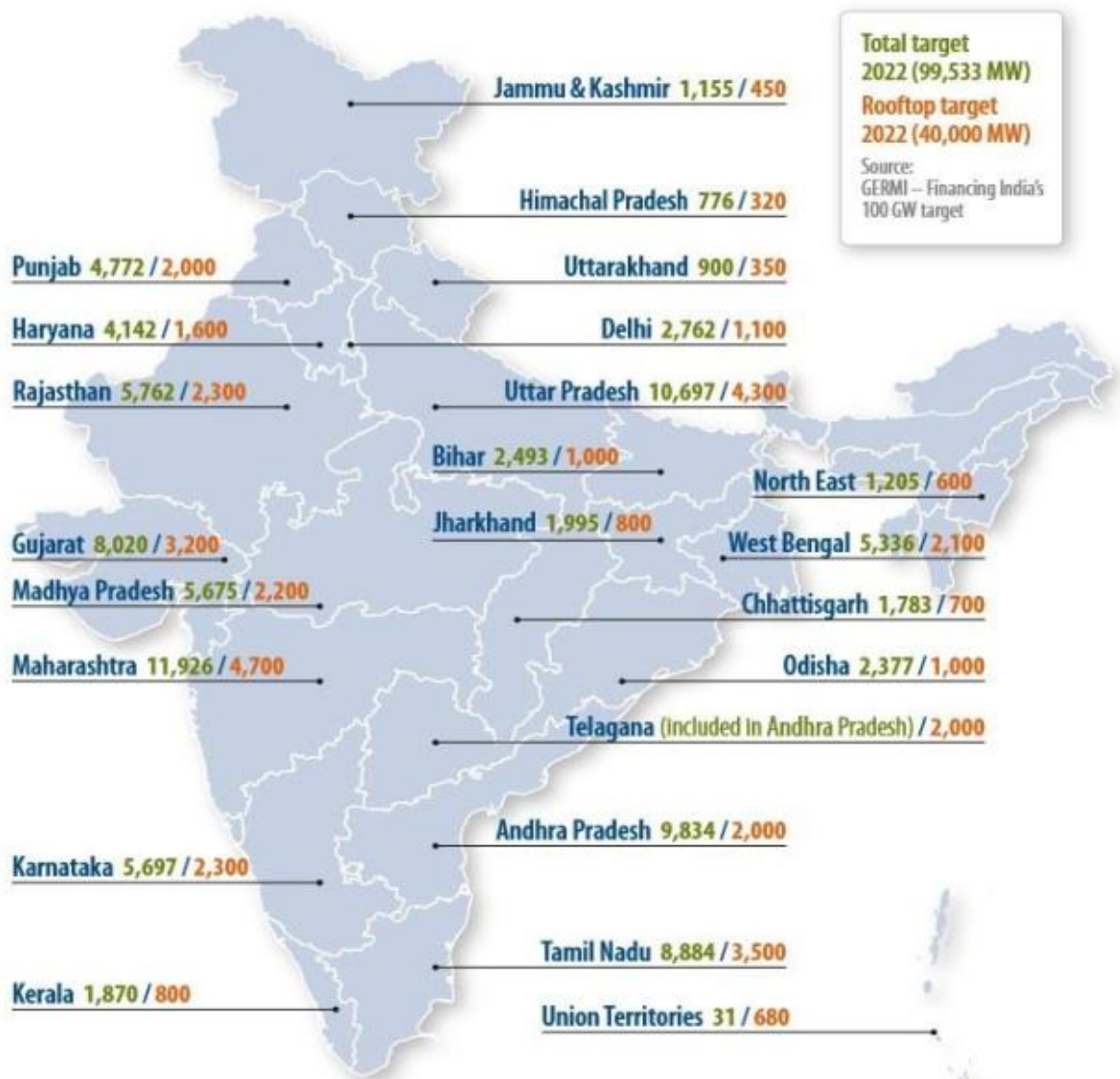


Image-1 : All India Installed Capacity (MW) Reg wise as on 31/05/2022



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

3.2.1 Schemes Available & capacity per Scheme

There are broadly two types of scheme categorised for solar power generation

- Ground Mounted Solar Plant
- Rooftop type Solar Power Plant.

Ground Mounted Solar Power Plant is Megawatt Scale large Plants comprising capacity generally greater than 1MW developed either on Govt Land/Leased Land, Uncultivated Land etc.

Capacity of each scheme is Shown in Above Image.

Rooftop Type is small sized categorised sanctioned load consumption-based system.

3.3 Installed Solar Capacity in Rajasthan State:

3.3.1 Installed and under construction

The history of power development in Rajasthan goes back to the year 1949, when 19 princely states merged to form Rajasthan. At that time, electric power was confined to very few towns and electricity was considered a luxury. The total number of towns and villages electrified at that time did not exceed 42 and the installed generating capacity was only 13.27 MW. However, with the formation of Rajasthan State Electricity Board (RSEB) on 1st July 1957, power sector in Rajasthan received priority and power projects began to mushroom all over State. Under new Power Reforms Undertaken by State Government on 19th July 2000, RSEB was unbundled.

Installed solar capacity in Rajasthan is 14454 MW whereas approx. 2500-3000 MW capacity is under construction. Initially, solar developers have proposed to install solar power plants in Rajasthan on project specific tariff basis. However, before such tariff is determined by RERC, they migrated at CERC approved tariff to Jawaharlal Nehru National Solar Mission Projects handled by RVUNL for effecting supply of solar power bundled with unallocated power of operated thermal power stations. In this process, numbers of potential sites for solar projects were identified.

TOTAL RE (Up to July 2022)	19116.80 MW
SOLAR	14454.70 MW
(a) GROUND MOUNTED SOLAR CAPACITY	11821.87 MW
(b) ROOFTOP	814.02 MW
(c) OFF GRID	438.41 MW
(d) SOLAR HYBRID	1380.40 MW



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3.3.2 Schemes Implemented:

The Ministry of New and renewable Energy (MNRE) has drawn a scheme to set up number of solar parks across various states in the country, each with a capacity of Solar Projects generally above 500 MW. The Scheme proposed to provide financial support by Government of India to establish solar parks with an aim to facilitate creation of infrastructure necessary for setting-up new solar power projects. Solar Energy Corporation of India (SECI), a central public sector enterprise under MNRE, has been implementing various schemes to develop solar parks in the country. As per the policy, these solar parks will be developed in collaboration with the State Governments. SECI will handle funds to be made available under the scheme on behalf of GOI.

As part of Solar Park development scheme, land required for development of Solar Power Projects with cumulative capacity generally 500 MW and above will be identified and acquired and various infrastructure like transmission system, water, road connectivity and communication network etc. will be developed. The parks will be characterized by well-developed proper infra-structure where the risk & gestation period of the projects will be minimized. At the state level, the solar park will enable the states to bring in significant investment from project developers in Solar Power sector, to meet its Solar Purchase Obligation (SPO) mandates and provide employment opportunities to local population. The state will also be able to reduce its carbon footprint by avoiding emissions equivalent to the solar park's generated capacity.

3.3.3 RPO Obligations & Current Status

In Rajasthan, RPO (for wind and bio-mass based generation) was specified with effect from 1.4.06 for Distribution licensees vide amendment to RERC (Power Purchase & Procurement Process of Distribution Licensee) Regulations, 2004. It was specified for Captive power plants and Open Access consumers with effect from 1.4.07 vide RERC (Renewable Energy Obligation) Regulations, 2007. Thereafter, separate RPO for Solar energy was specified for distribution licensees vide notifications dated 24.5.11, 20.2.14 and order dated 7.6.17 and 11.1.19 Separate RPO for solar energy was specified for captive power plant and open access consumers of 10 MW and above vide amendment to RERC (Renewable Energy Obligation Regulations notified on May 30, 2014 and order dated 07.06.2017 and 11.1.19.

The Solar RPOs of specified year is as under:

FY 2011-12 [#]	0.50%
FY 2012-13 [#]	0.75%
FY 2013-14 [#]	1.00%
FY 2014-15 [@]	1.50%
FY 2015-16 [@]	2.00%



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FY 2016-17@	2.50%
FY 2017-18@	4.75%
FY 2018-19@ (revised)	4.75%
FY 2019-20@	6.00%
FY 2020-21@	7.25%
FY 2021-22@	8.50%
FY 2022-23@	9.50%
FY 2023-24@	10.50%

for distribution licensees only

@ for distribution licensees, CPP and open access consumers of 10 MW & above

In addition to the general RPO mandated on utilities, particularly Distribution Companies (DISCOMs), many national and state governments around the world have also started specifying the solar-specific RPO, often known as the 'solar carve-out'. However, the success of such cases rest on the implementation of such obligations, which are usually in the form of enforcements. Many states in India, through their respective State Electricity Regulatory Commissions, have also defined their RPOs. As per the current scenario, electricity utilities of only seven to eight states have been able to meet their RPO targets. Other states fail to attain their RPO targets due to lack of uniformity in providing longer trajectory for the RPO targets. With the new RPO targets a positive impact on pricing and tariff for electricity of solar and other renewable energy resources is visible

3.4 Scheme of Solar Park & Advisability for the Solar Park:

The scheme for "Development of Solar Parks and Ultra Mega Solar Power Projects" was rolled out by Ministry of New & Renewable Energy on 12-12-2014. Under this scheme, it was proposed to set up at least 25 Solar Parks and Ultra Mega Solar Power Projects targeting over 20,000 MW of solar power installed capacity within a span of 5 years starting from 2014-15. The capacity of the Scheme has been enhanced from 20,000 MW to 40,000 MW vide this Ministry's order dated 21-03-2017. These parks are proposed to be set up by 2021-22.

The scheme envisages supporting the States/UTs in setting up solar parks at various locations in the country with a view to create required infrastructure for setting up of solar power projects. The solar parks provide suitable developed land with all clearances, transmission system, water access, road connectivity, communication network, etc. The scheme facilitates and speed up installation of grid connected solar power projects for electricity generation on a large scale. All the States and Union Territories are eligible for getting benefit under the scheme.



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The capacity of the solar parks shall be 500 MW and above. However, smaller parks are also considered where contiguous land may be difficult to acquire in view of difficult terrain and where there is acute shortage of non-agricultural land. The solar parks are developed in collaboration with the State Governments and their agencies, CPSUs, and private entrepreneurs. The implementing agency is termed as Solar Power Park Developer (SPPD). There are 8 modes for selection of SPPD (detailed discussed in the respective section).

At the state level, the solar park will enable the states to bring in significant investment from project developers in Solar Power sector, to meet its Solar Purchase Obligation (SPO) mandates and provide employment opportunities to local population. The state will also be able to reduce its carbon footprint by avoiding emissions equivalent to the solar park's generated capacity.

State Wise Solar Park Implementation under Phase-I & Phase- II are as under:-

Some Other Solar Park State Wise (Commissioned & under construction)

A) Bhadla Solar Park (Rajasthan)

Bhadla solar park is a 2.25GW solar complex being developed in Bhadla village in Jodhpur district of Rajasthan, India. The project is being developed in four phases, with Rajasthan Solar Park Development Company Limited (RSPDCL) developing the first two phases, Saurya Urja Company of Rajasthan developing phase three, and Adani Renewable Energy Park Rajasthan developing phase four.

Following are the projects developed in Bhadla Solar Park.

Table 1: Project at Bhadla Solar Park, Karnataka

Sl.No	Project Developer	Capacity (MW)
1	SB Energy	600
2	Hero Futures	300
3	NTPC	260
4	ACME Solar	200
5	Azure Power	200
6	Charisma Energy/Sunseap	140
7	Engie	140
8	Avaada Power	100
9	Fortum Solar	70
10	Vector Green	70
11	Renew Power	60
12	Phelan Energy	50
13	Electrotherm	5
14	Aditya Birla	20
15	LNB Group	5



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Sl.No	Project Developer	Capacity (MW)
16	Roha Dyechem	25
	Total	2245

B) Pavagada Solar Park (Karnataka)

The Pavagada Solar Park, located in Tumakuru district of Karnataka, has been developed by the Karnataka Solar Park Development Corporation Limited (KSPDCL), a joint venture between the SECI and the KREDL. Initially, the plan was to build a solar park of capacity 2,000 MW across 13,000 acres of land. Furthermore, an additional capacity of 50 MW was added later.

Table 2: Project at Pavagada Solar Park, Karnataka

Sl.No	Project Developer	Capacity (MW)
1	Tata Power	400
2	ReNew Power	350
3	Fortum Solar	350
4	Avaada Energy	300
5	SoftBank Energy	200
6	Adani Green Energy	150
7	ACME Solar	100
8	Azure Power	100
9	Rattan India	50
10	Karnataka Renewable Energy Development Ltd (KREDL)	50
	Total	2050

C) Dholera Solar Park (Gujarat)

The Dholera solar park is divided into two phases of 1 GW and 4 GW. The solar park is expected to cover an area of more than 27,000 acres to develop 5 GW of solar projects. Out of the total 1000 MW capacity under Phase 1 the auction process for 300 MW was completed earlier. In March 2020, GUVNL retendered 700 MW of solar projects located in the state's Dholera solar park (Phase IX). The tender was initially announced in January 2019 with a lower tariff ceiling of Rs 2.75 per kWh and a total capacity of 1 GW.

Table 3: Project at Dholera Solar Park, Gujarat

Sl.No	Project Developer	Capacity (MW)
1	Tata Power	200
2	GIPCL	50
3	Vena Power	100
4	Tata Power	100
5	Renew Power	200



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6	SJVN Limited	100
7	TEQ Green Power (O2 Power)	200
	Total	1000

The Auction for balance capacity of 4 GW is expected to be tendered soon.

D) Other Major Solar Parks

Listed below are the Solar Power Plants in India along with the location, capacity and developer details.

Table 4: Major Solar Parks in India

State	Solar Park	Cap acity (MW)	Solar Power Parks Developer (SPPD)
Andhra Pradesh	Ananthapuram-I Solar Park	1500	AP Solar Power Corporation Pvt. Ltd. (APSPCL), JVC of SECI, APGENCO and NREDCAP
	Kurnool Solar Park	1000	
	Kadapa Solar Park	1000	
	Ananthapuram-II Solar Park	500	
	Hybrid Solar Wind Park	160	
Arunachal Pradesh	Lohit Solar Park	20	Arunachal Pradesh Energy Development Agency (APEDA)
Gujarat	Radhnesada Solar Park	700	Gujarat Power Corporation Limited (GPCL)
	Harsad Solar Park	350	
	Dholera Solar Park Ph-I	1000	
	Dholera Solar Park Ph-II	4000	Solar Energy Corporation of India (SECI)
Himachal Pradesh	Kaza Solar Park	1000	JVC of SJVN & Govt of HP
Jharkhand	Floating Solar Park	150	Solar Energy Corporation of India (SECI)
Karnataka	Pavagada Solar Park	2000	Karnataka Solar Power Development Corporation Pvt. Ltd. (KSPDCL), JVC of KREDL & SECI
Kerala	Kasargod Solar Park	105	Renewable Power Corporation of Kerala Limited (RPCKL), JVC of SECI
Madhya Pradesh	Rewa Solar Park	750	Rewa Ultra Mega Solar Limited (RUMSL), JVC of MPNRED & SECI
	Mandsaur Solar Park	250	
	Neemuch	500	Rewa Ultra Mega Solar Limited (RUMSL), JVC of MPNRED & SECI



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State	Solar Park	Capacity (MW)	Solar Power Parks Developer (SPPD)
	Agar	550	Rewa Ultra Mega Solar Limited (RUMSL), JVC of MPNRED & SECI
	Shajapur	450	Rewa Ultra Mega Solar Limited (RUMSL), JVC of MPNRED & SECI
	Omkareswar Floating Solar Park	600	Rewa Ultra Mega Solar Limited (RUMSL), JVC of MPNRED & SECI
	Chhattarpur Solar Park	950	Rewa Ultra Mega Solar Limited (RUMSL), JVC of MPNRED & SECI
	Barethi Solar Park	550	NTPC
Maharashtra	Sai Guru Solar Park (Pragat)	500	M/s Sai Guru Mega Solar Park Pvt. Ltd. (formerly M/s Pragat Akshay Urja Ltd.)
	Patoda Solar Park (Paramount)	150	M/s Paramount Solar Power Pvt. Ltd. (formerly M/s K. P. Power Pvt. Ltd.)
	Dondaicha Solar Park	250	Maharashtra State Electricity Generating Company Ltd. (MAHAGENCO)
Manipur	Bukpi Solar Park	20	Manipur Tribal Development Corpn. Ltd. (MTDCL)
Meghalaya	Solar Park in Meghalaya	20	Meghalaya Power Generation Corporation Ltd (MePGCL)
Mizoram	Vankal Solar Park	20	Power & Electricity Department
Odisha	Solar Park by NHPC	40	NHPC Limited
	Solar Park by NHPC	100	NHPC Limited
Rajasthan	Bhadla-II Solar Park	680	Rajasthan Solar Park Development Company Ltd. (RSDCL)
	Bhadla-III Solar Park	1000	M/s Surya Urja Company of Rajasthan Ltd (SUCRL) JVC of State Govt
	Bhadla-IV Solar Park	500	M/s Adani Renewable Energy Park Rajasthan Limited (AREPRL) JVC of State Govt
	Phalodi-Pokaran Solar Park	750	M/s Essel Surya Urja Company of Rajasthan Limited (ESUCRL) JVC of State Govt



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State	Solar Park	Capacity (MW)	Solar Power Parks Developer (SPPD)
	Fatehgarh Phase-1B Solar Park	421	M/s Adani Renewable Energy Park Rajasthan Limited (AREPRL) JVC of State Govt
	Nokh Solar Park	925	Rajasthan Solar Park Development Company Ltd. (RSDCL)
Uttar Pradesh	Solar Park in UP	440	Lucknow Solar Power Development Corporation Ltd. (LSPDCL) JVC of UPNEDA & SECI
	Jalaun Solar Park	1200	BSUL

3.4.1 About Solar Park

Solar power projects can be set up anywhere in the country, however the scattering of solar power projects leads to higher project cost per MW and higher transmission losses. Individual projects of smaller capacity incur significant expenses in site development, drawing separate transmission lines to nearest substation, procuring water and in creation of other necessary infrastructure. It also takes a long time for project developers to acquire land, get change of land use and various permissions, etc. which delays the project. To overcome these challenges, the scheme for “Development of Solar Parks and Ultra-Mega Solar Power Projects” was rolled out in December 2014 with an objective to facilitate the solar project developers to set up projects in a plug and play model.

The solar park is a concentrated zone for development of solar power generation projects and provides developers an area that is well characterized, with proper infrastructure and access to amenities and where the risk of the projects can be minimized. Solar Park will also facilitate developers by reducing the number of required approvals.

3.4.2 Development of Solar Park

Implementation Agencies

- SECI will be Nodal agency on behalf of MNRE appointed from Govt. of India (GOI).
- State Govt. will designate an agency for development of Solar Park known as Solar Power Park Developer (SPPD).



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

3.4.3 Solar parks are envisaged to be developed in below Modes/ Scheme

Mode-1

The State designates a nodal agency which takes on the development & management of the solar park. This agency could be a State Government Public Sector Undertaking (PSU) or a Special Purpose Vehicle (SPV) of the State Government.

Mode-2

The State designated nodal agency and Solar Energy Corporation of India (SECI) sets up a Joint Venture Company for the development & management of solar park. Both, SECI and the State designated nodal agency will have 50% equity each. State government may also allow more than one agency for the joint venture provided total equity from State Government remains 50%.

Mode-3

The State designates SECI as the nodal agency and SECI undertakes the development and management of solar park on behalf of State Government on mutually agreed terms.

Mode-4

The private entrepreneurs may carry out the development and management of the solar park. They may have equity participation from State Government.

The private entrepreneurs may be selected in following manner:

- **Mode 4A:-** If the land belongs to the Government or its agency, then Solar Power Park Developer (SPPD) would be selected based on open bidding & the bidder with the lowest quotation of development and O&M cost would be selected.
- **Mode 4B:-** If the land is to be provided by the SPPD itself, then SPPD with the lowest quotation of developed land per MW and O&M charges per MW would be selected.

• Mode- 5

Central Public-Sector Companies (CPSUs) like NTPC & SECI may directly approach the Ministry for development of solar parks. They may do so in the following manner:

- **Mode 5A:-** CPSUs having their own land may approach Ministry directly for setting up of solar parks. It may develop the park on its own or through an SPD contractor. CPSUs will be the SPPD. Further CPSUs can also develop their own plant in the park.
- **Mode 5B:-** CPSUs having their own land may select an SPPD based on open bidding and select the lowest bidder.
- **Mode 5C:-** CPSUs who don't own land may float a tender and invite third party for the selection of SPPD.

Mode-6



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

The private entrepreneurs may set-up a solar park without Central Finance Assistance (CFA) from the Ministry. The minimum capacity of the park shall be 100 MW, though 50 MW parks may be allowed in areas with difficult terrain or with scarce non-agricultural land. The private entrepreneurs may submit the proposal along with the Detailed Project Report (DPR) and documents in support of 100% land possession. Post examination of the DPR the principle approval will be awarded. Although the connectivity and LTA with CTU may be issued after financial closure, award of works for road, water and internal transmission. The solar park shall be completed in 18 months; otherwise, their allotted grid connectivity may be cancelled and might be given to someone else.

Mode-7

In order to address the two most critical elements such as land & power evacuation infrastructure for solar Parks a new mode-7 is being implemented by developing of RE Park through SECI.

Mode-8

The SPPD of UMREPP may be any CPSU/State PSU /State govt Organisation or their Subsidiaries. A JV Company (JVC) between two or more of the above entities may also act as the SPPD.

RVUNL 1310 MW Solar Park has been Approved by MNRE under the mode- 8.

4 Project Details:

4.1 Site Details:

RVUNL is planning to develop Solar Park in Tehsil Pugal District Bikaner to accommodate 2000 MW in phase manner (1310MW installing capacity in phase-I, 690 MW in Phase-II) under three villages namely Ramsar Chhota, Bandarewala & Barala. Currently 1310 MW Phase-I has been proposed in Ramsar Chhota & Bandarewala considering 540Wp type Modules with fixed tilt axis module mounting structure(mms) arrangement requiring estimated land as 2.31 Hectare/MW. This has been considered looking to the current market scenario that now days 500-600 watts modules are commercially manufactured & supplied and secondly, tracking system for Pugal (28.2740 N- & 72.88 E) would not be much more beneficial. Since most of the solar power plants located in Rajasthan is installed with fixed tilt technology.

The Government land available in village Ramsar Chhota & Bandarewala tehsil Pugal, district Bikaner has been provided by Revenue Department, Collectorate Office Bikaner, Government of Rajasthan, with all Khasra list & land acquisition has been completed by RVUNL. Initially for Phase-1 1310 MW RVUNL have been considered 55 no's khasra land parcels with a land size of 2294.41 Hectares in Ramsar Chhota, and 90 no's of khasra plot in Bandarewala with a land capacity of 732.518 hectares. District Collector Bikaner has allotted land to RVUNL under lease (for 30 years) on the DLC rate of that area. Payment for land has been made. Annual lease charges will be payable as recurring charges.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.1.1 Site Location & Land Details:

The proposed RVUNL solar park is situated in the town of Pugal at village Ramsar Chhota & Bandarewala within the Bikaner District for phase-I. Pugal Tehsil is located 432 km northwest to Jaipur, the state capital of Rajasthan District. Sites receiving good amount of solar radiation and closer to the STU/CTU are usually selected for the setting up a solar park. According to census 2011 information the sub-district code of Pugal Block (CD) is 00474. Total area of Pugal tehsil is 2,412 km². Pugal tehsil has a population of 67,163 peoples. Pugal has a population density of 28 inhabitants per square kilometre. There are about 11,977 houses in the sub-district. When it comes to literacy, 41.23% population of Pugal tehsil is literate, out of which 51.01% males and 30.21% females are literate. There are about 58 villages in Pugal tehsil, which you can browse from Pugal tehsil villages list (along with gram panchayat information) below.

State Govt. may provide Government waste/non-agricultural land to speed up the process. But good solar radiation and proximity to the CTU/STU are not the only criteria for selection of land. The cost of land is a very important factor too. The cost of land should be kept as low as possible in order to make attractive to the developers. Therefore, the site is selected in such a manner that inexpensive land can be made available. If land cannot be made available at one location, then land which are in proximity are chosen for the development of solar park. The Land for RVUNL Solar Park has been considered with all such factors & aspects.

Table 5: Tehsil Overview

Description	Details
Sub-District	Pugal
District	Bikaner
State	Rajasthan
Total Area	2,412 km ²
Total Population	67,163
Density	28/km ²



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

Proposed Site Location

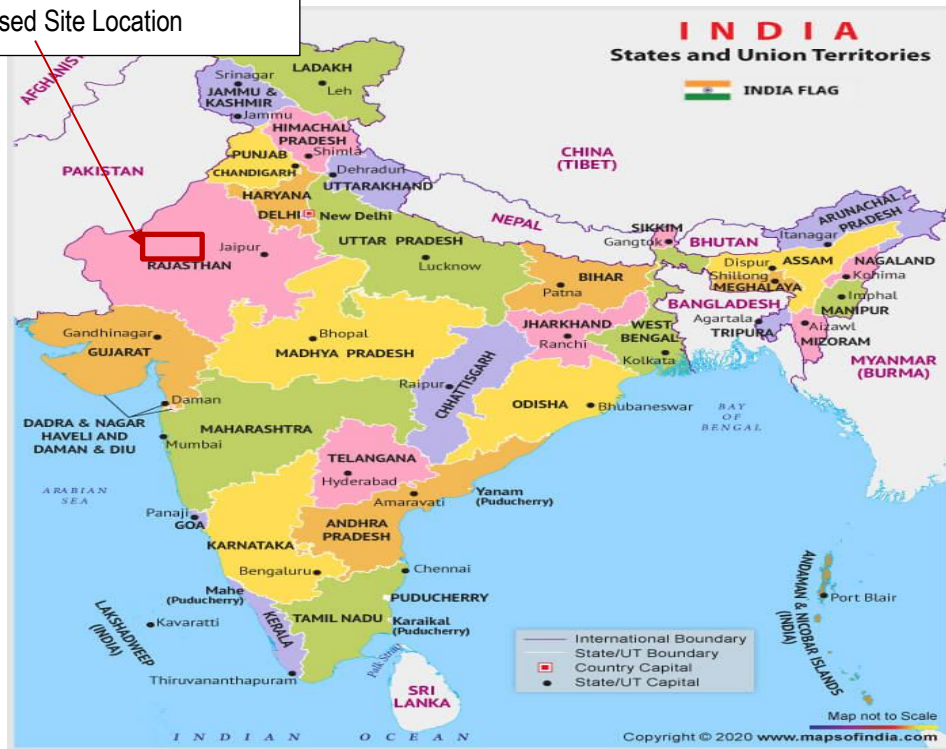
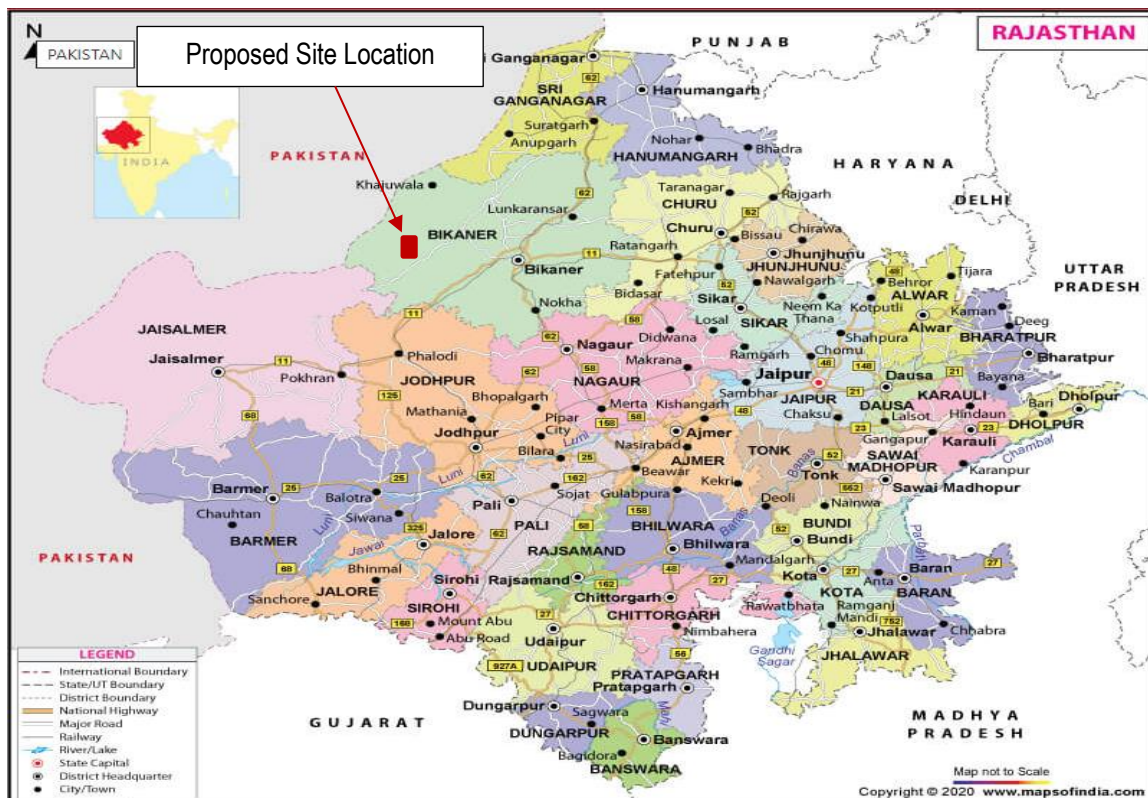
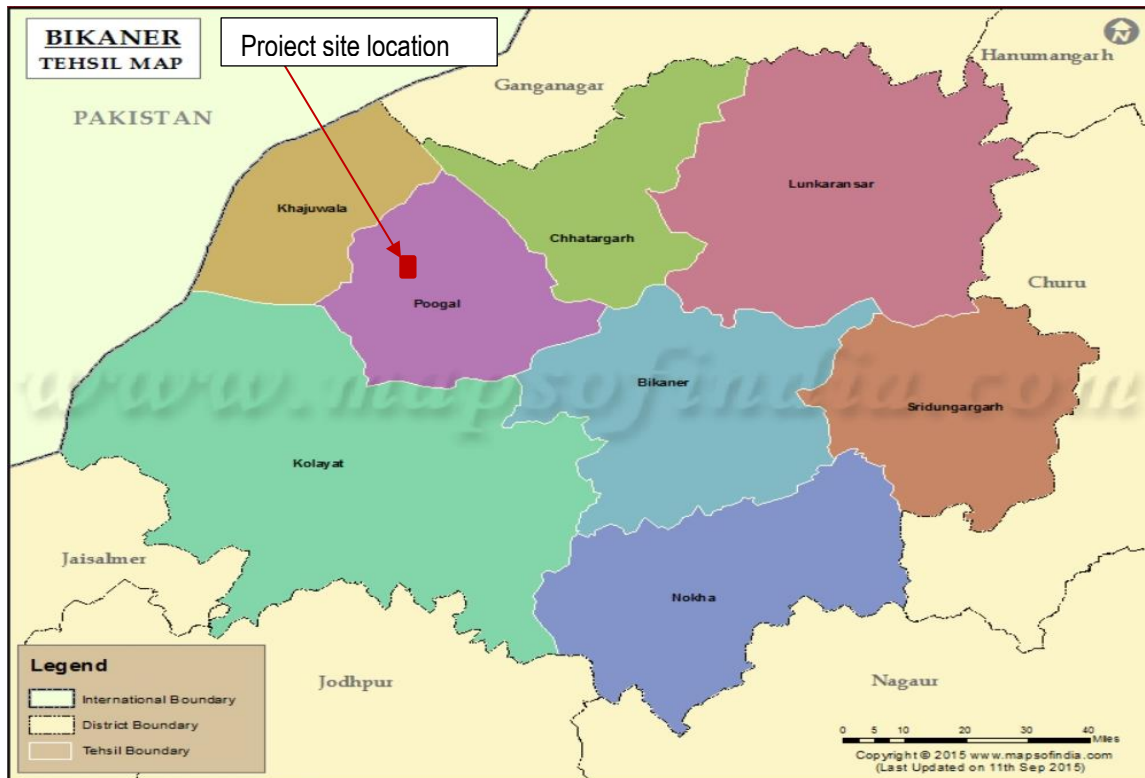


Figure 2: Project Location





Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.



Detail of Site Location

Parameter	Location
Site Location	Ramsar Chhota, Bandarewala (Phase -1, 1310 MW)
Tehsil	Pugal
District	Bikaner
Latitude & Longitude	28.30, 72.88
Elevation (m)	177 M
Area of Land (Hectares)	3026.99 Hectares (Phase- 1)
Power Capacity	2000MW (AC) (1310MW- Phase I) Ramsar Chhota & Bandarewala (690 MW- Phase II) Barala

4.1.2 Land & Land Usage

The changing environmental conditions and human interventions have caused increase in the spread and concentration of soil salinity over the past several decades. The geomorphic processes of erosion, sediment transport, deposition and the extent and condition of tidal wetlands greatly influence the usage of land in the proposed area. Since area is prone to high and low tides suitable measures shall be taken for the optimum usage of the land.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

The land use pattern of an area affects vegetation, land quality, local weather & quality of life. It is very important to understand the land use pattern of any area & the dynamics of its shift overtime. This determines the ensuing per unit load on agriculture land, forest land, periphery areas to cities & factors responsible for land degradation.

6.95% land in Rajasthan land is categorised under barren & uncultivable. For development of this RVUN solar park, Such factors has been considered.

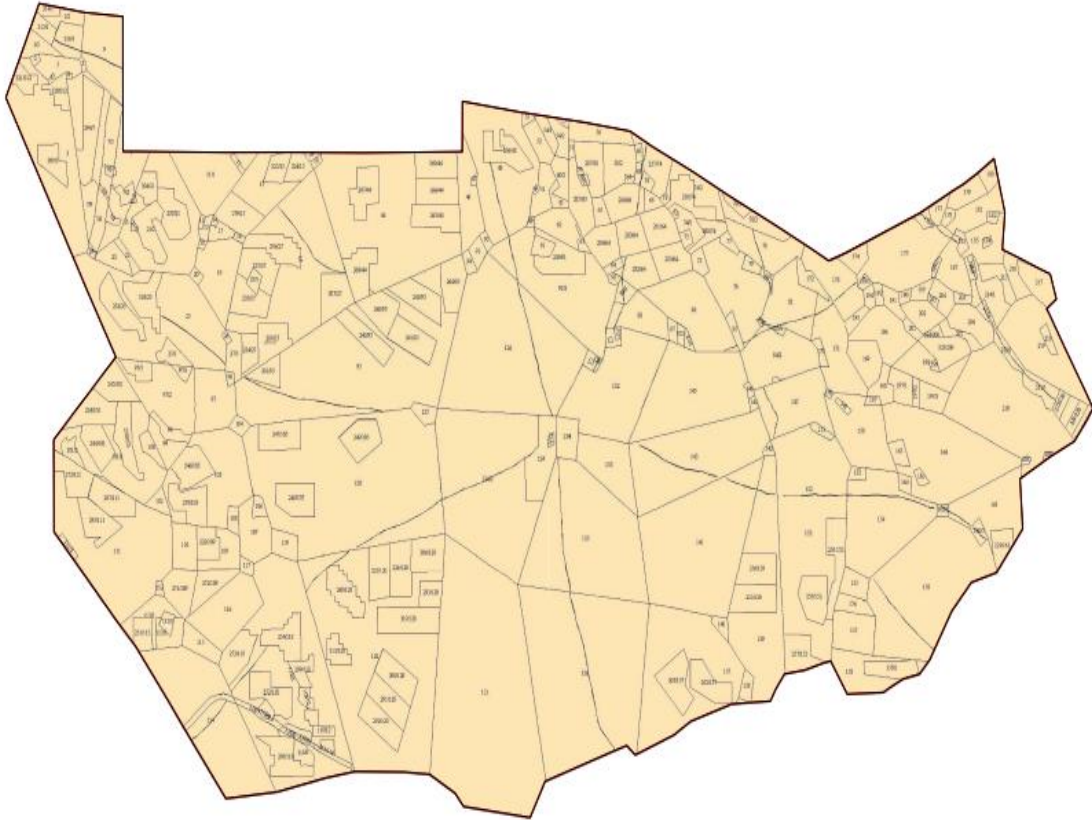
The entire land parcel identified for the development of solar park for both phases is spread across three villages in Pugal tehsil. The land breakup for all three villages is as below:

Table 6: Land Details for Solar Park

Village	No of Land Parcels based on khasra	Land Area (Ha)	Land Area (Aces)	Proposed Solar Capacity (AC MW)
Ramsar Chhota (Phase-I)	55	2294.41	5669.61	995 MW
Bandarewala (Phase-I)	90	732.518	1810.05	315 MW
Bandarewala (Phase-II)	212	821.42	2029.73	250 MW
Barala (Phase –II)	289	997.22	2464.18	440 MW
Total	646	4846.31	11975.49	2000 MW



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.



RMSAR CHOTA

Figure 3: Village Map of Ramsar Chhota



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

BANDERWALA

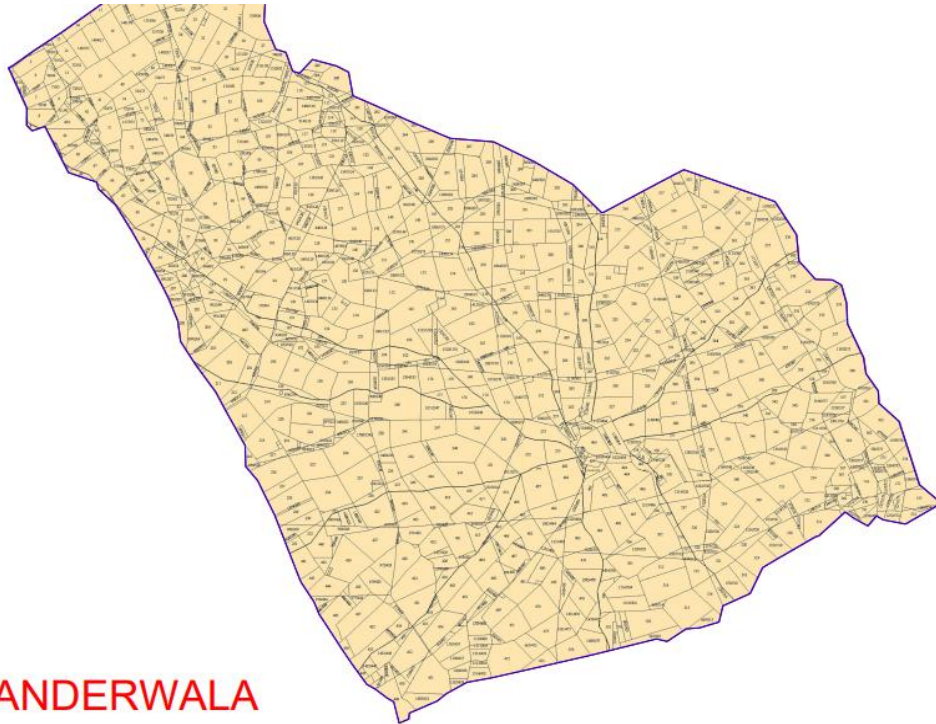


Figure 4: Village Map of Banderwala

BARALA

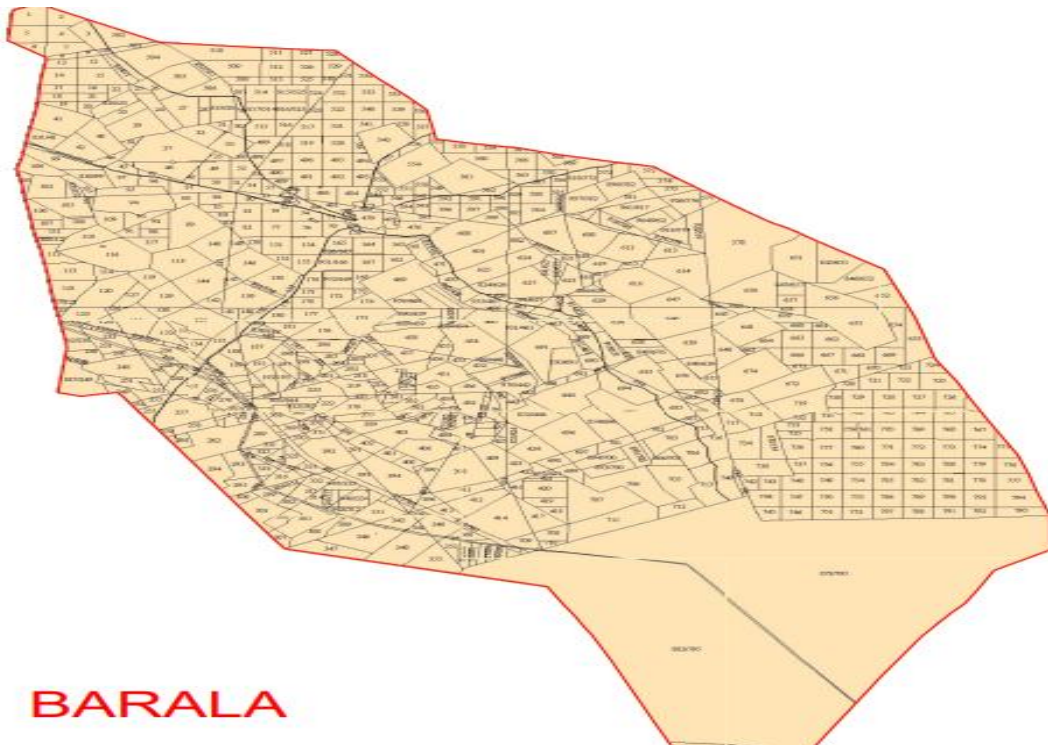
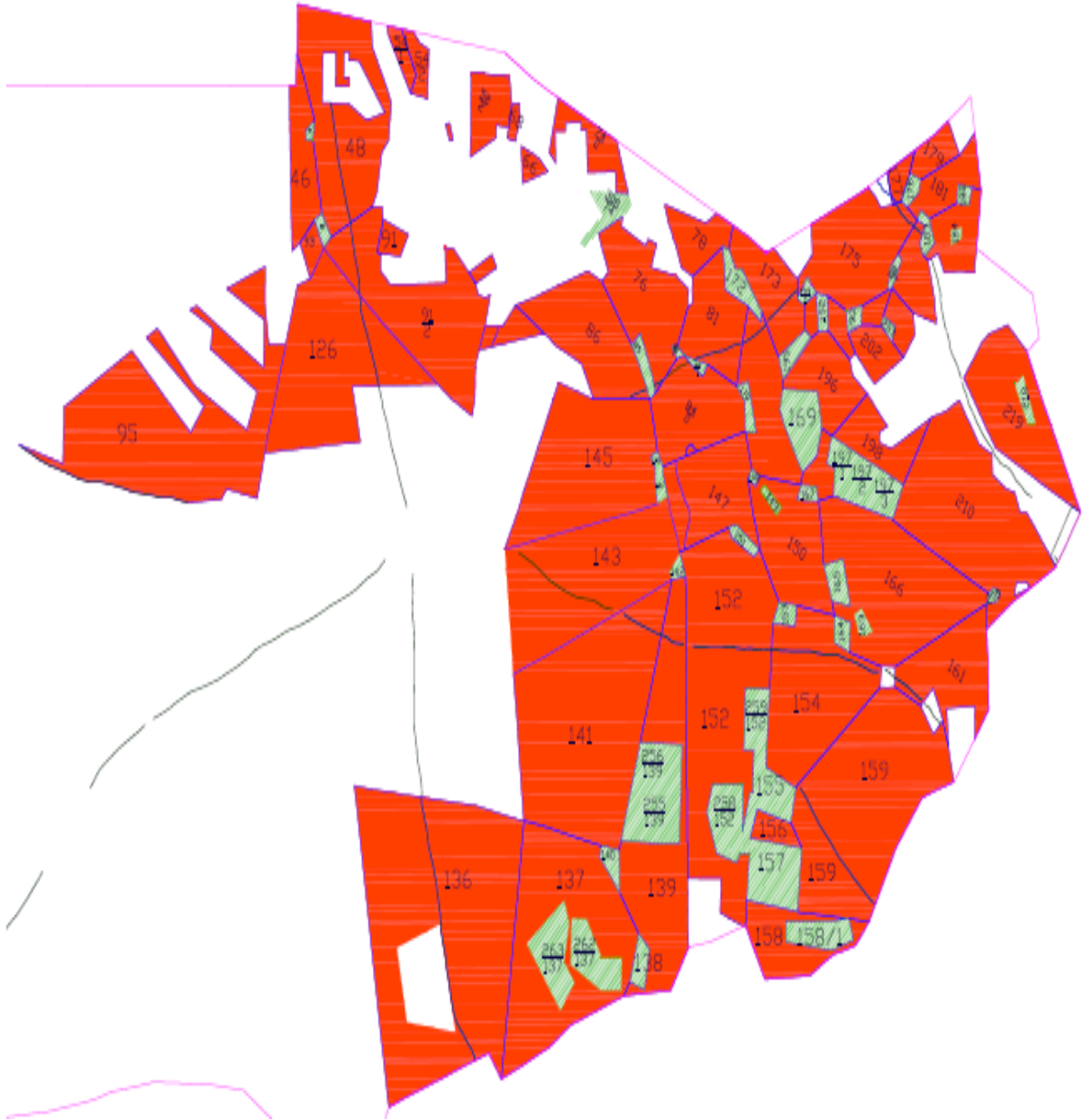


Figure 5: Village Map of Barala



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.



Village - Ramsar Chhota



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

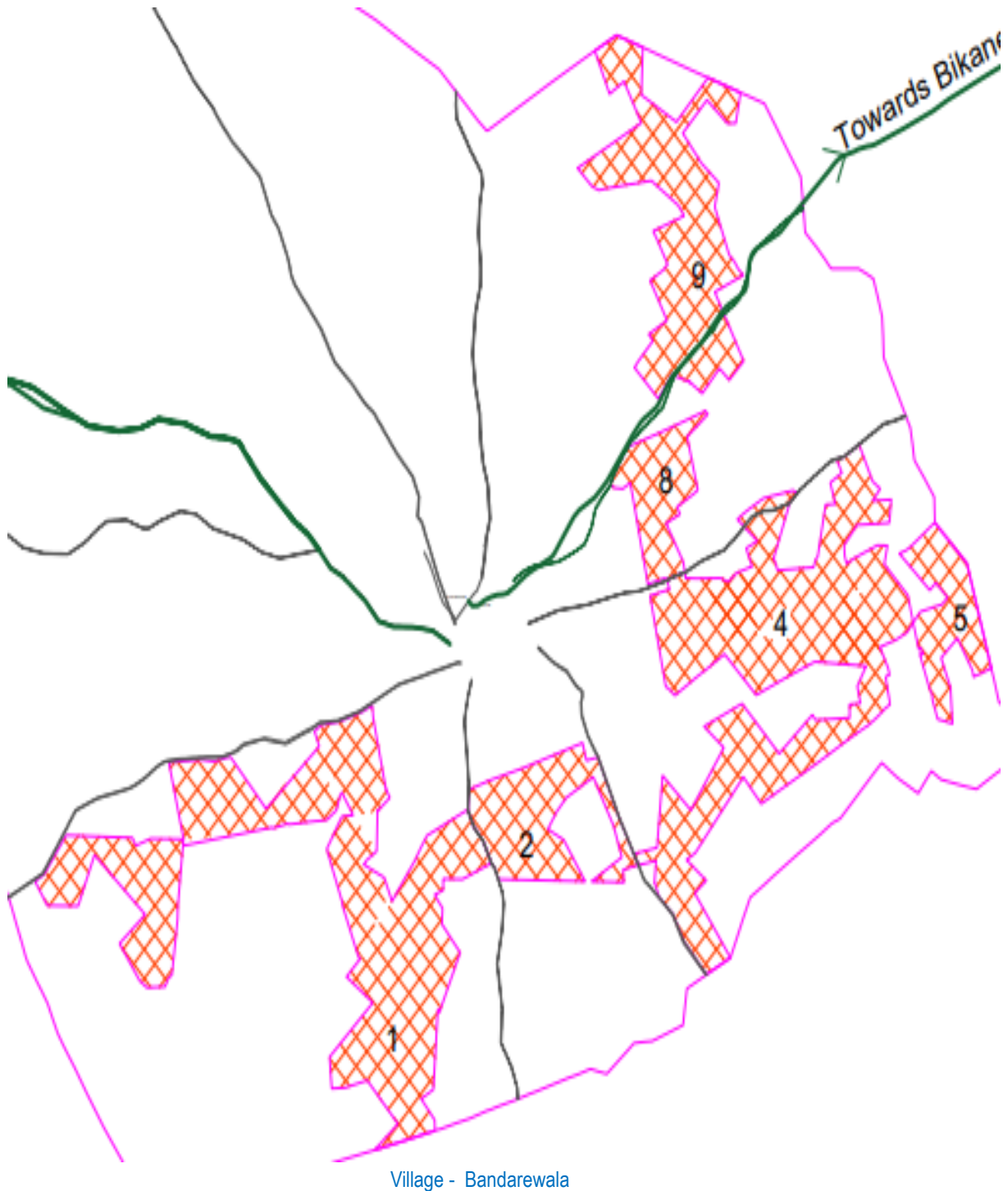


Figure 6: Land Layout of Ramsar Chhota & Bandarewala for Phase -I (1310 MW)

From the above layout, it can be clearly viewed that the red coloured Large solid filled patches of land in Ramsar Chhota & Red Colour Hatched in bandarewala village for which land acquisition work has been completed by RVUNL for development of RVUNL Solar Park for Phase -1 (1310MW).



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.1.3 Total Power capacity

RVUNL Solar Park has been proposed for 2000 MW (AC) in phased manner. Currently 1310 MW (AC) has been proposed in phase- I at Village Ramsar Chhota & Bandarewala at tehsil- Pugal, Distt- Bikaner.

4.1.4 Size & Number of Plots.

The entire land parcel identified for the development of solar park of 1310 MW capacity is spread across two village under Pugal tehsil, village - Ramsar Chhota & Bandarewala.

The land allotted by Revenue Department, District Collectorate Bikaner under Lease of 30years. Development of RVUNL 1310 MW (Phase-1) Solar Park has been proposed in Ramsar Chhota & bandarewala Village which comprises of 2294.41 hectare of land with 55 No's of Khasra Plots & 732.58 hectares of land with 90 no's of khasra plot respectively which is further subcategorised as 3 Numbers of plot.

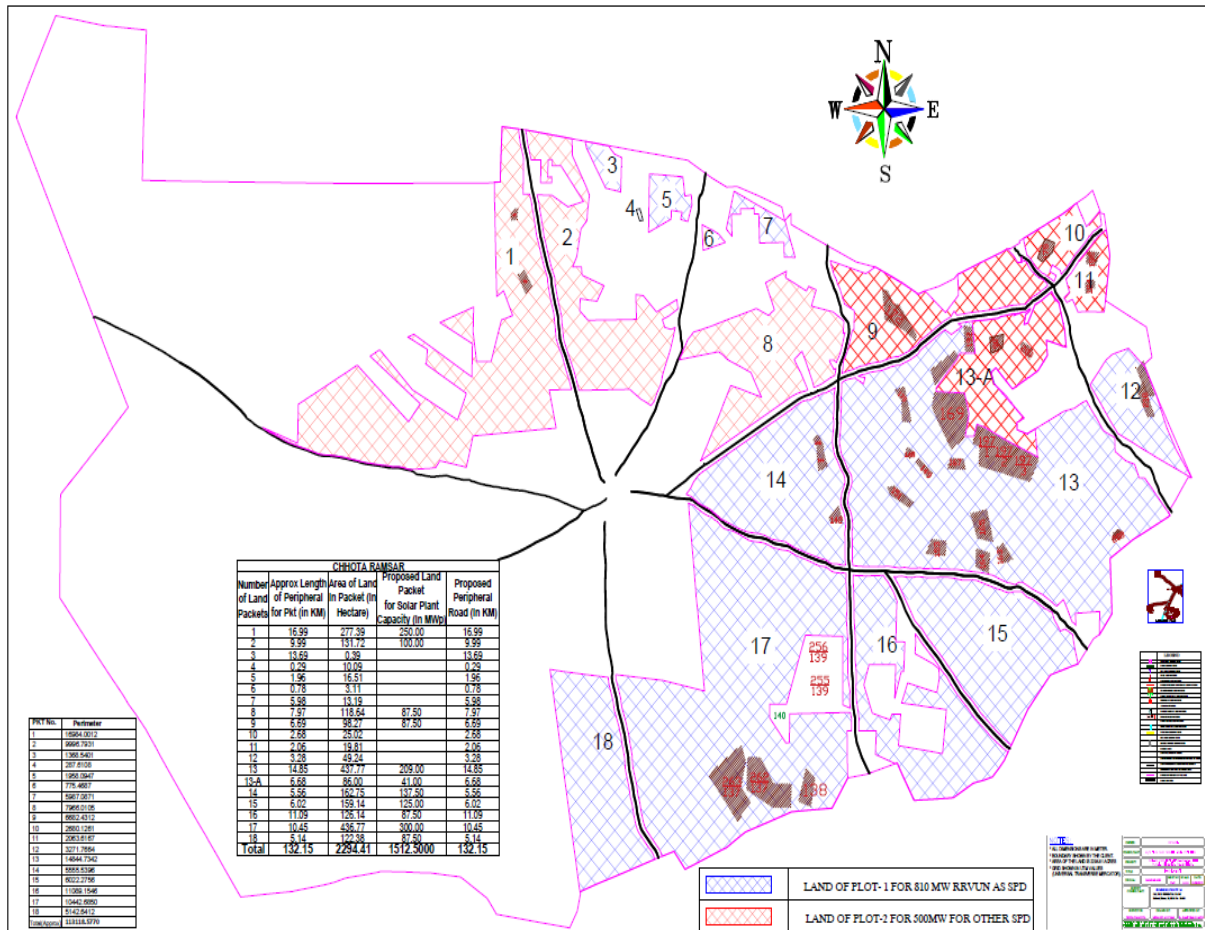
Village	No of Land Parcels/ Plots	Land Area (Ha)	Land Area (Aces)	Proposed Solar Capacity (MW)
Ramsar Chhota (Phase-I)	55	2294.41	5669.61	995 MW
Bandarewala (Phase-I)	90	732.58	1810.20	315 MW

Land of 1310 MW has been further divided in 3 no's. of plot comprising one plot for 660 MW capacity, second for 335 MW & third for 315MW capacity. Land for all the plots are identified & tabulated below. For reference subdivision of land plot drawing has been provided below the table.

Land category	Proposed Solar Capacity (MW)	Land Area (Ha)	Land Area (Aces)
Plot -1 (Ramsar Chhota)	660 MW	1537.56	3799.39
Plot- 2 (Ramsar Chhota)	335 MW	756.85	1870.21
Plot- 3 (Bandarewala)	315 MW	732.58	1810.20



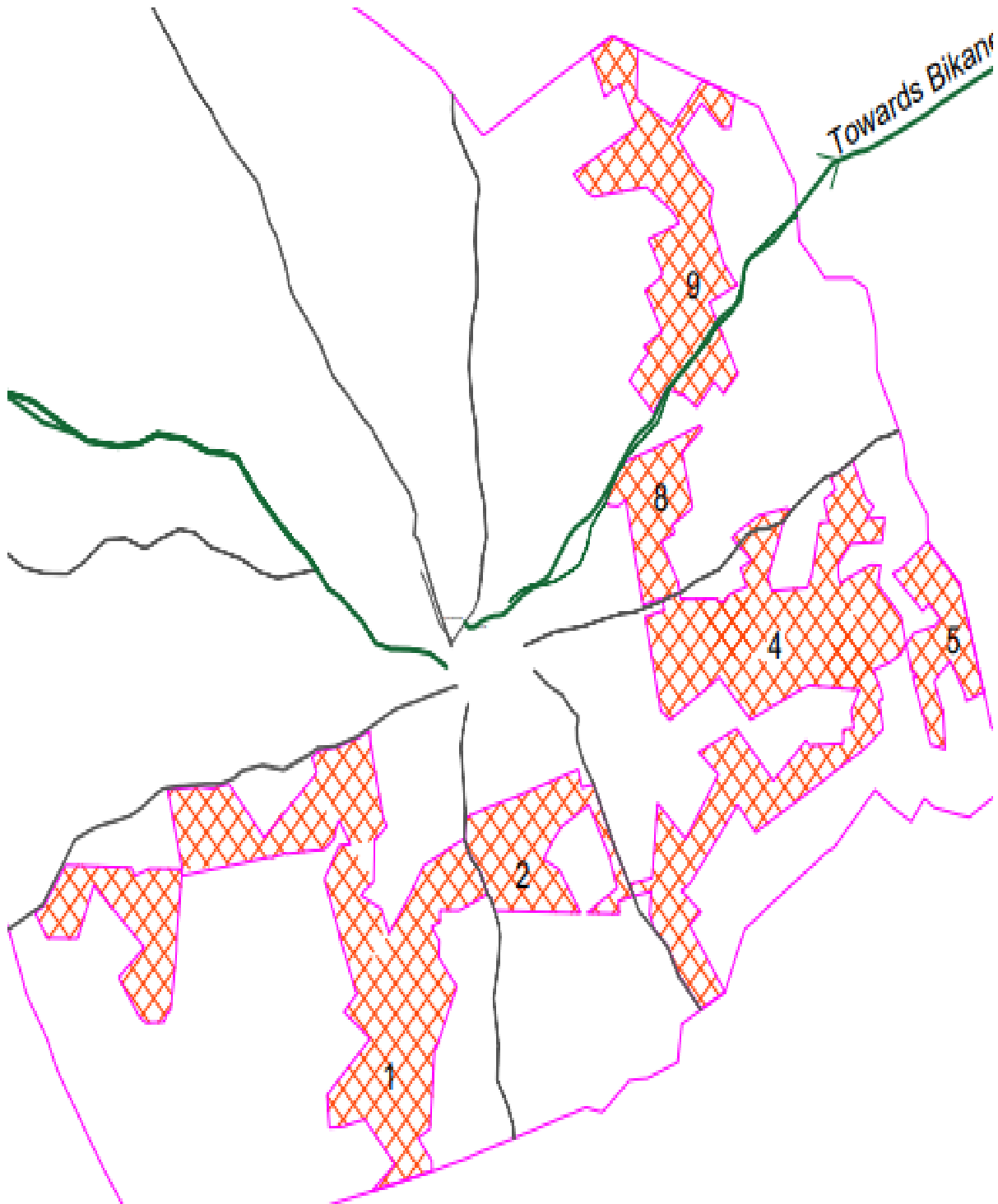
Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.



Number of Land Packets	Approx Length of Peripheral for Pkt (in KM)	Area of Land In Packet (In Hectare)	Proposed Peripheral Road (In KM)
1	16.99	277.39	16.99
2	9.99	131.72	9.99
3	13.69	0.39	13.69
4	0.29	10.09	0.29
5	1.96	16.51	1.96
6	0.78	3.11	0.78
7	5.98	13.19	5.98
8	7.97	118.64	7.97
9	6.69	98.27	6.69
10	2.68	25.02	2.68
11	2.06	19.81	2.06
12	3.28	49.24	3.28
13	14.85	437.77	14.85
13-A	6.68	86.00	6.68
14	5.56	162.75	5.56
15	6.02	159.14	6.02
16	11.09	126.14	11.09
17	10.45	436.77	10.45
18	5.14	122.38	5.14
Total	132.15	2294.41	132.15



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.



Details of Land Sub- division based on packet division of plots for Ramsar Chhota & Bandarewala

Land division for allotment of solar plots to SPDs.

Blue coloured Patches has been considered for Plot – 1 (660 MW)



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

Red coloured patches has been considered for Plot – 2 (335 MW)

Red Coloured patches has been considered for Plot- 3 (315 MW)

4.1.5 Internal transmission Infrastructure requirements within park & upto sub-station of STU/ CTU.

The Power Generated from Solar power developers will be evacuated in two parts.

1. Collecting and Transmitting power at 33kV level from each Block/ ICR to Interfacing Point or Solar Park Pooling sub-station of RVUN as first part.
It is responsibility of SPD to take care of this part.
2. Transmission from Interfacing point to Pooling sub-station & PSS to the Grid Sub Station of RVPN at 400kV is second part .
It will be facilitated and taken care by RVUNL. It is responsibility of SPPD (RVUNL) to take care for the development & maintenance of this part.

4.1.6 External transmission Infrastructure

New Grid Substation 400kV expandable to 765kV infrastructure has been sanctioned by RVPNL Considering this solar park for power evacuation & further transmission as required. This New GSS has been planned by RVPNL near to its existing Surasar Substation 132kV for RVUNL Solar Park. This newly Proposed GSS has been considered with further expansion.

4.1.7 Road Infrastructure

Existing Main Roads passing through solar park will be widened to 10-meter width (7.5 meters BT and 1.25-meter shoulders either side) and strengthened.

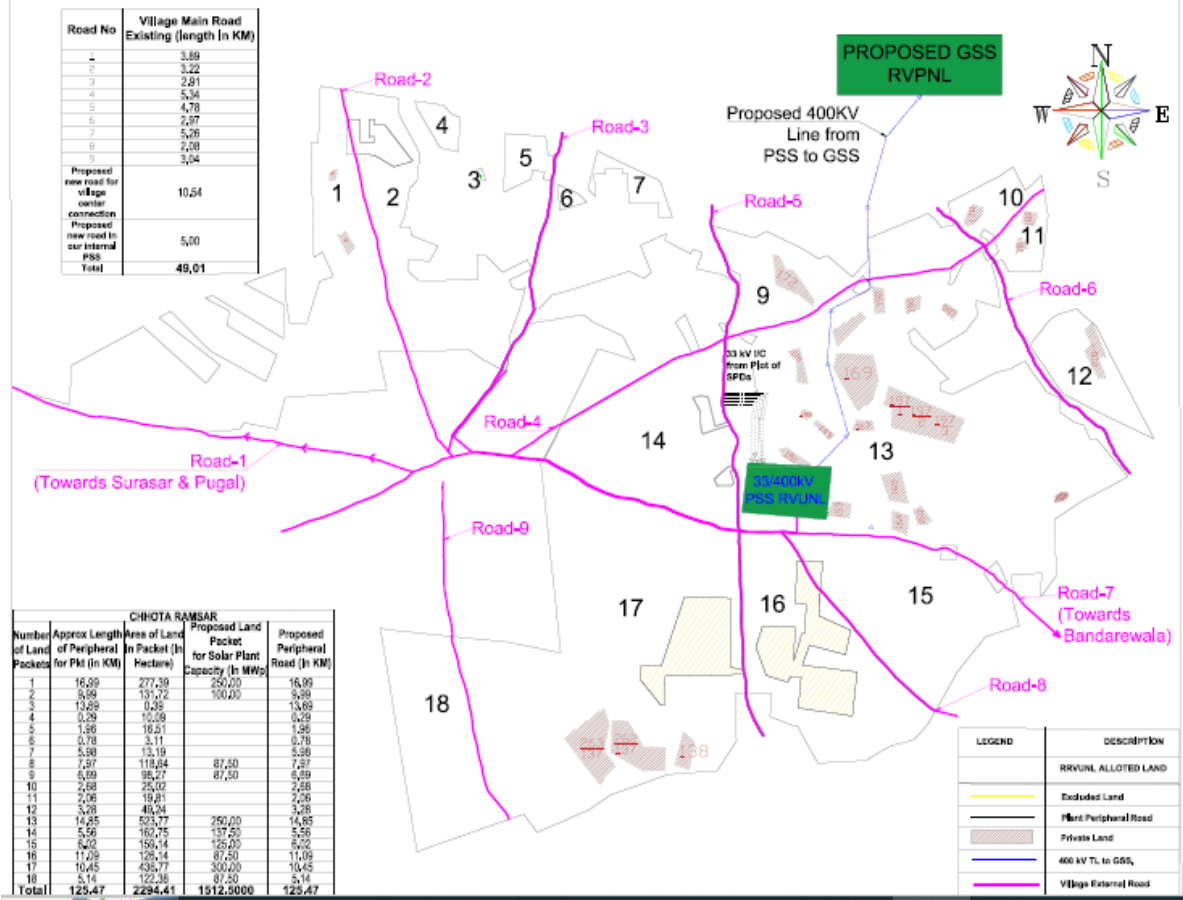
New Road Connecting to PSS will be constructed of width 10 meter including shoulder both side & its maintenance will be done by SPPD's.

New Internal Peripheral Secondary Road inside the Boundary of Solar Park & as required for the local villager's land coming next to our land patches will be developed by SPPD up to 7.5 mtr wide (5.5 mtr WBM & 1 mtr shoulder each side) & its maintenance will be done by SPPD's.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

LAYOUT OF EXTERNAL BT ROAD & INTERNAL PERIPHERAL ROAD FOR RVUN SOLAR PARK 1310MW AT CHHOTA RAMSAR



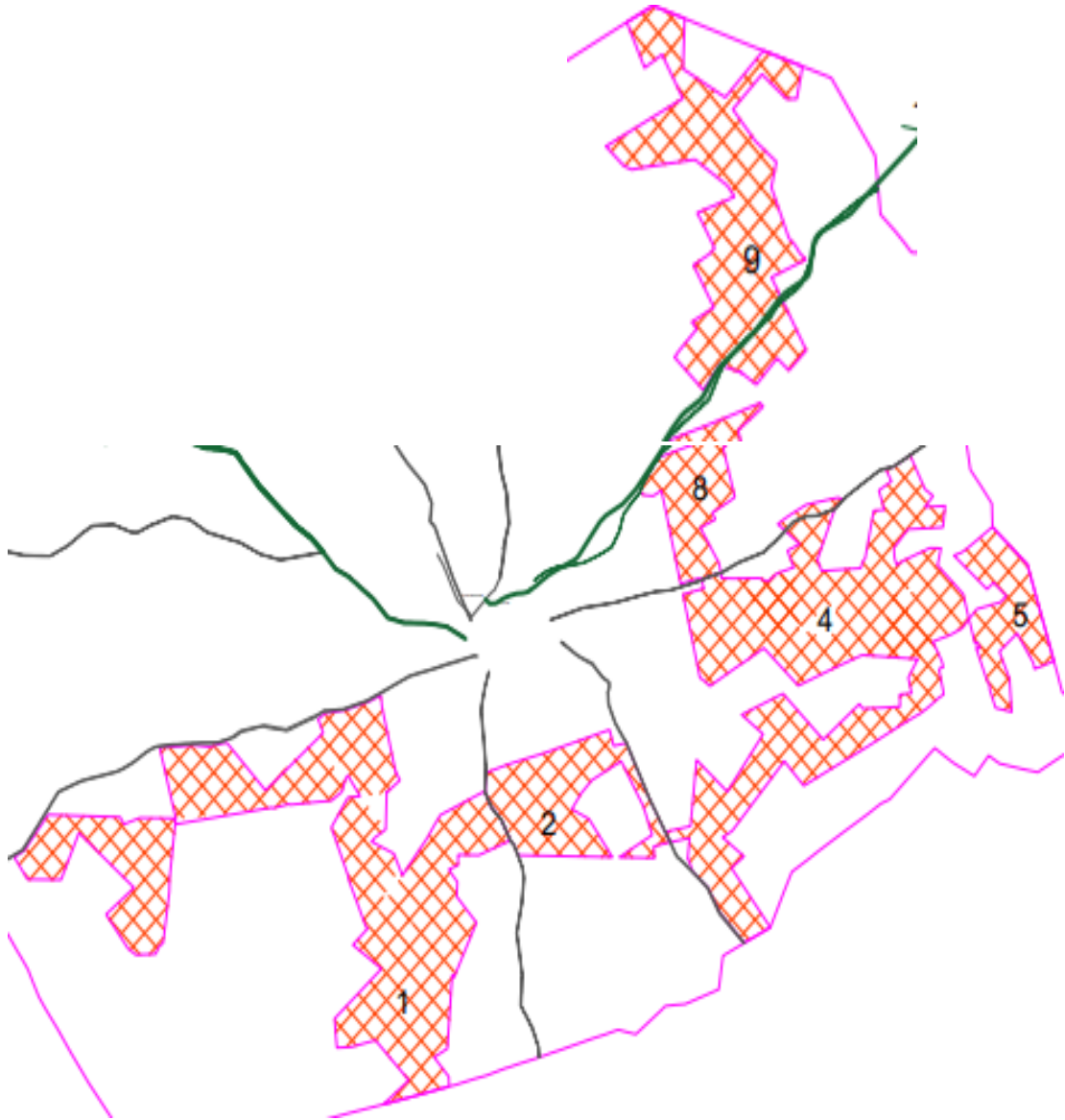
Road Line Diagram of Ramsar chhota village

Road No	Village Internal-Road Existing (length In KM)
1	3.89
2	3.22
3	2.91
4	5.34
5	4.78
6	2.97
7	5.26
8	2.08
9	3.04
Proposed new road for village center connection	
10.54	
Proposed new road in our internal TSS	
5.00	
Total	49.01

Details of existing road length strengthening and proposal of new construction



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.



BANDAREWALA		
Number of Land Packets	Approx Length of Peripheral for Pkt (in KM)	Area of Land In Packet (In Hectare)
1	18.34	272.93
2	5.27	55.52
4	16.16	223.75
5	3.90	36.06
8	3.22	35.44
9	8.22	108.88
Total	55.11	732.58



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.1.8 Water Pipeline / Supply/ Reticulation & Recirculation

To meet the water requirement Water will be pumped from Indira Gandhi canal (panna lal pumping Station).

For the maintenance of canal, the supply of water will be closed for 30 days. For fulfilling the water requirement in this canal closure duration, the storage of raw water reservoir of 100 lakh liters capacity has been proposed.

Treated water will be supplied in solar park through Overhead Tanks with suitable pipeline to the nearest boundaries of SPD.

Further SPD can plan for their internal storage and circulation up to module area or as per their requirements.

4.1.9 Common facilities Planned

Land of entire Solar Park will be facilitated with interconnected 10mtr wide bitumen road including shoulders. Internal road is proposed with 7.5mtr wide as WBM Roads. Water Storage system & supply system has been proposed considering the closure of canal of 30 days with proper treatment.

Portable type Security Cabins & Structural Elevated watch tower has been proposed for prevention of trespassers. CCTV & Monitoring system has been Planned for Remote security.

Mobile internet/connectivity enhancement with other safety measures and fire firefighting system has been Considered in the proposal of solar park development by SPPD (RVUNL).

4.1.10 CSR Activities

Indicative activities to be taken up by RVUNL in a phased manner with a mutual understanding with the local communities and keeping in consideration the financial availability, feasibility and practicality of implementation are given below:

(i) Infrastructure development Considered for the nearby Communities

(ii) Sustainable livelihood activities in the Communities

Detailed description is discussed in 4.5.7 under CSR activity.

4.2 Ownership & Location of Solar Park

Ownership of solar park is available with RVUNL for 1310MW solar park at village Ramsar Chhota.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.2.1 Mode of Approach for Location

4.2.2 Connectivity

Road

The site is best approached from Jamsar – Attarabad Road through National Highway (NH) 62, which connects the site to the National NH 11 from Bikaner to Fatehpur. NH 11 which connects the site to NH 52 Fatehpur, and Jaipur is 175 km.

Table 7: Road Connectivity to Nearest Township

Parameter	Location
Nearest Tehsil	Pugal, 46.2 km
Nearest District Headquarters	Bikaner, 45.4 km
Nearest State Headquarters	Jaipur, 384 km

Airport

The nearest airport to the site is Bikaner Airport which can be best approached via NH 11. The site is also well connected to airports at Bikaner and Jodhpur. Jodhpur airport is the nearest airport to facilitate cargo transportation. The distances to the airports from the site are as follows:

Table 8: Distance to Nearest Airport

Location	Airport Type	Distance
Bikaner	Domestic	48.2 KM
Jaipur	Domestic & International	398 KM
Jodhpur	Domestic	300 KM
Jaisalmer	Domestic	329 km

Port Connectivity

The nearest Major port to the Project Site is Kandla port in Gulf of Kutch in Gujarat, which is located about 903 km to the south of the Site. The Kandla port and Bikaner is well connected by NH 141, NH 41, NH 27, NH 62, and Gujarat State Highway SH 54 and hence, movement of cargo, machineries, equipment, etc. using these corridors up to the project site is convenient.

Table 9: Distance to Nearest Port

Location	Port Type	Distance
Deendayal Port Trust (Kandla)	Domestic & International	903 KM
Mundra	Domestic & International	957 KM



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.2.3 GPS Coordinates of Boundary of land.

Plot No's.	Directions	Coordinates (Northing & Easting)	
Plot –1 (660 MW)	North side of land	28.194689 N	72.542258 E
	South side of land	28.165036 N	72.570280 E
	East side of land	28.181029 N	72.582877 E
	West side of land	28.171389 N	72.544836 E
Plot – 2 (335 MW)	North side of land	28.194689 N	72.542258 E
	South side of land	28.183075 N	72.562010 E
	East side of land	28.192644 N	72.580068 E
	West side of land	28.182723 N	72.530445 E
Plot- 3 (315 MW)	North side of land	28.46556 N	73.091388 E
	South side of land	28.29194 N	73.008056 E
	East side of land	28.28112 N	73.077778 E
	West side of land	28.29667 N	73.023612 E

4.2.4 Nature of Land -

The Maximum portion of land acquired for entire Solar Park has been leased by land revenue Department by District Collectorate Bikaner for 30 years to SPPD RVUNL. However, if any small patches of private land coming in between of allotted land will be purchased based on the mutual agreement of private landowner & RVUNL.

The land has been classified under the Silty Poorly graded & Silty Sand as per Preliminary Geotech Investigations.

4.2.5 Land Ownership Status-

The Maximum portion of land is of Govt & has been leased by Rajasthan State Govt Revenue Deptt, Collectorate Office Bikaner and allotted to Rajasthan Rajya Vidyut Utpadan Nigam Limited (RRVUNL) for a period of 30 years & extendable if required. Further private land coming in between the allotted land can be purchased after the mutual agreement with landowner & RRVUNL.

4.2.6 Land Acquisition Process-

Land acquisition has been completed by RRVUNL (SPPD) & Land Demarcation work is under process.

4.2.7 Land Allocation process to SPDs

Park Phase-I 1310 MW comprises of 3026.99 hectare which is categorised under three no's of plot detailed with 660MW in plot 1, 335MW in plot 2, 315MW in plot 3 detailed described below.



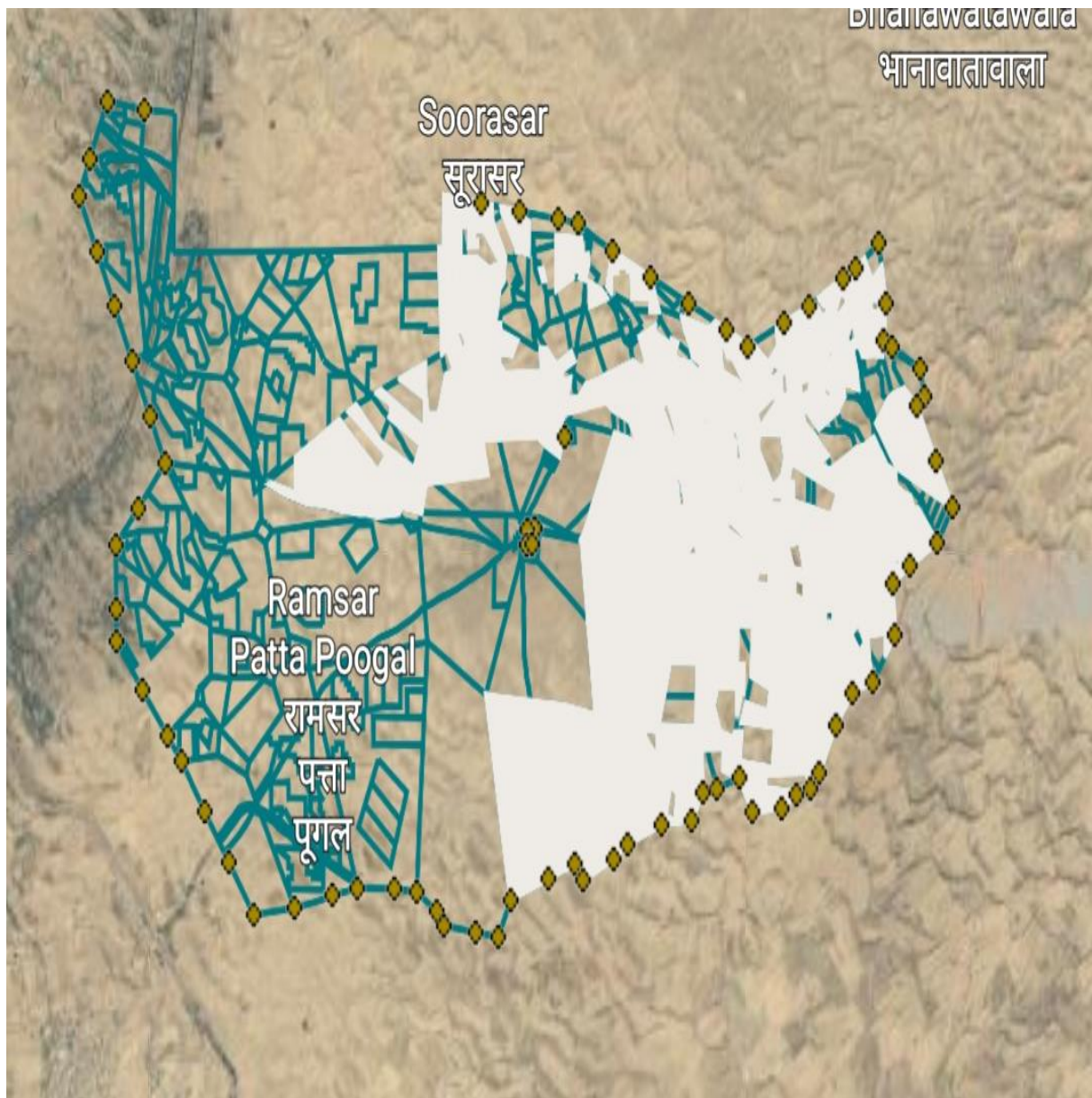
Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

Plot 1 has been summarised for a cumulative capacity of 660 MW for which land size has been considered for 1537.56 Hectares.

Plot -2 has been Considered for a cumulative capacity of 335 MW for which land size is 756.85 hectare.

Plot- 3 has been considered for a cumulative capacity of 315 MW for which land size is 732.58 hectares.

4.2.8 Google Image with Boundaries





**Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at
Bikaner, Rajasthan.**

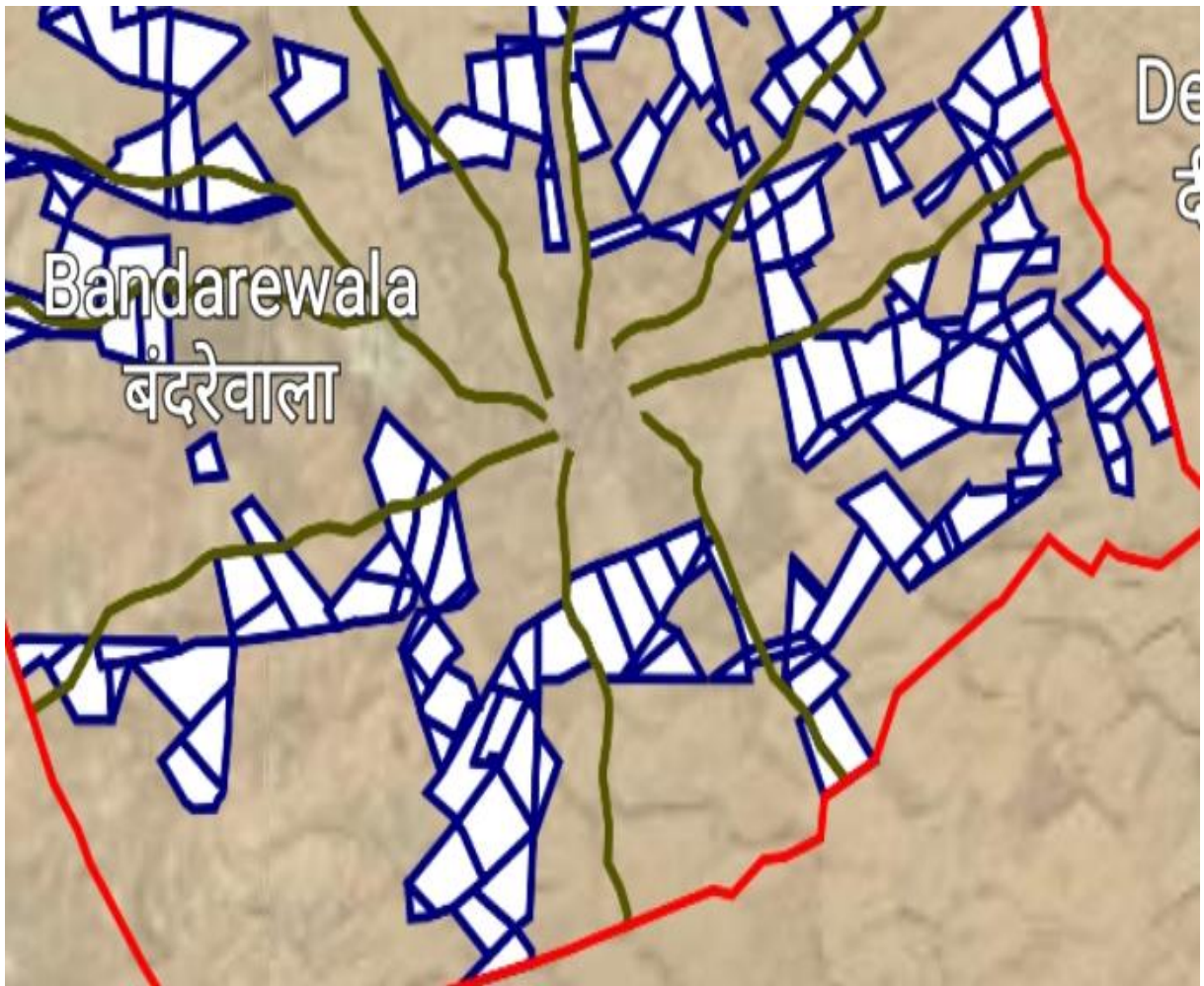


Figure 7: Google Earth Image of land in bandarewala (White Colour Shade)



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.2.9 AutoCAD or Technical drawing

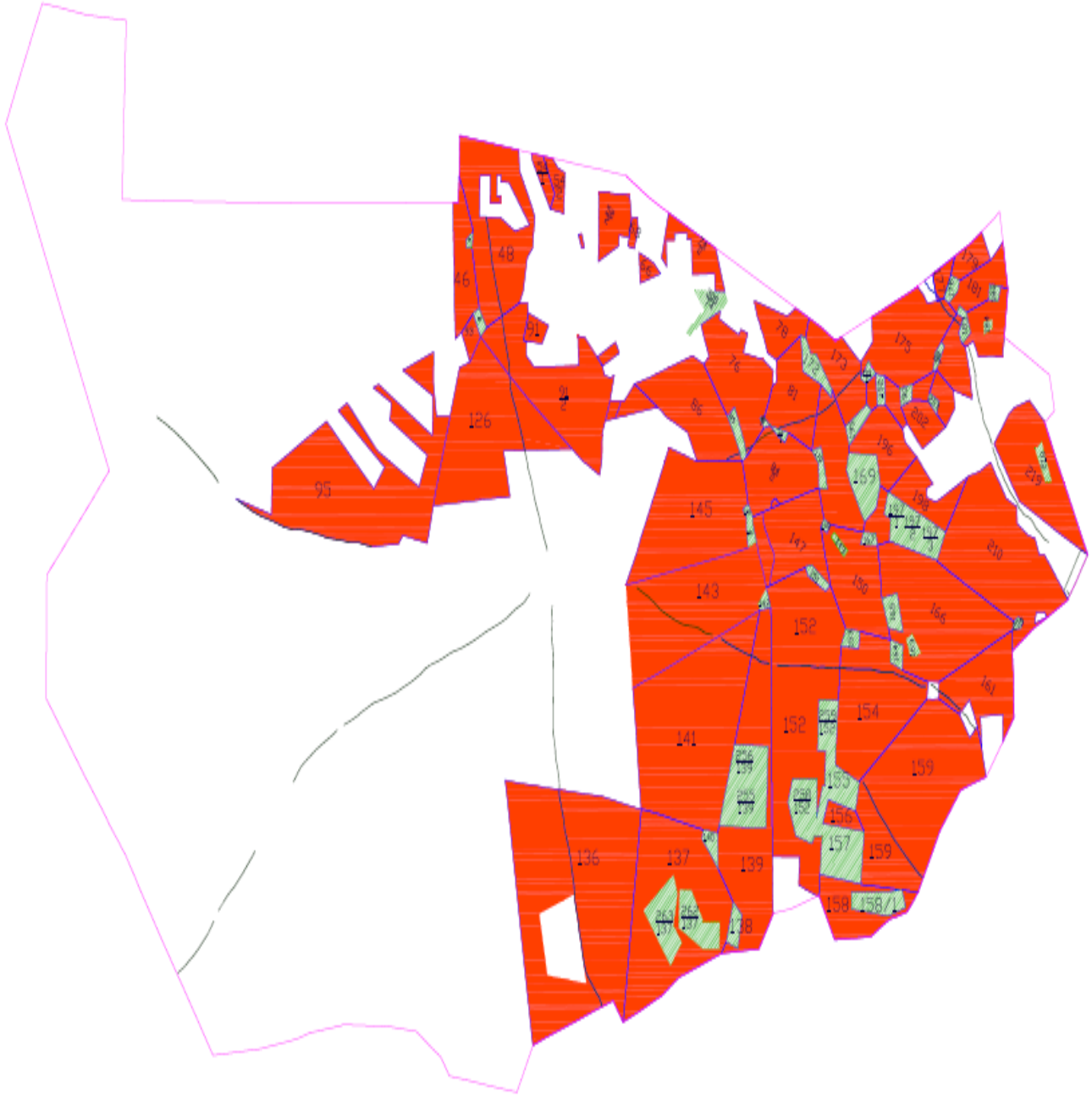


Figure 8: AutoCAD Image of Land.



Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

4.3 Solar Irradiation & Weather Data

4.3.1 Solar Radiation Including GHI

India being a tropical country receives good sunshine over most parts of the country, and the number of clear sunny days in a year is also quite high. It receives solar energy equivalent of more than 5,000 trillion kWh per year; however, the equivalent solar energy potential is about 6,000 million GWh of energy per year. In India, the daily GHI is around 5.0 kWh/m² in north-eastern and hilly areas to about 7.0 kWh/m² in western regions and cold desert areas with the sunshine hours ranging between 2300 and 3200 per year. In most parts of India, clear sunny weather is experienced for 250 to 300 days a year. The annual GHI varies from 1600 to 2200 kWh/m². Around 25-30 percent annual diffuse irradiance (DHI) component has been indicated by IMD in most of the Indian locations. A GHI map for India, developed by The Energy and Resources institute (TERI) based on the data of IMD (1980, 1982) is presented in below Figure.

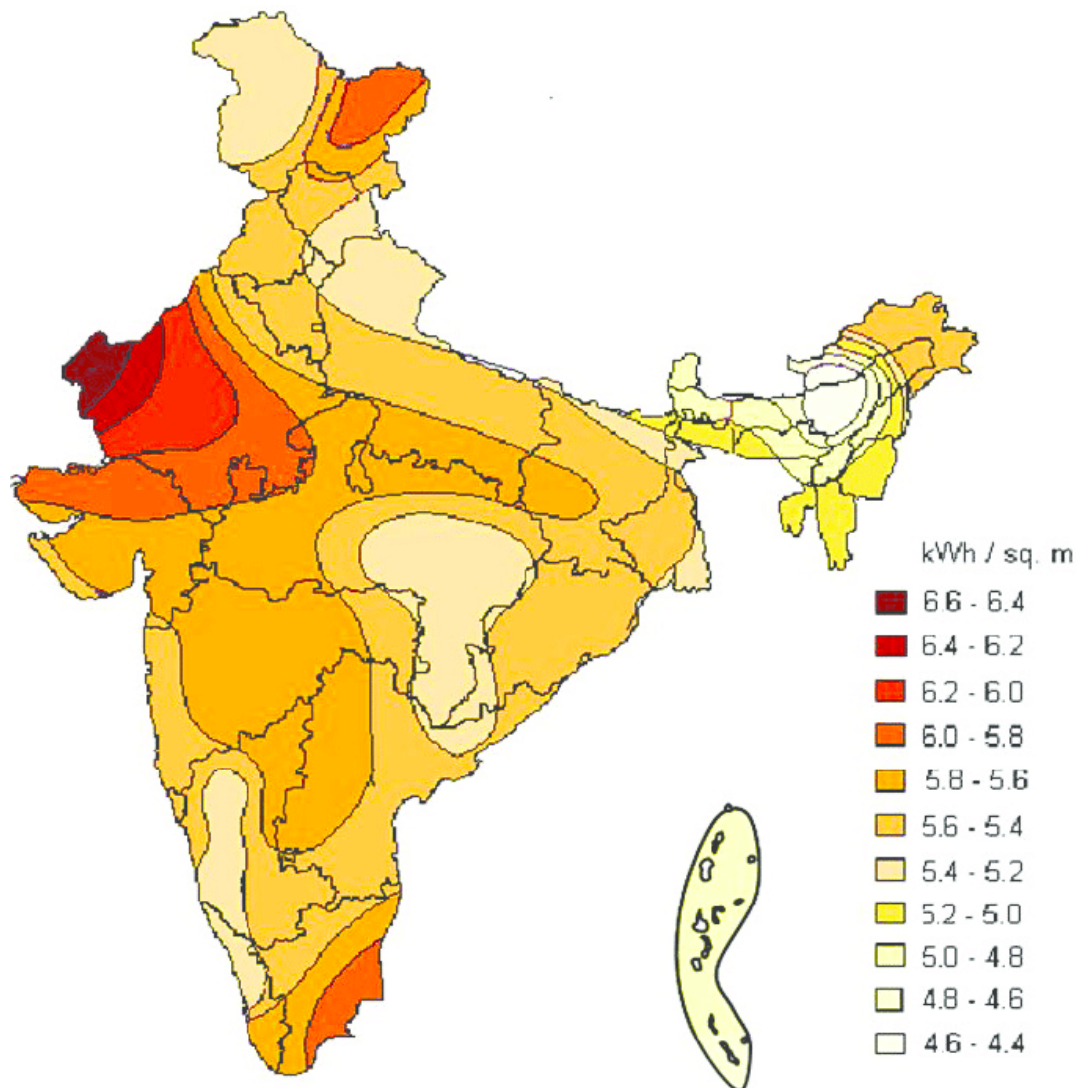


Figure 9: Global Horizontal Irradiance Map of India (Source: TERI)



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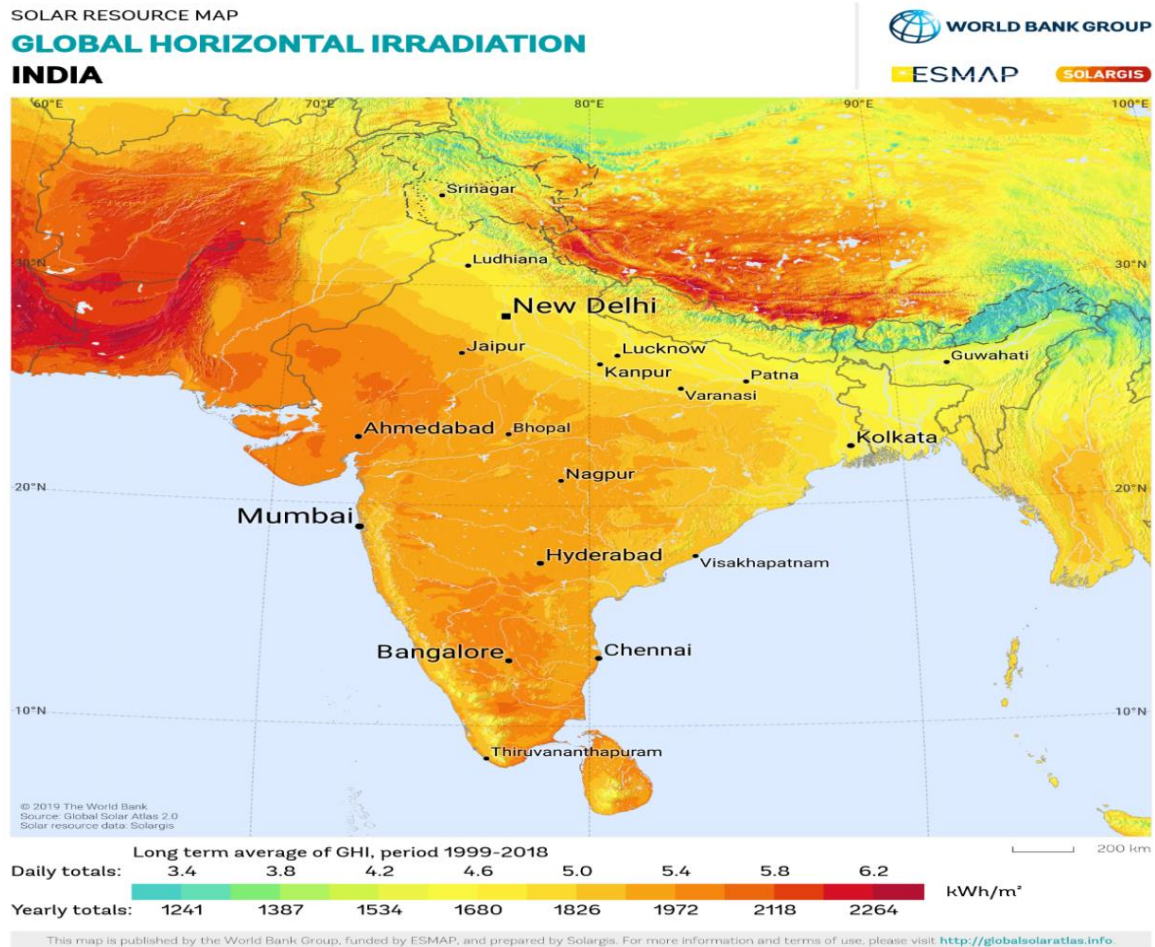


Figure 10: Global Horizontal Irradiance Map of India (Source: Solargis)

GHI is essential for stationary type of solar energy collectors/ PV modules; however, DNI is required for concentrating collectors (CSP and CPV systems). It is well established that one year of measurements could be more than 15% above or below the long-term average whereas averaging five years still could lead to deviations larger than 10%. Therefore, at least 10 years of data should be considered to gather reliable information on solar resources

4.3.2 Solar Radiation Databases

Availability of long-term ground solar radiation (GHI and DNI) data over the potential locations (where waste land is available) is one of the major barriers towards rapid deployment of solar energy projects in India. Hence the dependence of the project developers and lenders is mainly towards the satellite data or interpolated data.

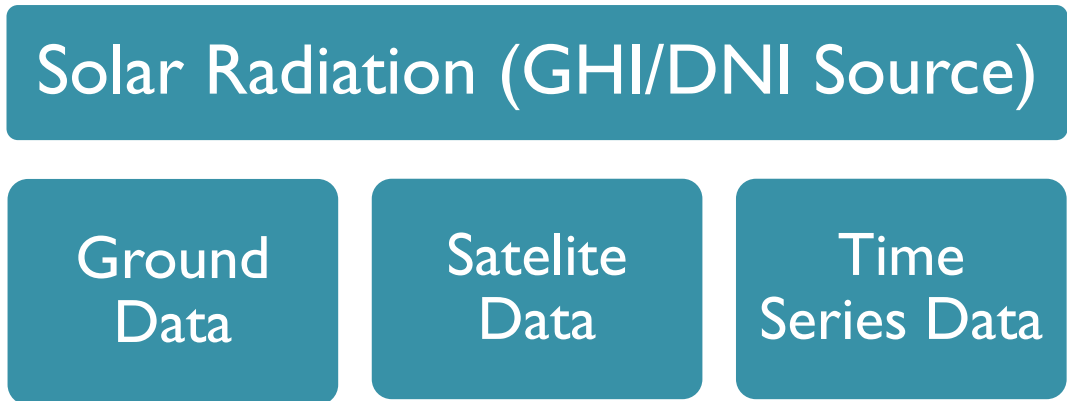


Figure 11: Different Available Source for Meteo Data

4.3.2.1 Ground Measured Data

These are the databases which are sourced from ground station measurements. The measuring station consists of radiation measuring instruments installed as per standard guideline and such instruments record data round the year which are stored in data logger. In India, IMD databases are based on ground stations.

Recently the Centre for Wind Energy Technology (C-WET11) of MNRE has implemented around 51 weather stations across the country for GHI, DI and DNI measurement under solar radiation resource assessment (SRRA) programme of MNRE under Phase-I; however, 60 AWS are under implementation under Phase-II.

Although these data are considered best for any analysis, but unavailability of ground stations at all the potential locations limit their usefulness. The accuracy of the measured data is challenged by the accuracy of instruments and maintenance practices including cleaning, calibration, and quality check procedures. Long term solar radiation data is available only for a few locations of the country. ISHRAE weather database is based on the ground measurements of IMD and statistical approach.

4.3.2.2 Databases based on satellite images

These are the databases which are sourced from the satellite measurements. These databases measure the Albedo¹² and calculate the radiation at any reference location. The accuracy depends upon the spatial and temporal resolution¹³ along with other factors. Databases like NASA, NREL, 3TIER, DLR, SolarGIS belong to this category and can provide solar radiation in context of India. In case of unavailability of ground data, these data are the first point of reference to judge the solar potential at any location. Although the values are often too optimistic, these data do help in estimating an overall potential with a certain uncertainty. The key benefits of satellite data are spatial-continuous and time-continuous data with invariant uncertainty.



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4.3.2.3 Derived databases and system integrating data

These are the databases which are based on reanalysis of the collected data. The basic idea of reanalysis is to use a frozen state of the art analysis system and assimilate data using historical data. The databases namely Meteonorm (6.0, 7.0), PVGIS etc. belongs to this category. These databases use software and statistical tools to process the data collected from satellite and ground measurement to minimize the error in reporting the actual radiation potential at any location.

4.3.3 Solar radiation over Pugal

The Solar Park is proposed to be developed in Tehsil- Pugal, District-Bikaner. The proposed site is spread across three villages in the same tehsil of the Bikaner district. The total land availability for the proposed site considered for phase-1 (1310MW) is 3026.99 ha. The solar radiation data is 100% interpreted satellite data. Solar irradiation at the solar park location for phase-1 is > 1800 kWh/m². The average monthly GHI and DNI data for the period of 1991 to 2010 from SolarGIS and Meteonorm source is tabulated below.

Table 10: Solar Irradiance at Site Location – SolarGIS and Meteonorm

Month	Ramsar Chhota GHI (kWh/m ²)	
	SolarGIS	Meteonorm
January	116	110
February	134	133
March	187	166
April	201	195
May	206	205
June	182	194
July	170	179
August	177	173
September	177	167
October	165	146
November	124	123
December	112	103
Year	1951	1893

Key Observation

- Proper access to all three villages is available from Bikaner (District Headquarters) and Pugal (Tehsil Headquarters).
- Solar irradiation at the location is > 1800 kWh/m².
- No eco sensitive areas such as wildlife sanctuary, Dams, Reservoirs is reported in the identified location.



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- Upon Initial Observation, the soil has been categorised as soft / loose soil/ silty sand (Desert Sand).
- Variation in land profile may happen during sand shifting in seasons.
- Land Patches found have significant undulations.
- LV, HV & EHV Line crossing observed near to the identified land.
- Forest Area is reported near to identified project land.
- Proposed Project land is found to be in Zone II Seismic Zone which is least active.
- Proposed Project Land is in Low-risk area in terms of Flood Hazard
- Proposed Project Location is Identified as High Wind Zone ($V_b \sim 47$ m/s) and necessary precaution for structural design shall be considered.

The proposed project site is in under 'Hot deserts, Arid' category based on the climate at the region

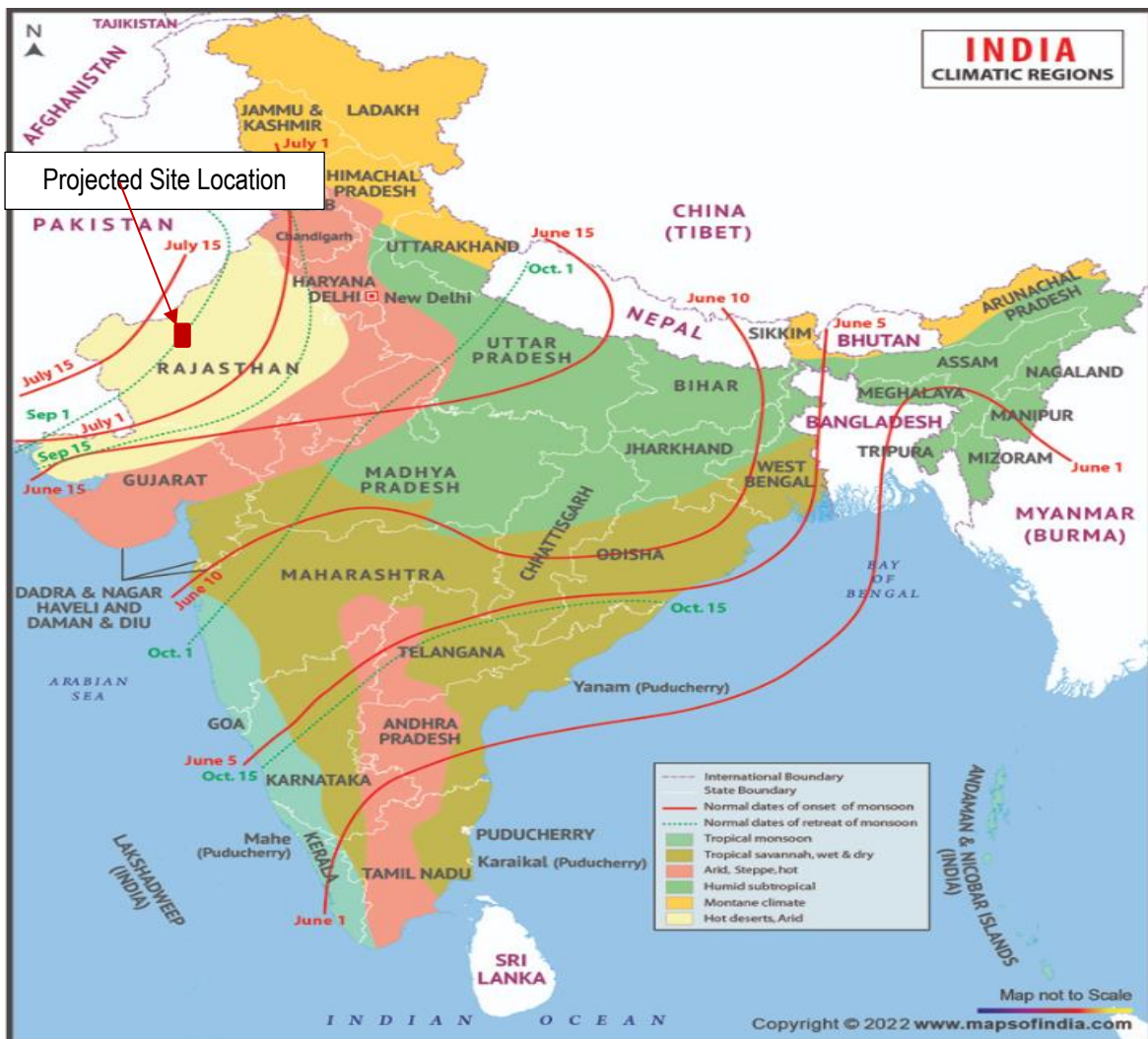


Figure 12: Climate Region Map



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4.3.4 Seismology

The site falls in Seismic Zone III of the seismic zone map of India, which indicates Moderate Damage Risk Zone which is liable to MSK VII. As, the site lies under moderate risk zone, developers need to take adequate measures while installing the Solar Panels to ensure least damage during earthquake.

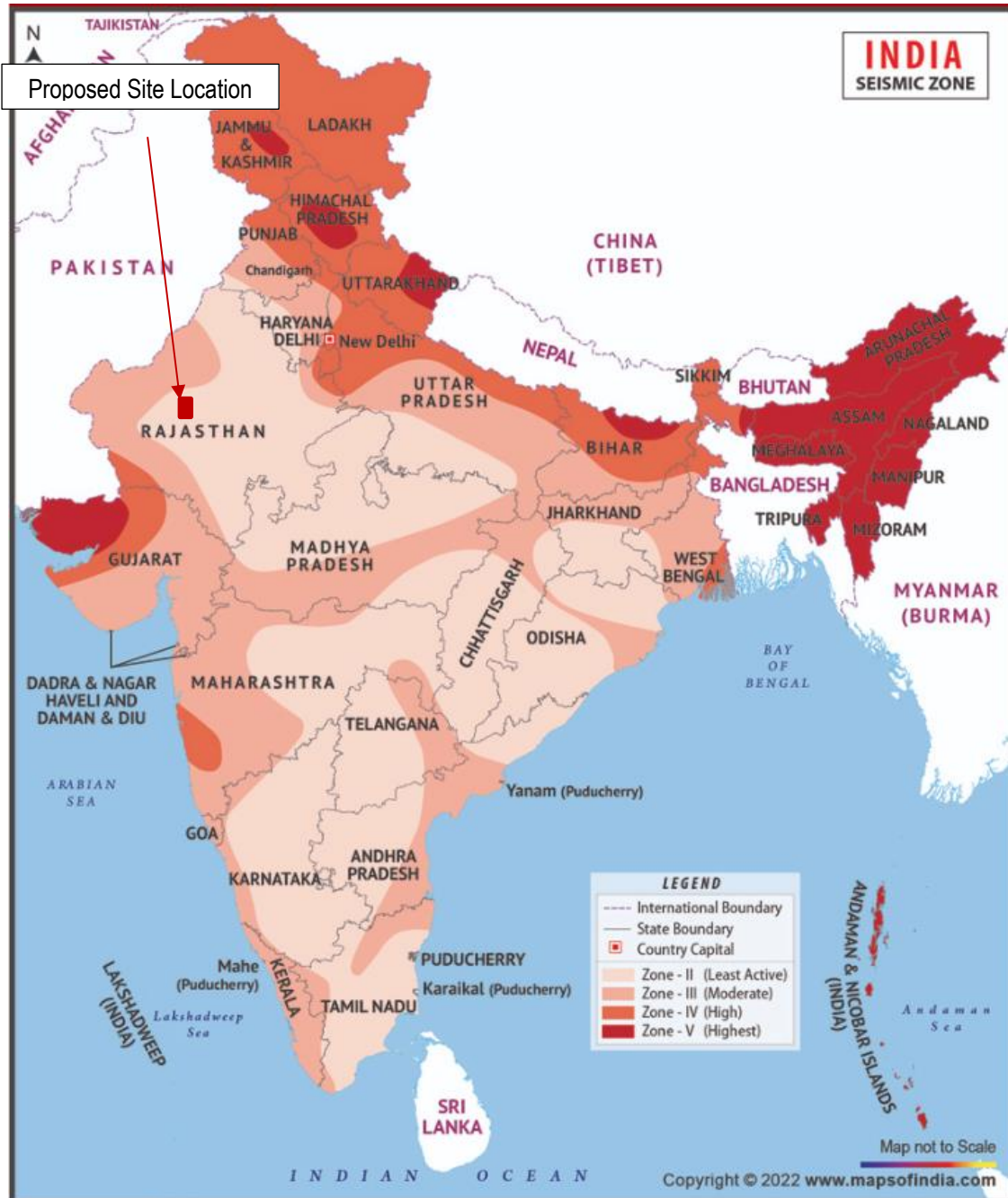


Figure 13: Seismic Zone Map

The proposed project site is in the Zone -3 (Moderate)



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4.3.5 Climate & Meteorological Conditions

4.3.5.1 Temperature

The average monthly Temperature data for the period of 2000 to 2009. The temperature data is interpreted from Bikaner (38 km), Hissar (273 km), Jodhpur (222 km), Jaisalmer (266 km), Delhi (392 km), New Delhi (401 km) locations.

Table 11: Monthly Average Temperature Data

Month	Average Ambient Temperature [°C]
January	15.5
February	19.9
March	25.9
April	31.1
May	35.6
June	35.3
July	33.4
August	32.3
September	31.7
October	28.9
November	22.8
December	17.4
Year (Average)	27.5

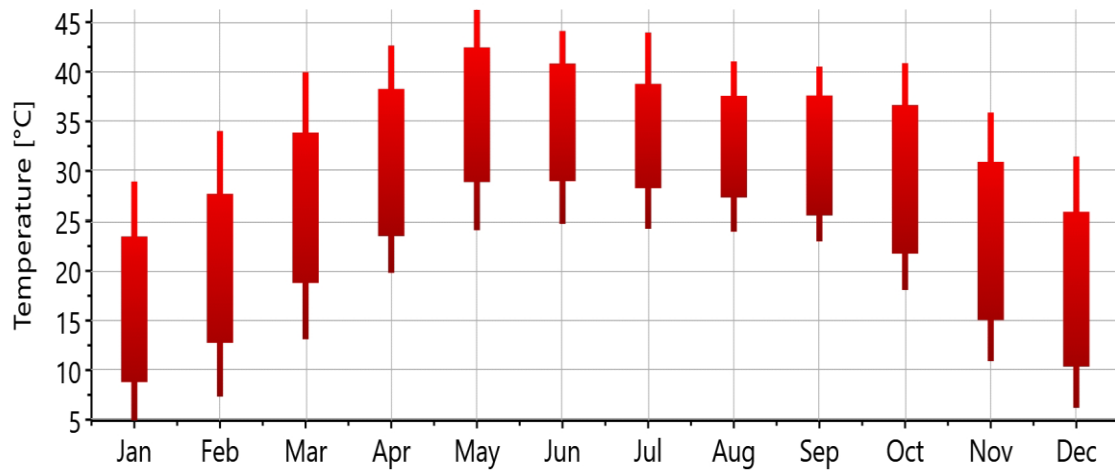


Figure 14: Monthly Minimum & Maximum Temperature at Project Location



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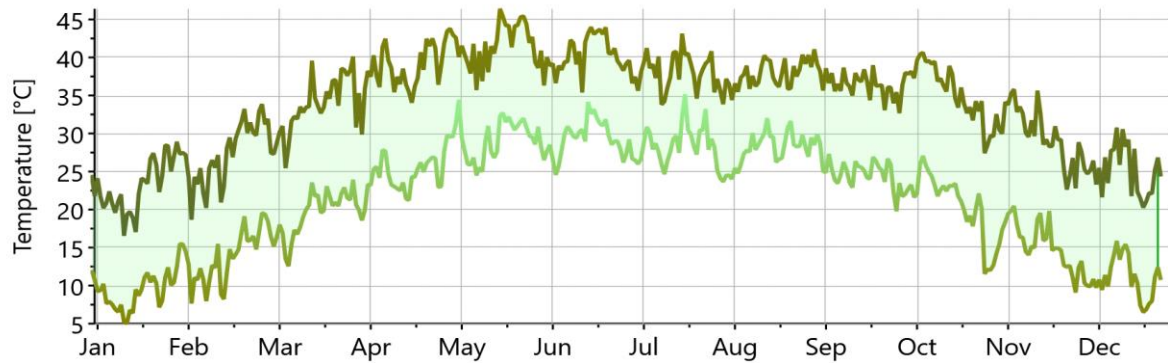


Figure 15: Daily Average Temperature at Project Location

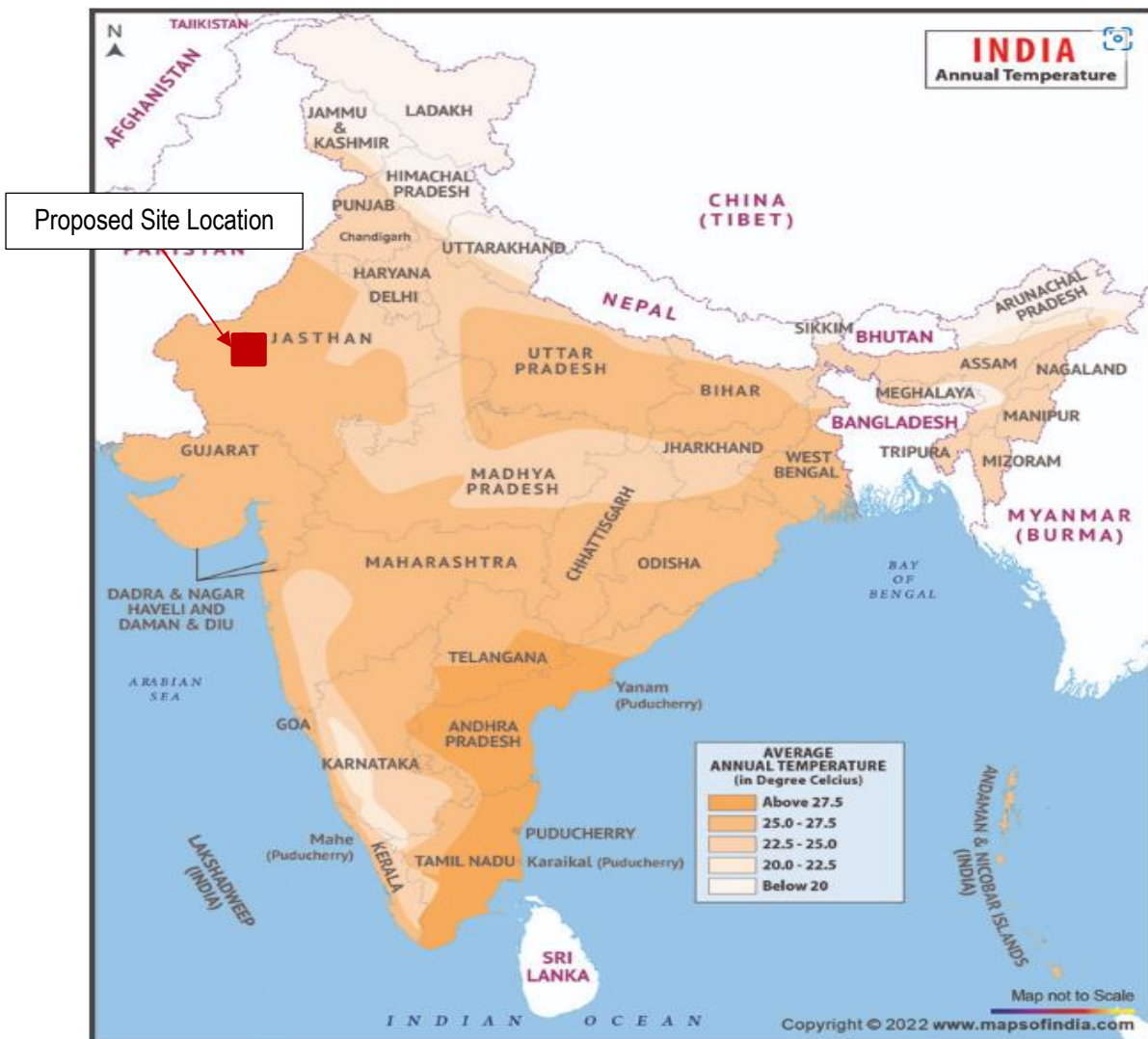


Figure 16: Annual Average Temperature Map

Annual average temperature zone of proposed project site is in the category of 25 to 27.5 °C.



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4.3.5.2 Rainfall

The average annual rainfall is 580 mm. between the driest and wettest months, the difference in precipitation is 223 mm.

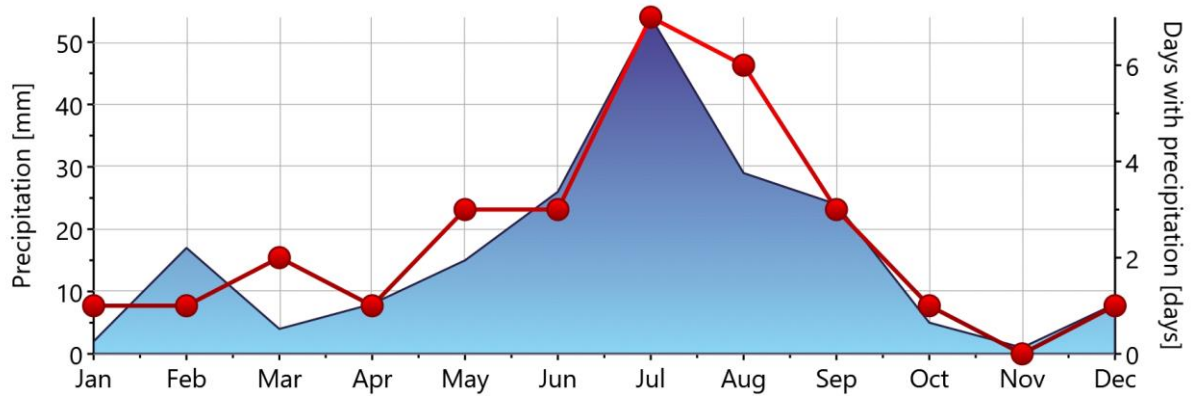


Figure 17: Precipitation Chart for Project Location

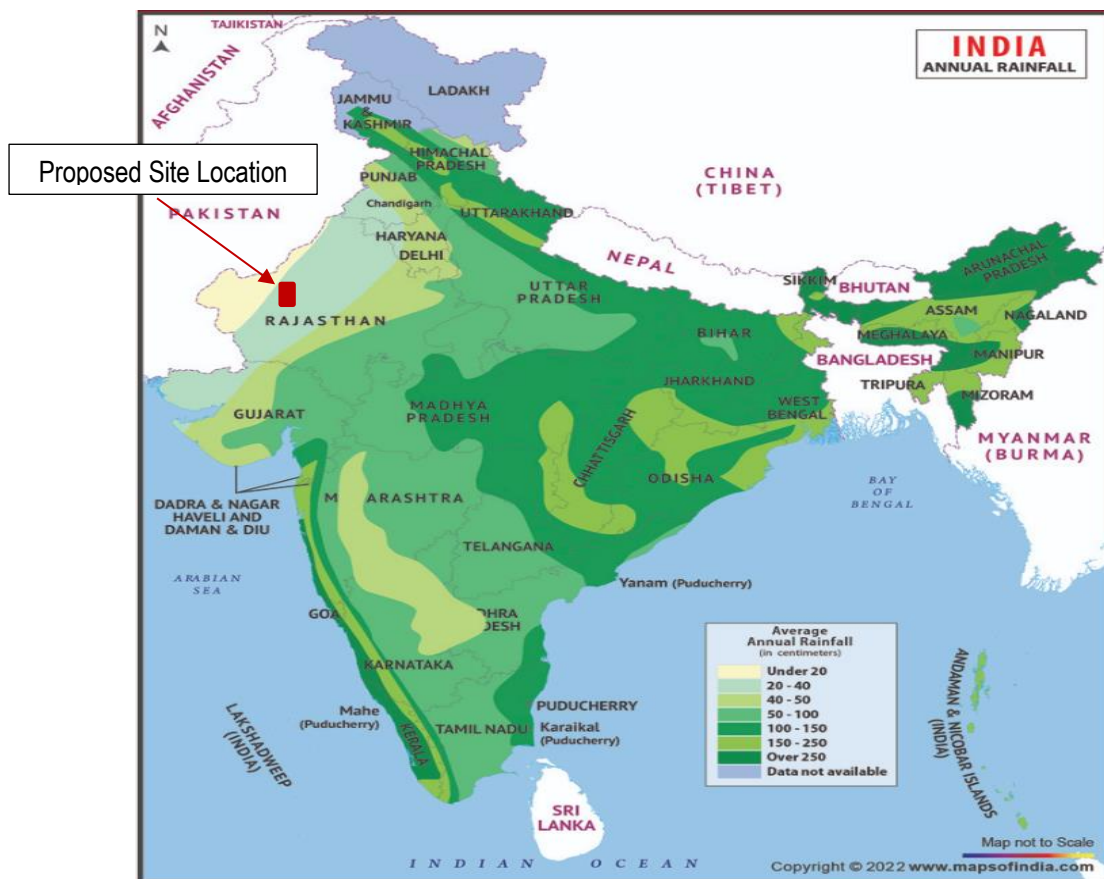


Figure 18: Annual Rainfall Map

The annual average rainfall at the proposed project location is about 20 to 40 cm.



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4.3.5.3 Wind

The wind speeds in the region are light to moderate with some strengthening during the southwest monsoon. The wind speed is generally high during the period from April to August. Wind pattern indicates that wind blows from northeast in post monsoon and winter season, from southwest in monsoon and from northwest in summer season.

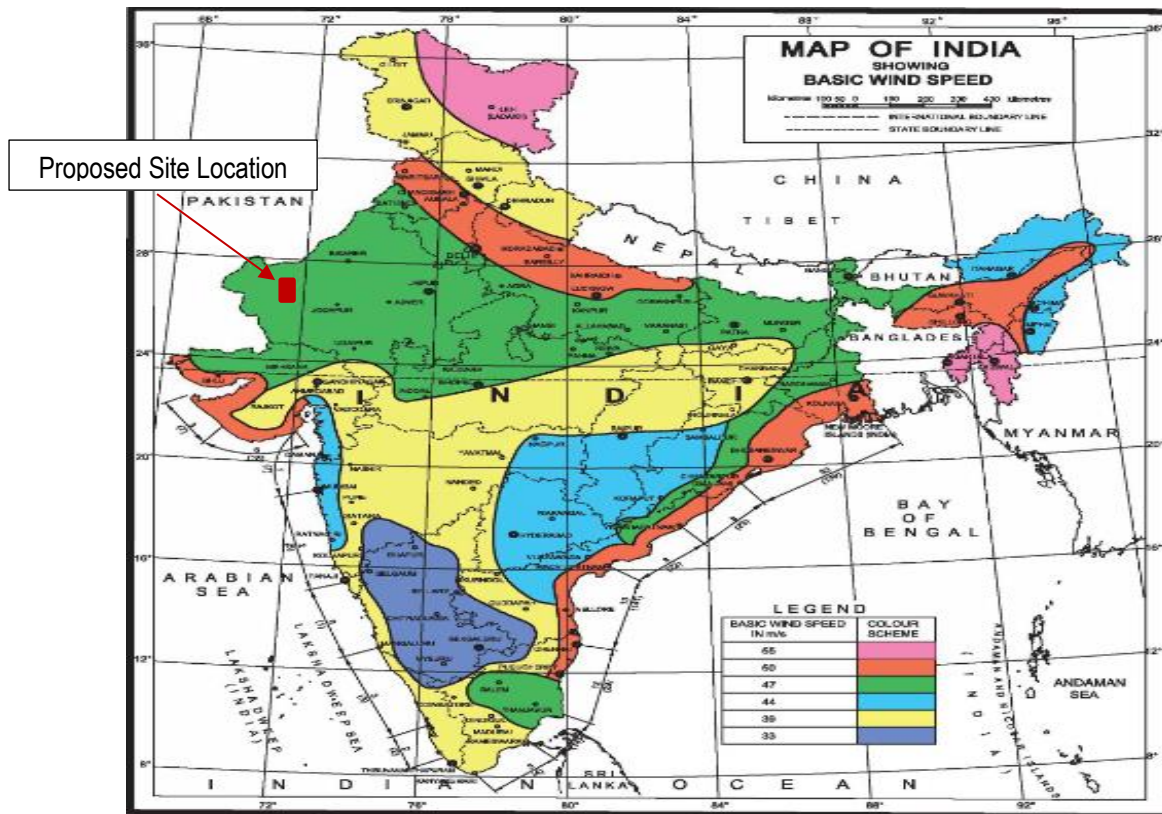


Figure 19: Wind Zone Map as per IS 875

Table 12: Monthly Average Wind Speed

Month	Wind velocity [m/s]
January	0.7
February	1
March	1.09
April	1.21
May	1.91
June	2.09
July	1.9
August	1.5
September	1.09
October	0.81
November	0.5



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Month	Wind velocity [m/s]
December	0.6
Year	1.2

4.4 Annual Energy Yield Assessment.

4.4.1 Simulation Using reputed PV Software

For solar park, energy yield cannot be exactly determined as PV modules selection by different solar project developers may differ. Further the D.C. to A.C conversion ratio may differ based on alignment of modules, inverter, cabling, transformers, Module installation (fixed tilt, seasonal tilt, angle of tilt), modules cleaning and maintenance practice and outage of transmission lines. Considering Average value of 20% CUF, annual yield will be 1.799 million kWh/MW/ annum and for Solar Park of 1310 MW it will be 2358 million unit/year.

Global Horizontal insolation is considered for non-concentration type collector and hence for PV modules. GHI to annual yield can be determined as under. (Calculations are for 1 MW output (A.C.).

Month	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	MWh	MWh	Ratio
January	115.80	50.80	13.60	146.70	142.60	118406.54	114536.58	0.88
February	133.70	50.70	17.20	161.60	157.20	128027.01	123819.24	0.86
March	187.10	68.80	23.50	209.50	203.70	160943.87	155554.95	0.83
April	201.30	85.20	29.60	209.50	203.40	157143.31	151960.31	0.81
May	205.50	108.20	34.40	202.10	195.70	149637.13	144907.90	0.81
June	182.20	105.30	36.50	175.90	170.30	129898.05	125829.30	0.80
July	170.00	103.90	35.20	164.90	159.50	122684.56	118767.35	0.81
August	177.00	94.20	34.20	179.00	173.50	133446.42	129116.19	0.81
September	176.60	73.50	32.30	191.30	185.80	143071.16	138440.12	0.81
October	165.40	61.10	27.00	193.90	188.60	147433.11	142612.77	0.83
November	124.30	52.20	20.10	155.30	151.00	122028.04	117952.40	0.85
December	112.20	47.10	14.89	145.90	141.90	117369.59	113492.34	0.87
Year	1951.10	901.00	26.58	2135.60	2073.20	1630088.78	1576989.44	0.83

4.4.2 Orientation & Tilt Angle of Solar PV Module.

Orientation of solar plant will be true south; Modules will be fixed on true north south facing so that maximum coming rays can be used, and tilt angle will be 18 degree.



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4.4.3 Capacity Utilization factor (CUF)

For solar park, yield cannot be exactly determined as PV modules utilized by different developers may differ. Further the D.C. to A.C ratio may differ based on alignment of modules, inverter, cabling, transformers, Module installation (fixed tilt, seasonal tilt, angle of tilt), modules cleaning and maintenance practice and outage of transmission lines. Considering Average value of 20% CUF, annual yield will be 1.79 million kwh/MW/ annum and for Solar Park of 1310 MW it will be 2358 million unit / year.

4.4.4 Annual Degradation

Degradation will be depended on module specification mentioned by manufacturer. Now a days modules are available in market of degradation factor of 0.55 annually.

4.5 Land Assessment of Solar Park.

4.5.1 Geotechnical Assessment of Site

The objectives of Geotechnical Survey were to assess soil profile of the site as well as to find out electrical resistivity of the soil. Geotechnical investigation includes surface & subsurface exploration, soil sampling, and laboratory analysis. Geotechnical investigations have substantial importance in assessing requirement of foundation to be laid down preventing human and material damage due to the earthquakes, foundation cracks, and other catastrophes.

Soil resistivity is a measure of soil resistance to the flow of electricity. It is a critical factor in design of the grounding system in an electrical substation, or for lightning conductors. To attain the desired objectives.

Subsequently boreholes/trial pits were drilled to assess the soil profile and electrical resistivity of the soil.

4.5.1.1 Geotechnical Analysis:

I. Bore Hole Investigation

- a. For geotechnical investigation work, auger boring used as per IS 1892-1979 clause 3.6.1.
- b. Drilling bore holes of size 100 mm/Nx size using auger boring up to refusal, collecting necessary disturbed/undisturbed soil samples wherever strata changes.
- c. Standard penetration test was conducted in the bore holes at regular intervals of 1.5 m or at every change of strata, whichever is earlier as per IS 2131-1981.
- d. Disturbed and undisturbed samples were collected at different depths/wherever strata



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changes as per IS 1892-1979.

- e. Disturbed samples recovered were sealed in polythene labelled and sent to the laboratory for laboratory test.

II. Standard Penetration test (SPT)

Standard Penetration Test (SPT) was conducted as per IS: 2131 – 1981. SPT split spoon sampler of standard dimensions was driven into the soil from the borehole bottom using 63.5 kg hammer falling from 75 cm height.

The SPT weight was mechanically lifted to the specified height and allowed to fall freely on the anvil with the use of cat-head winch with one to one and half turn of the drum. Blow counts for the penetration of every 15 cm were recorded and the N is reported as the blow counts for 30 cm penetration of the sampler leaving the first 15 cm penetration as seating drive. When the number of blows exceeded 50 to penetrate the first or second 15 cm length of the sampler, the SPT N is regarded as more than 100 as described in IS : 2131 - 1981. The test is terminated in such case and a record of penetration of the sampler under 50 blows or more is made. SPT refusal is recorded when there is no penetration of the sampler at any stage and also when a rebound of the sounding system is recorded. The N values are appended in bore log sheet.

III. Laboratory testing of extracted samples

The following laboratory tests were carried out on soil samples collected at site as applicable:

- i) Grain size analysis
- ii) Liquid limit,
- iii) Plastic limit
- iv) Plasticity index
- v) Bulk Unit Weight
- vi) Direct Shear test
- vii) Classification of Soil type.

Results of Soil Investigation conducted at Ramsar Chhota village are as below



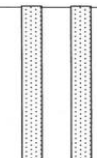
Detailed Project Report for 1310 MW Phase-I RVUNL Solar Park at Bikaner, Rajasthan.

Report No.: GM/22/198B

Date: 25/08/2022

Project: Geotechnical investigation report for setup of 2000MW Solar Park & Installation of 810MW Solar project in tehsil- Pugal, Distt - Bikaner, Rajasthan.

Bore Log Chart

Ref. No. SRF/22/805A			Ref. Dated: 09.08.2022			Date of Field Testing: 06.08.2022				ULR No. TC74832200000947F					
						Ground water table: - Not encountered				Bore Hole No.:- 01 of 18					
Depth from NSL (m)	Nature of Sample		Grain Size Analysis			Consistency Characteristics (%)			Bulk unit weight	Direct Shear Test		S.P.T Value (Observed) (N Value)	IS: Classification		
			Gravel (%)	Sand (%)	Silt & Clay (%)	L.L.	P.L.	P.I.	Y (t/m³)	C* (t/m²)	φ°				
1.00	SPT		0.00	53.01	46.99	NP	NP	NP	1.85	0.00	30.50	15	SM		
1.50	SPT		0.00	55.12	44.88	NP	NP	NP	1.86	0.00	31.00	16	SM		
2.00	SPT		0.00	59.64	40.36	NP	NP	NP	1.86	0.00	31.00	16	SM		
3.00	SPT		0.00	68.26	31.74	NP	NP	NP	1.90	0.00	31.25	18	SM		
4.50	SPT		0.00	70.49	29.51	NP	NP	NP	1.93	0.00	32.00	21	SM		
6.00	SPT		0.00	83.36	16.64	NP	NP	NP	2.03	0.00	34.00	37	SM		

Report No.: GM/22/198B

Date: 25/08/2022

Project: Geotechnical investigation report for setup of 2000MW Solar Park & Installation of 810MW Solar project in tehsil- Pugal, Distt - Bikaner, Rajasthan.

Ref. No. SRF/22/805A			Ref. Dated: 09.08.2022			Date of Field Testing: 07.08.2022			ULR No. TC74832200000947F					
						Ground water table: - Not encountered			Bore Hole No.:- 03 of 18					
Depth from NSL (m)	Nature of Sample	Symbolic Representation	Grain Size Analysis			Consistency Characteristics (%)			Bulk unit weight	Direct Shear Test		S.P.T Value (Observed) (N Value)	IS: Classification	
			Gravel (%)	Sand (%)	Silt & Clay (%)	L.L.	P.L.	P.I.	Y (t/m³)	C' (t/m²)	φ°			
1.00	SPT		0.00	77.78	22.22	NP	NP	NP	1.74	0.00	29.75	10	SM	SILTY SAND
1.50	SPT		0.00	87.19	12.81	NP	NP	NP	1.78	0.00	30.25	12	SM	
2.00	SPT		0.00	93.06	6.94	NP	NP	NP	1.88	0.00	31.25	17	SP-SM	
3.00	SPT		0.00	88.88	11.12	NP	NP	NP	1.98	0.00	32.00	26	SP-SM	POORLY GRADED SAND – SILTY SAND
4.50	SPT		0.00	94.44	5.56	NP	NP	NP	2.01	0.00	33.00	31	SP-SM	
6.00	SPT		0.00	93.96	6.04	NP	NP	NP	2.03	0.00	33.50	36	SP-SM	

IV. Geological hazards & Seismic Activities

The site is least prone to any potential geological hazard like landslide, flooding & erosion, subsidence, primary ground rupture, strong ground motion & liquefaction etc.

V. Ground water

Water table at Pugal and nearby area is quite deep (more than 50 meters) and underground water is available in limited quantity. The water available is saline which is neither potable nor useful in module washing. Its extraction is not advisable since the levels of the aquifers will be affected resulting in drying of the soil.

4.5.1.2 Foundation Assessment

The Preliminary Geo Technical Investigation (GTI) work has been carried out at Ramsar Chhota & bandarewala village and samples from boreholes, trial pits were collected for Initial Soil Investigation & chemical analysis of collected samples. Based on preliminary technical analysis report details of soil is hereunder.



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Table 13: Safe Bearing Capacity as per GTI

Sl.No	Depth from NSL (M)	Safe Bearing Capacity		Recommended Safe Bearing Capacity T/m ²
		Shear Criteria T/m ²	Settlement Criteria T/m ²	
1	1	6.83	10.85	6.5
2	1.5	10.00	14.53	7.5
3	2	14.83	19.96	10
4	3	20.50	25.24	15
5	4.5	27.08	28.71	20
6	6	32.67	31.43	20

Soil properties was categorised as poorly graded Sand, poorly graded Silty Sand up to entire 6.0 m depth and found almost same for all bore holes. Rock was not encountered at different level of depth & is recommended to consider the safe bearing capacity as per general shear criteria at 6.0 m depth as per IS 12070 with zero core recovery was 20 T/m².

Bearing capacity of the soil is suggested as 20 T/m². Foundations for PV mounting structure will require depth of more than 1.5mtrs.

4.5.1.3 Earth works Assessment.

The Land in RVUNL Solar Park has Significance of undulation. Earthwork will require for levelling the undulations which can be done easily as the soil is sandy desert.

For development of solar project at the allotted land in RVUN solar park if land levelling, dressing is required than this work will be carried out by the solar project developers (SPD's).

4.5.1.4 Soil Resistivity Analysis

Soil resistivity measurements were carried out using wenner's 4 pin method as per IS-7808(1975) specification. In this investigation computerized resistivity monitor was deployed to obtain information about resistance values at different depths.

In this system electric current are injected into ground and resistance offered by different litho units (soil/weathered zone/ Rock horizon/Rock type) are obtained into apparent resistivity and total resistance. Resistivity response of soil depends primarily on the constituents of that soil clayey soil tends to have low resistivity owing to the presence of conductive minerals sandy soil show comparatively higher resistivity owing to dominance of conductive minerals based on calibration it was found that the resistivity varies with change in lithology. The thickness of alluvium over rock surface also affects the resistivity of deeper formation. Each spot measurement made recorded and work carried out by competent trained and qualified personal.



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The study area situated near village Ramsar Chhota, Bikaner represents plain Topography and area covered by thick alluvial formation and intercalation of shale and clay material making poor quality of ground water. Soil cover in the study area varies from 0-20 mt then underlined by clay and sand. First aquifer is formed of Sand and Gravel after it there may be possibility of intercalation of clay. The presence of such clayey material making this aquifer highly saline in nature so in this area ground water is saline.

(Spot-1)

Depth (Meter)	Litho logy	Discharge	Quality
0-20	Sand	---	---
20-60	Gravel, Sand and Kankar	--	--
60-120	Clay Gravel, Sand with saturation	Low	Saline
120-150	Clay Sand with saturation	Low	Saline

(Spot-2)

Depth (Meter)	Litho logy	Discharge	Quality
0-15	Sand	---	---
15-70	Gravel, Sand and Kankar	--	--
70-130	Clay Gravel, Sand with saturation	Low	Saline
130-150	Clay Sand with saturation	Low	Saline

(Spot-3)

Depth (Meter)	Litho logy	Discharge	Quality
0-20	Sand	---	---
20-80	Gravel, Sand and Kankar	--	--
80-130	Clay Gravel, Sand with saturation	Low	Saline
130-150	Clay Sand with saturation	Low	Saline

(Spot-4)

Depth (Meter)	Litho logy	Discharge	Quality
0-20	Sand	---	---
20-80	Gravel, Sand and Kankar	--	--
80-130	Clay Gravel, Sand with saturation	Low	Saline
130-150	Clay Sand with saturation	Low	Saline

Observation:

Spot 01

According to geophysical data it is inferred that at this spot the water table encountered at the depth of 60mt with Low saturation with Clay, Gravel & Sand is dominant upto depth of 120mt. after the depth of 120mt clay is dominant and making water quality saline and quantity very low.

Spot 02

According to geophysical data it is inferred that at this spot the water table encountered at the depth of 70mt with Low saturation with Clay Gravel & Sand is dominant upto depth of 130mt. after the



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depth of 130mt clay is dominant and making water quality saline and quantity very low. `

Spot 03

According to geophysical data it is inferred that at this spot the water table encountered at the depth of 80mt with Low saturation with Clay Gravel and sand is dominant upto depth of 80mt. after the depth of 130mt clay is dominant and making water quality saline and quantity very low.

Spot 04

According to geophysical data it is inferred that at this spot the water table encountered at the depth of 80mt with Low saturation with Clay Gravel and sand is dominant upto depth of 80mt. after the depth of 130mt clay is dominant and making water quality saline and quantity very low.

Recommendations

In this area water quality is saline due to presence of clay material in it. The aquifer of the area is sand and Gravel but contaminated with clay. The nature of clay is showing saline quality of ground water.

4.5.2 Topographic Survey

Topographic Survey is used to identify and map the contours of the ground and existing features on the surface of the earth or slightly above the earth's surface (i.e. trees, buildings, streets, walkways, utility poles, retaining walls, etc.). Generally, the purpose of topographic survey is to prepare a base map of the proposed site showing perimeter boundary for the design of buildings/colonies/roads/highways/ plants/sub-stations etc.

For carrying-out topographic survey of the proposed site, Drone & DGPS System was utilised. Also, locations of over-ground features like approach road, BT road, WBM road, kuccha road, electrical lines, sub-stations, canal etc. were noted. While carrying-out topographic survey 08 temporary benchmarks (TBMs) points were also generated on some permanent features like roadside milestone, on water tanks & Permanent Structures etc. Based on the survey, the site map was drawn and contours using Auto Plotter software were generated at 1 meter interval.

Based on the topo survey it has been observed that the site has undulations. Land is sloping towards north-east & south-west direction. The site has lesser/slope undulation in center of plot areas.

One Bituminous Road maintained by General Reserve Engineer Force (GREF/BSF) Constructed by IGP from Pugal Road Crossing to village Amarpura is passing near by the proposed solar park at Ramsar Chhota. Besides this BT Road parallelly Indira Gandhi Canal is constructed and



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Kachcha Rasta from village Ramsar Chhota & bandarewala to other villages are passing through the proposed park.

4.5.3 Hydrological Study

Ramsar Chhota & Bandarewala is situated in Thar Desert belt where water is a scarce commodity. Water table at both villages and nearby area is deep (more than 40 meters) and underground water is available in limited quantity. The water available is saline which is neither potable nor useful in module washing. Its extraction is not advisable since the levels of the aquifers will be affected resulting in drying of the soil. The nearest water resource is Indira Gandhi Canal (Panna Lal Pumping Station) which is at about 18 km distance. Water can be transported from upstream sump of this Pumping Station. Latter is preferred as no head works is involved, pipeline will be laid-down along with village internal road and it will not affect water supply of Village lift canal. A total of 450 cusec water has been allocated to power sector by the Indira Gandhi Nahar Board, out of which 58 cusec water is reserved for solar power plants. Water requirement of RVUNL Solar Park will be 0.2 cusecs as worked out. It is planned that water pipeline to be laid along the road from Pannalal Pumping station to village Ramsar Chhota will be used after treatment for below works.

- a) Drinking water supply to employees and staff working in solar park and the local villages.
- b) Washing of PV modules.
- c) Green belt development.

4.5.4 Preparation of Land

Proposed solar park land has undulations at many places which would require levelling. Levelling of land in solar plots will be done by SPD's and levelling of land for roads and water infrastructure and park pooling substation will be done by SPPD's (RVUNL). Solar power developer (SPD's) will construct boundary wall for solar plots, their internal inverter control room/main control room and water infrastructures for their upto their module areas.

SPPD's will prepare the land for PSS, road & water infrastructure, green belt from storage tank to nearest boundary point of solar project developer.

Based on the topographic survey it has been observed that the site has undulations. Land is sloping from North-West to South- East side. The site has lesser undulation in Central side whereas North and North- West side has higher undulation (sand dunes).

The perimeter for Solar Park 1310MW phase-1 is around 165.47 running kilometres which covers all the 3plots and one solar park pooling substations (PSS 33/400kV Level).

Fencing will be constructed by SPPD's.



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Green area/belt will be developed on Separate plots as well both sides of the internal roads of the park and at the PSS by SPPD's.

4.6 Infrastructure Development

Solar Park infrastructure is based broadly on three no's of Plots with further sub divided in 25 no's of packet having one 33/400 kV PSS for power evacuation. Due to existing B.T. Road and Kacha Rastas, plot has been subdivided in no of packet. Park layout is based on following considerations:

- (i) Existing Roads passing through solar park will be widened and strengthened by SPPD's to 10 meter wide (7.5 meters BT and 1.25 meter shoulders either side).
- (ii) Internal peripheral roads will be developed by SPPD's of WBM type with 7.5m wide only on the entire peripheral of solar park. Interconnecting road in the solar module area /blocks or inverter room will be developed by SPD's.
- (iii) Raw water reservoir to meet water requirement will be provided on southern most part of the proposed Solar Park. Treatment plant with overhead tank will be provided to meet daily requirement of water. SPPD's will provide the water to the nearest boundary point touching with storage tank. Further pipeline infrastructure for their internal connectivity/ module cleaning or up to any emergency storage tank will be carried out by SPD's.
- (iv) Sewage disposal shall be through septic tanks in individual plot that will be constructed by developer and in PSS/ Switchyard/ Admin Buildings that will be constructed by RVUNL.
- (v) Vacant space along Road & 400/33 kV lines will be utilised for Greenery development.

The infrastructure is so planned keeping in view that if extension of the Solar Park is planned and/or nearby a Solar Park comes-up the same infrastructure can be utilised for the extension/new Solar Park/Solar Projects.

4.6.1 Electrical Infrastructure.

The potential of solar and wind power is concentrated mostly in few solar and wind rich States. In fact, 9 States in India, account for more than 77% of the RE capacity addition by 2022. The solar and wind power generated in these States may not be consumed fully by them and Power from RES from these States needs to be transmitted to the load centres in other states. For such transmission and integration of RE power with conventional power, Green Corridors have been conceived. Green Energy Corridor conceived in Draft National electricity plan (vol -2 transmission) Dec16 for Western – Northern region consists of 765 kV Double Circuit line from Bhuj pool (Gujrat) - Banaskantha (Gujrat) - Chittorgarh (Raj.) – Ajmer (Raj.) – Bikaner (Raj.) – Moga (Punjab). This has been modified to have a spur 765 kV double circuit line from Bikaner (Raj)- Bhadla(Raj.) - Fateh Garh (Raj) and LIL of double circuit Ajmer (Raj.) – Bikaner line to Bhadla-II. This green corridor is being constructed



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by Power Grid Corporation of India. Their 765 kV Bhadla-I substation is at Nure-ki- Burj close to existing 400 kV Bhadla substation of RVPN Further, 400 KV new Switching Substation line for Twin Moose SCTPS- Bikaner Line has been sanctioned by RVPN in the vicinity of Solar park.

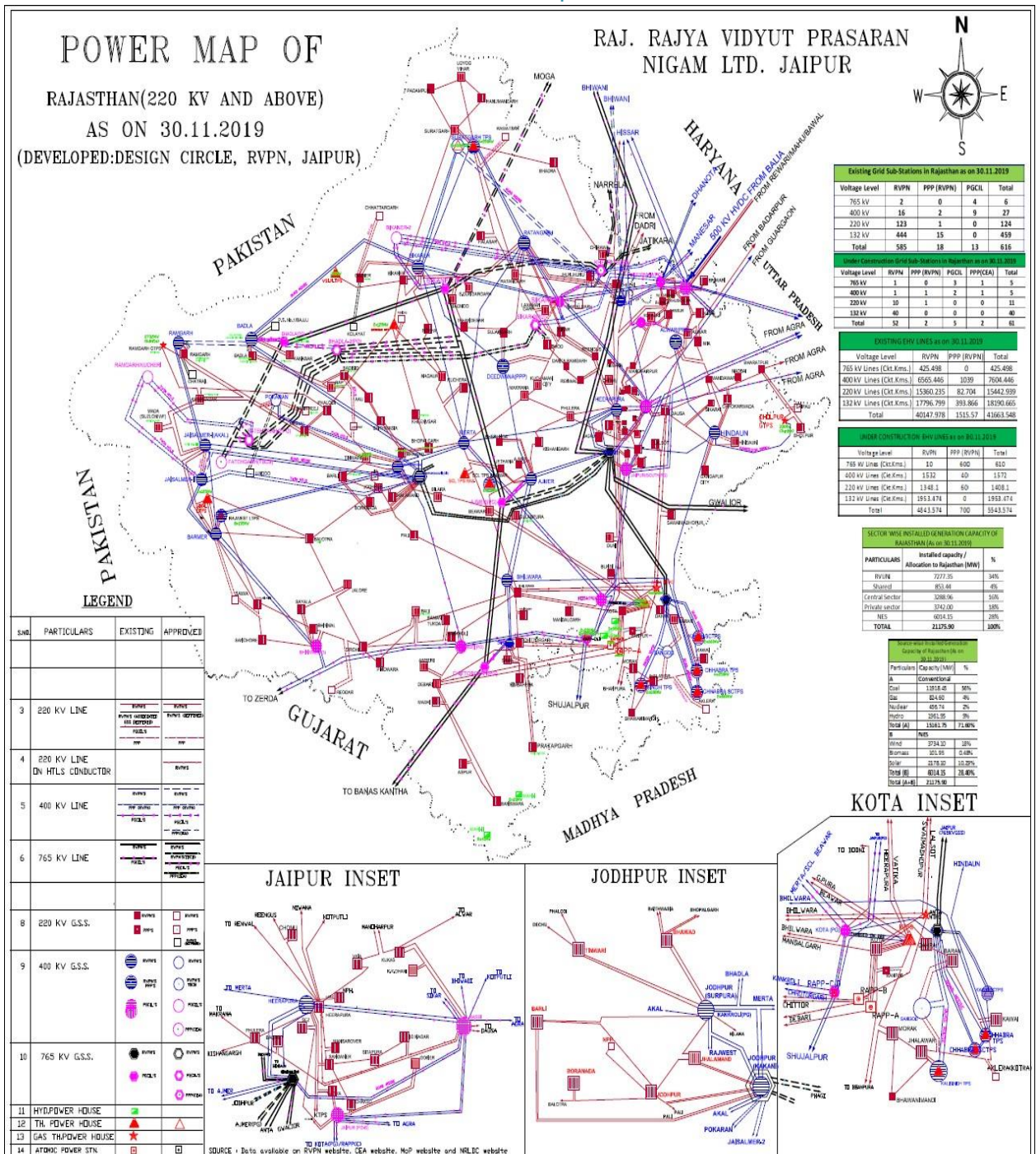
The Power evacuation from RVUN Solar Park will be through 400 kV lines on Double Circuit route. Double circuit will be from our 400 kV PSS (pooling sub stations of the Solar Park) to Rajasthan Rajya Vidyut Prasharan Nigam Ltd (RVPNL) newly proposed 400 kV GSS near to Surasar existing Substation and there will be interconnection between RVUNL PSS Line & RVPNL GSS.

SPD will Construct transmission line of 33kV to RVUNL PSS at their care & responsibility, SPPD (RVUNL) will Construct line of 400kV from RVUNL PSS to RVPN GSS for interfacing.



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Power Map





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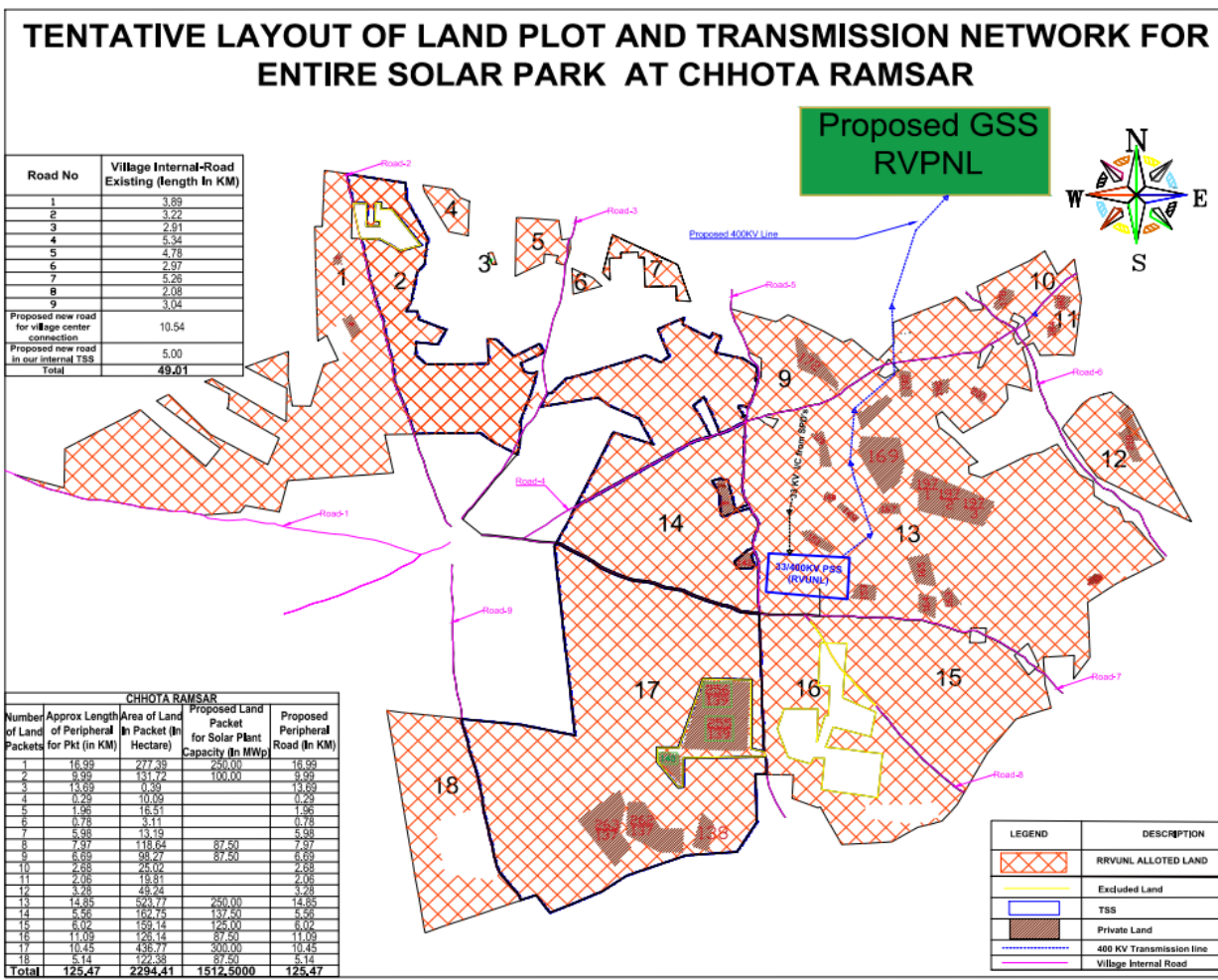


Figure: Transmission line network 400 kV in blue Coloured

Existing Electrical Infrastructure/ Nearby Substation Details.

GSS- 1	132/33 kV grid substation (GSS) at RD-710 RVPN (Village Surasar), Pugal Road i. Transformer Capacity: 25 MVA and no provision for capacity augmentation ii. Distance from project land: 20 KM
GSS- 2	220 /132 kV Substation - Chhatargarh i. Transformer Capacity: 160 MVA and Yes there is provision for capacity augmentation plan up to 150 MVA only ii. Distance from project land: 45 KM iii. Substation i/c feeder - 132 kV Loonkarasan Line 1 & 2, 132 kV Kajuwala
GSS- 3	220 /132 kV Substation - Kajneer i. Transformer Capacity: 2500 MVA and No details available ii. Distance from project land: 45 KM iii. Substation i/c feeder – Data not available

Based on the site visit and the details of the nearest GSS, it is evident that the nearby GSS doesn't have adequate redundancy to accommodate the power generated from the RVUNL solar park.



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Hence, a dedicated GSS with a voltage level 400kV shall be constructed near the proposed project location for power evacuation of entire 2000 MW. Hence the new proposal of 400/765 kV substation from RVPN has been done which will be developed for the power evacuation from the proposed RVUNL Solar Park. Grid Substation shall be developed by RVPN. The detail of the proposed substation is shared in the below table.

Table 14: Proposed RVPN Grid Substation Details

S.No	Description	Information
1	Location of the proposed grid substation	N 28°21'31.54356" & 72°57'7.75728"
3	Voltage level of substation	Proposed for 400/765 kV.
4	Status of land for transmission line (TL) from project location till substation	Land from project location to substation for transmission line will be of mixed category. (Govt. + Private land).
5	Status of Approvals & permits for substation and Connectivity philosophy	Under Process with RVPN for power evacuation approval and bay allocation from their proposed substation in the Pugal Area.

4.6.2 Control Infrastructure

Setting-up Control infrastructure is in the scope of Park Developer for optical fibre communication between 400KV PSS and 400 KV GSS of RVPNL for voice communication, requisite data transmission and differential protection on this network.

Forecast, Scheduling and Monitoring

Forecasting of generation is the prerequisite of scheduling the generation. Scheduling and Real time monitoring of power generation is the operational requirement of the electrical grid system. Accurate Solar power forecasting will help reducing the uncertainty of variable renewable generation and will enable grid operators to more efficiently commit or de-commit other generators to accommodate changes in solar generation and to prepare for / provide for operating reserve to meet extreme events in which renewable generation is unusually high or low. Clouds and dust storm are the primary cause of variability for solar generation, aside from the predictable changes during the day and throughout the year. The ability to accurately forecast solar power depends on the character of cloud cover, including the amount of water or ice in clouds and aerosols. To assess near-term impacts of approaching clouds on solar generation and wind speed leading to dust storm, sky imagers and satellite images can be used. For longer periods, weather models can be used to determine how clouds may form and change. Solar Park generators and REMC /system operators need to partner with IMD and ISRO for this purpose.

Power system of the country has been integrated and operates as all India grid. Grid operation is monitored and controlled by National Load dispatch centre (NLDC), Regional load dispatch centres (RLDCs) and State load dispatch centres (SLDCs). For operation of solar power plants in Rajasthan,



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relevant load dispatch centres are Northern Regional Load Despatch Centre (NRLDC) at New Delhi and Rajasthan State Load dispatch Centre (SLDC) and Renewable Energy Management Centre (REMC) at Jaipur.

Under section 28(3) of the Electricity Act, NRLDC is

- (i) Responsible for optimum scheduling and despatch within the northern region (NR) consisting of states of Rajasthan, Uttar Pradesh, Uttarakhand, Haryana, Punjab, Himachal Pradesh, Jammu & Kashmir, Delhi and union territory of Chandigarh.
- (ii) Monitoring of grid operation of NR.
- (iii) Responsible for carrying out real time operations for grid control and despatch of electricity within the NR through secure and economic operation of the regional grid in accordance with the Grid Standards and the Grid code i.e. Indian Electricity Grid Code (IEGC) specified by Central Electricity Regulatory Commission (CERC) under section 79.

Likewise, SLDC under section 32(2) is responsible for scheduling, dispatch, monitoring of grid operation, real time operation of the Rajasthan State Grid in accordance with the Grid Standards and the State Grid Code. (i.e. Rajasthan Electricity Grid Code (REGC) specified by Rajasthan Electricity Regulatory Commission (RERC) under section 86). The supply of electricity to users utilizes open access to transmission system which is governed by open access regulations. IEGC, REGC and open access regulations (vide para 7), provides generation stations for conveying day ahead scheduling on 15 minute's time block basis by specified time in morning to SLDC and SLDC finalizing the schedule of injection / drawal from Northern Region as per State's load requirement and availability of transmission system and conveying it to NRLDC. Grid management requires generating stations to maintain discipline by not deviating from schedule finalized by LDCs. Any deviation is subject to penal charges as per relevant deviation settlement regulations. Accordingly all solar power generating Stations of RVUNL Solar Park will be governed by the provisions of Rajasthan Electricity Regulatory Commission (Forecasting, Scheduling, Deviation Settlement and Related Matters of Solar and Wind Generation Sources) Regulations, 2017. They also require integration of schedule for each pooling station of solar park through Qualified Coordinating Agency (QCA) or one of the generator may be authorised to act as QCA and supplying the same to REMC and SLDC. QCA will settle the deviation settlement charges and effect its de-pooling among generators and effecting due recovery.

Scheduling of power will require forecast of availability of next day's solar insolation. The forecast will be carried out based on meteorological data for next day from metrological laboratory, NREL, NASA satellite data, input from REMC etc. arranged by them, and data base available with them and using appropriate software.

Real-time monitoring will be done by SLDC. For this purpose generation data will have to be electronically conveyed by individual generators at interval of 1-2 minute's as may be decided



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by SLDC. ABT compliant meters to have capability of transmitting such data to optical fibre communication system. Generators (SPPD) will ensure that meters provided on 400kV lines will be capable of transmitting real-time data required by SLDC on specified protocol and shall have requisite compatibility.

RERC has notified Rajasthan Electricity Regulatory Commission (Forecasting, Scheduling, Deviation Settlement and Related Matters of Solar and Wind Generation Sources) Regulations, 2017 in September 2017 was effective from 1st January 2018 but has been made effective from 1st Oct 18 with some relaxation in deviation charges from 1.10.18 to 31.3.20 through RERC order of 27.09.19. Accordingly Solar power generators having Individual or combined capacity of 5 MW and above whether connected to the State Grid independently or through pooling stations and/or solar parks are governed by it. Accordingly, the wind and solar generator or their Qualified Coordinating Agency (QCA) shall provide SLDC with a schedule based on its own forecast and such schedule shall be used as reference for deviation settlement. QCA will be a mutually agreed agency registered with SLDC, to act as a coordinating agency on behalf of Solar Generators connected to a pooling station and may be one of the generators. Besides this, solar generators shall be metered by interface metering with a provision for recording and storing all the load survey and billing parameters for every 15-minute time block. Monthly meter readings shall be forwarded to the SLDC in addition to data acquisition through SCADA for energy accounting and that Data telemetry shall be adopted at the inverter or plant level as considered appropriate by SLDC. Full data telemetry and communication facilities shall be provided by the wind and solar generator or QCA. Solar generators, represented via Qualified Coordinating Agencies (QCAs), shall mandatorily provide to the SLDC, in a format prescribed by SLDC, the technical specifications at the beginning and whenever there is any change. The data relating to power system output & parameters and weather-related data as applicable, or any other data as directed by SLDC shall be mandatorily provided by such generators or QCA to the SLDC in real time. All solar power developers / producers in the solar park shall provide these data to SLDC themselves or through QCA and shall supply these to RVUNL for reference and passing these to Renewable Energy Management Centre (REMC) as and when established and operational for RE power forecasting and monitoring. SPD's shall install instruments and test module for measurement of GHI and DNI and, two anemometers for wind speed and direction and two cloud cover (okta) measuring instruments and transmission of their data at 15 minute's interval as above.

CERC has notified Central Electricity Regulatory Commissions (Deviation settlement and related matters) regulations 2014. It has been amended by 2nd amendment in Aug 15 to incorporate provisions relating to solar or wind generation. These regulations apply to regional entities (i.e. persons who are in the Regional Load Dispatch Centre control and whose metering and energy accounting is done at regional level) and users i.e. a person such as a generating Company including captive generation plant or transmission licensee or distribution licensee or bulk consumer, whose electrical plant is connected to Inter-state transmission system at 33 kV or higher voltage.



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Metering:

ABT compliant meters, as per guidelines of CEA & RVPN, capable of providing real time data to the Load Dispatch Centre shall be provided on all incoming feeders (400 kV lines) at RVPN's 400/765 kV dedicated GSS for solar park. These shall comply with Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006 as amended from time to time and specific requirement of RVPN and shall be installed at the locations decided by RVPN duly tested and sealed by RVPN. In addition, Main and Check ABT meters and telemetry system, compatible with specification of RVPN for STOMS (Smart Transmission Operation and Management System) centre, established by RVPN, will be provided by Generators (SPPD) on all incoming feeders (400 kV lines) at RVPN 400/765 kV RVPN GSS.

4.6.3 Road Infrastructure

Existing Main Roads passing through solar park will be widened to 10 meters width (7.5 meters BT and 1.25-meter shoulders either side) and strengthened. New 10mtr wide BT road will also be constructed for RVUNL PSS access & inter connection with existing roads by SPPD's.

SPPD's will develop Internal Peripheral Secondary 7.5mtr wide WBM road (5.5mtr WBM & 1mtr shoulder each side) in the entire peripheral boundary of the solar park. Interconnection road between Solar Block/ Module area, Inverter control room & main control room need to be developed by SPD's as per their requirement to reach to all the pockets as per their plant layout.

Internal connecting road and lighting within the pockets/block/ panel mounting area will be developed by SPD.

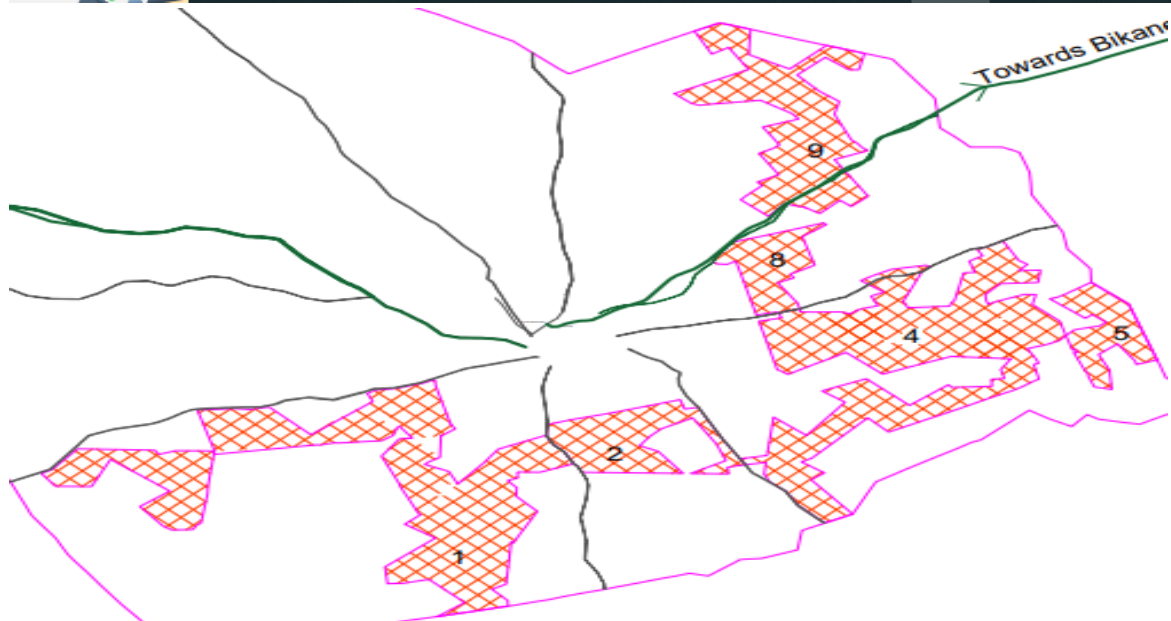
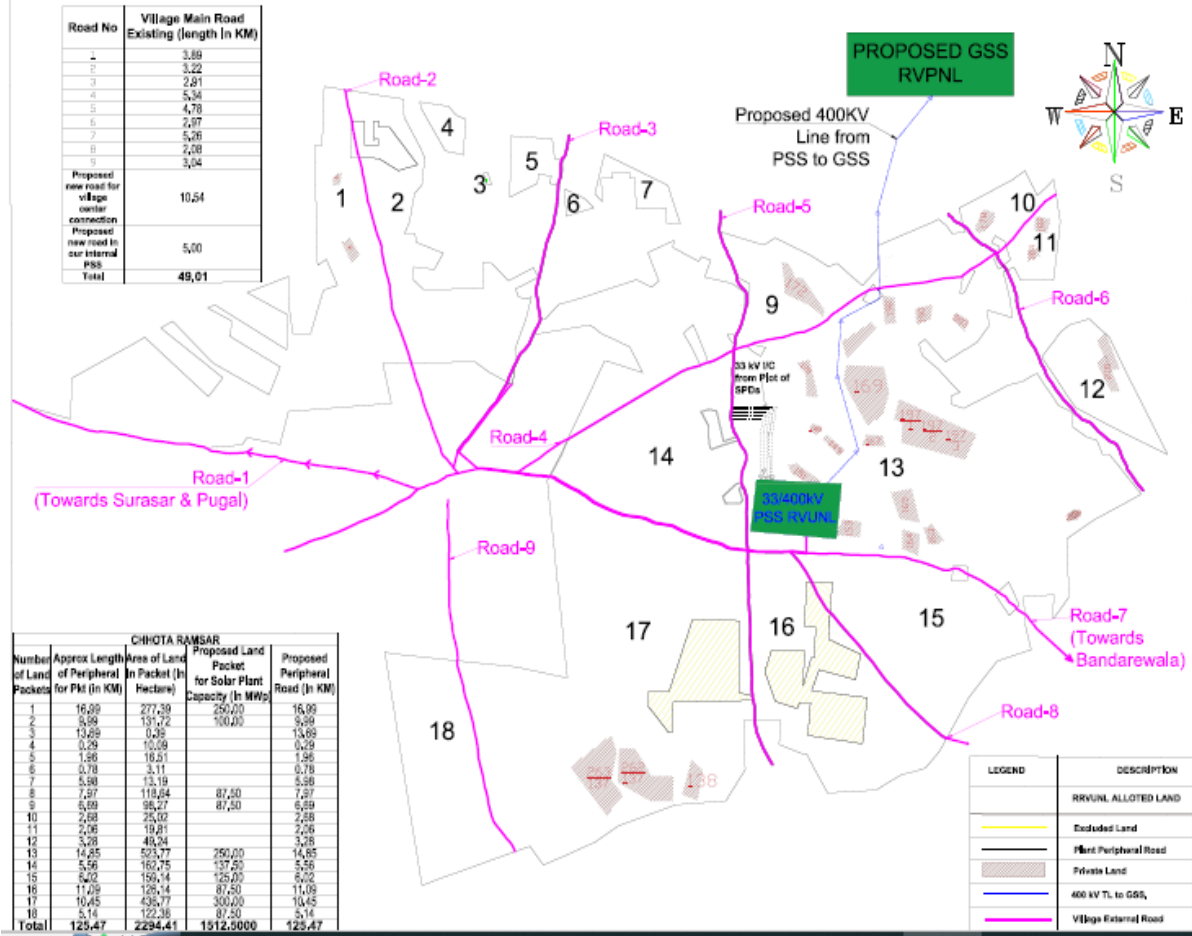
Tentative layout of Existing Road shown with Pink Colour & Internal peripheral Road shown with light black can be viewed in the below drawing which will be developed as discussed above.

Road No-1 & Road No-7 is connecting to the proposed PSS of RVUNL from Attarabad Main Road via Surasar Pugal to Bandarewala..



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LAYOUT OF EXTERNAL BT ROAD & INTERNAL PERIPHERAL ROAD FOR RVUN SOLAR PARK 1310MW AT CHHOTA RAMSAR



Road Map in RVUNL Solar Park.



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Road No	Village Main Road Existing (length In KM)	Number of Land Packets	Approx Length of Peripheral for Pkt (in KM)
1	3.89	1	16.99
2	3.22	2	9.99
3	2.91	3	13.69
4	5.34	4	0.29
5	4.78	5	1.96
6	2.97	6	0.78
7	5.26	7	5.98
8	2.08	8	7.97
9	3.04	9	6.69
10		10	2.68
11		11	2.06
12		12	3.28
13		13	14.85
14		14	5.56
15		15	6.02
16		16	11.09
17		17	10.45
18		18	5.14
Total	49.01	Total	125.47

BANDAREWALA		
Number of Land Packets	Approx Length of Peripheral for Pkt (in KM)	Area of Land In Packet (In Hectare)
1	18.34	272.93
2	5.27	55.52
4	16.16	223.75
5	3.90	36.06
8	3.22	35.44
9	8.22	108.88
Total	55.11	732.58

Length of Road in RVUNL Solar Park.

4.6.4 Water Infrastructure

The Village Ramsar Chhota & Bandarewala is situated in desert belt and as such water is a scarce commodity. Water table is quite deep (more than 40 meters below ground level) and ground water is available in limited quantity. Ground Water is saline in nature which is neither potable nor useful in module washing. The nearest water source is Indira Gandhi Nahar Pariyojana (IGNP) Canal. Nearest off-take point is at Pana Lal Canal Pumping station which is about 20 km West- South of Ramsar Chhota. Water supply for Solar Park is conceived for Drinking water, washing of the PV modules, green belt development, etc. The quantum of water required to wash a module is considered to be 2.5 litres per Module surface per washing with total 10 washings in a year. For

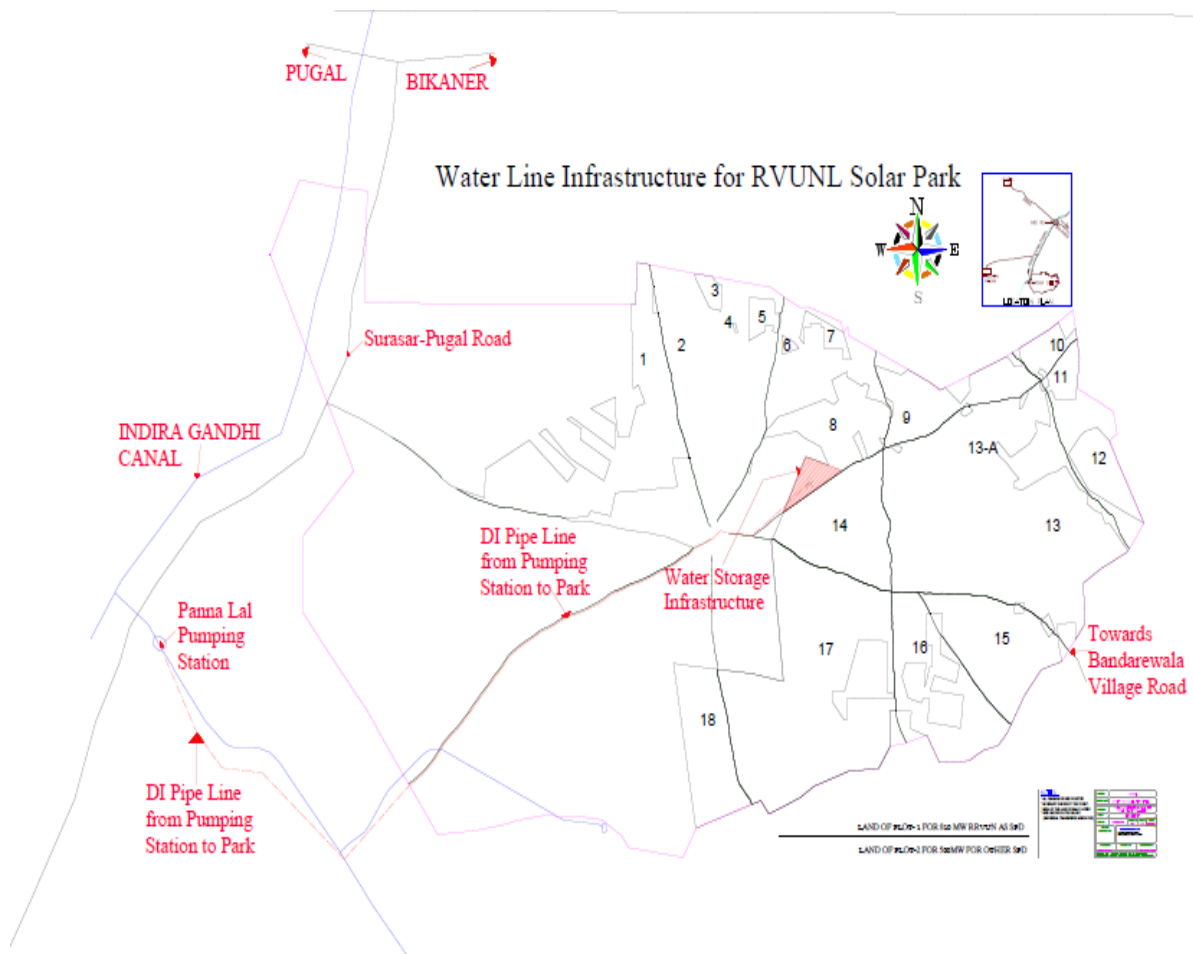


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drinking water, sanitations etc, the quantum of water required are considered at the standard norms of 70 litres per person per day. Water will be pumped from IGNP Panna Lal Pumping Canal at Ramsar Chhota to Solar Park's water reservoirs of capacity 100 lakh liters to store 30 days water for supply during canal closure. Capacity of pumping station and pipe line for raw water supply takes into account for requirement of filling reservoirs during canal closure, evaporation loss, reservoir leakage, drinking water supply to adjoining village (if required) in future.

Raw Water Coming from Panna Lal Pumping Station will be carried out through DI Pipe line of 150mm dia & will be conserved in newly proposed underground RCC Water tank Reservoir. Further This raw water will be processed through filtration plant & will be stored in Overhead RCC Water tank of capacity 2.25 lakh liters. Such treated Water will be supplied to SPD's to the Nearest boundary Point in the allotted plot & further will also be utilised for drinking & Sanitation Purpose in PSS & Admin Building of Solar Park. This entire Setup will be maintained by SPPD's.

SPD's has to develop the infrastructure of emergency storage & laying of pipeline from the provided tapping point near to the boundary for their solar plant blocks/inverter and main control room/nearby module cleaning areas in their respective allotted plots.



Line diagram for Water Infrastructure for RVUNL Solar Park.



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4.6.5 Green Belt

The main objectives of the green belt plantation is to reduce the impact of erosion, to stabilise dunes and to provide aesthetic enhancement of the project. The selection of plant types and planting arrangements will be decided based on the discussions with the Forest Department considering no / little shadow falling on solar modules.

Adaptability and suitability to agro-climatic reasons/ zones. Presently plantation of *Acacia tortilis* (Babul), at 5m x 5m spacing, along main roads and on northern boundary of solar park. *Cassia angustifolia* (Sonamukhi or Senna) and *Cenchrus ciliaris* (pruned to about 1 meter height) along the interior roads by keeping 1-2 meters away from boundaries of solar park.

4.6.6 Common furnishings

Solar Park will have monitoring system through security guards, elevated watch tower & portable Security Cabins. Proposal of surveillance through CCTV has also been considered.

4.7 CSR Activities.

The Section 135 of the companies Act provides that every company having net worth of rupees five hundred crore (or more), or turnover of rupees one thousand crore or more or a net profit of rupees five crore (or more) during any financial year shall constitute a Corporate Social Responsibility Committee of the Board. This committee will formulate a Corporate Social Responsibility policy indicating the activities to be undertaken by the company.

Activities to be carried-out utilising CDF will be as per RVUNLs Board may deem fit prioritised based on socio-economic backwardness. The broad activities may include the followings:

- i. eradicating extreme hunger and poverty
- ii. promotion of education
- iii. promoting gender equality and empowering women
- iv. reducing child mortality and improving maternal health
- v. combating human immunodeficiency virus, acquired immune deficiency syndrome, malaria and other diseases
- vi. ensuring environmental sustainability
- vii. employment enhancing vocational skills
- viii. social business projects

Indicative activities to be taken up by RVUNL in a phased manner with a mutual understanding with the local communities and keeping in consideration the financial availability, feasibility and practicality of implementation are given below:

- Infrastructure development in the Communities



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- Development of Schools in villages
- Development of Health care centres and hospitals
- Roads
- Common Irrigation and water storage structures
- Solar lighting system
- Sanitation and public health
- Drinking Water
- Promotion of sports and games
- Setting up of skill development centres

(b) Sustainable livelihood activities in the Communities

- Self-help groups (microfinance for women and farmers)
- Integrated agriculture development
- Integrated livestock development and animal husbandry
- Watershed management
- Microenterprise development
- Scholarships to meritorious students belonging to SC, ST, OBC, minority and disabled categories
- Skill training/vocational training assistance programs for youth and women
- Capacity building of the project affected persons to improve their employability

5 Solar Park Development

5.1 Solar Power Park Developer (SPPD)

Rajasthan Rajya Vidyut Utpadan Nigam limited (RVUNL) is the Solar Power Park Developer (SPPD) at Ramsar Chhota village which has been entrusted with the development of power projects under State Sector, in the state along with O&M of State-Owned Power Stations.

Govt of Rajasthan constituted the RVUNL under Companies Act-1956 on 19th July 2000. The RVUNL is since playing lead role in giving highest priority to power generation for manifold & rapid development of state.

RVUNL is undertaking of Govt. of Rajasthan holding CIN No- U40102RJ2000SGC016484 has authorized and has 8 Board of directors.

Registered office of RVUNL is at Vidyut Bhawan, Jyoti Nagar, Janpath, Jaipur -302 005, Rajasthan.



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5.1.1 Shareholders of the Company

RVUNL is undertaking of Govt of Rajasthan, 100% shares are available under Government of Rajasthan.

5.2 Time Schedule for Implementation

Ministry of New & Renewable Energy vide letter no. 30/26/2014-15/NSM dated 12th December 2014 has issued Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects in the country commencing from 2014-15 and onwards (i.e. from the year 2014 – 15 to 2018 – 19). Further, MNRE vide letter no. 30/26/2014-15/NSM dated 21st March 2017 has enhanced capacity from 20 GW to 40 GW by 2019-20. In principle approval of RVUN Solar Park has been conveyed under phase-II of MNRE scheme on 18.07.17 and its subsequent modifications & further new mode named as Mode-8 was introduced. Proposal for RVUN Solar Park was approved by MNRE on 24.02.2022 for 810MW & next 500MW was approved on 09.05.2022 merging 1310 MW vide letter No320/7/2019-NSM dated 02.08.2022 under Mode- 8 of Solar Park Scheme and planned to be set on 2294.41 hectares land with 1310 MW capacity. Land allotment has been completed for 2294.41 hectares. Power evacuation from RVUN Solar Park was conceived to be by connectivity to 400 kV Newly Proposed GSS of RVPNL near to Ramsar Chhota Village. DPR of RVUN Solar Park is amended for 1310 MW and connectivity up to 400 KV Newly Proposed GSS and commencement date has been declared from 31.03.2022 as per MNRE.

Activity time scheduled to be provided in respect of following milestones.

All activity schedules are depicted in below scheduling chart.



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Project Implementation Schedule

The tentative project schedule covering all the key milestones of the project development till commissioning is prepared and tabulated as below

Table 15: Tentative Project Schedule for Solar Park

Activity		Duration																									
	(Month)	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10	Month 11	Month 12	Month 13	Month 14	Month 15	Month 16	Month 17	Month 18	Month 19	Month 20	Month 21	Month 22	Month 23	Month 24	Month 25	Month 26
Preparation of DPR	2																										
Approval of DPR	1																										
Financial Closure	2																										
land Acquisition	3																										
Plot marking	4																										
Selection of Developer of solar projects	5																										
Tender for Electrical Infrastructure	4																										
Award of contract for electrical infrastructure	2																										
Commencement & Completion	15																										
Commissioning	2																										
Tender and award of contract for Road infrastructure & common facilities establishment	4																										
Construction Period (Civil & water infrastructure)	6																										
Greenery Development	4																										
Tender & award of contract for civil works	4																										



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Construction period - civil works	8
Allotment and handing over of solar plots to Developers	2
Construction phase of solar plants	12
Testing and commissioning of solar plants	2



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5.3 Solar Park Development Framework (Implementation, Finance & Rolled out)

The Solar Park considered under Mode-8 has the Scheme by Ministry which provides Central Finance Assistance (CFA) for development of internal infrastructure including cost of transmission to the CTU/STU Point, The CFA of Rs 20 Lakhs/MW or 30% of the cost of development of UMREPP's whichever is lower would be provided.

Any augmentation/Strengthening and or creation of external power evacuation infrastructure may be done by CTU.

If the SPPD or any of its individual promoters has a trading license, he may act as a trader of power being produced in the park for which he would be entitled to claim a margin of Rs 0.07/Unit. This Compensation may not be provided to RE Projects to be developed under EPC mode where the cost of Power would already have a factor of Return on Equity.

Commencement work of solar park has been declared by MNRE from 31.03.2022.

RVUNL is investing for total solar park cost & has considered the MNRE CFA grant. Debt is considered for cumulative requirement of funds. Debt will be funded through loan from holding company or loan from banks & financial institutions.

RVUNL will collect an amount from SPD's as onetime upfront charges & Annual O&M charges for the maintenance of solar park infrastructure.

The financial model and detailed development schedule (Capital Expenditure Schedule) is given separately in Project Financials Section.

6 Environmental & Social Impacts (Indicative).

6.1 Environmental Setting Around the Site.

The proposed infrastructure facilities for RVUNL Solar Park at Ramsar Chhota & Bandarewala village do not attract any environmental clearance as per EIA notification 2006 and 2009 issued by Ministry of Environment and Forest (MoEF). However, provisions of other Act, rules and regulations will have to be complied. Accordingly, the project developer shall require obtaining consent from competent authorities such as the Rajasthan State Pollution Control Board (RSPCB), for 'Consent to Establish' by submitting a Common Application (as per Schedule-I), clearances under Water (Prevention and Control of Pollution) Act, 1974, Air (Prevention and Control of Pollution) Act, 1981) and authorization under Hazardous Wastes (Management and Handling) Rules, 1989, as amended.

MNRE had commissioned a study to assess the environmental and socio- economic impacts of RE projects in India, particularly from solar PV and on- shore wind farms. Study team had visited solar PV farm in Anantapur district of Andhra Pradesh, which was one of the biggest solar farms developed on agriculture land, that is, close to human habitation and it was deemed to be capturing the impacts of a



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large solar farm on the local ecology. The study report, submitted in 2013, has expressed that since RE technologies have relatively (compared to other energy options) lower ecological footprint hence such projects should not be treated at par with conventional infrastructure projects and that as established by life-cycle assessment, RE projects have comparatively low environmental externalities that too are limited to project development phase. The RE projects don't generate solid or liquid effluents during operations and thereby pollution of land, surface water or ground water resources is not anticipated from such projects. During the operations phase the impacts on the surrounding environment are negligible and are reversible in nature and can be mitigated by proper Environmental Management plan.

The proposed infrastructure facilities for Solar Park at Ramsar Chhota & Bandarewala do not attract any environmental clearance as per EIA notification 2006 and 2009 issued by Ministry of Environment and Forest (MoEF). However, provisions of other Act, rules and regulations will have to be complied.

List of Clearances required for establishing RVUNL Solar Park is Tabulated in section 7.4 (Statutory & Legal framework).

6.2 Environmental Impacts (+ve & -ve, flora fauna)

Although, environmental clearance is not required even then, based on the baseline environmental features of the project area and the proposed engineering works, the impacts of the proposed activities on various environmental and social attributes of the project area are negligible & discussed details in below Sections.

Infrastructure projects are generally expected to bring in positive impacts to the project area/town. However, these projects are expected to induce negative impacts especially during construction phase, if due care is not taken. The environmental impacts caused due to infrastructure projects can be categorised either as primary impacts (induced directly by the project) or secondary impacts (which are indirectly induced and typically include the associated investment and changing patterns of social and economic activities due to the project). Typically, the potential direct impacts of such infrastructure projects are mainly during the construction phase and the most of the indirect impacts are during the operational phase.

The major activities during the construction of the roads, transmission corridors, pooling stations & water pipeline would include earthwork excavation, material movement, construction of building, provision of infrastructural facilities like construction of storm water drains, sewerage lines, septic tanks, electrical and communication network etc. The support activities would include employment of labour, material transport, use of construction equipment's etc. The activities under operation phase would include increase in the movement of public and private vehicles, public and commercial activities. These activities are expected to cause minor environmental problems during the construction and operation phase of the project, which are discussed in the following sections.

The environmental extremities during construction and operation phase are:



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Impact on Flora & Fauna

The ecological assessment based on secondary information from Botanical and Zoological Surveys of India gives information on the forest species in the area which belong to tropical thorn forests dominated by *Prosopis cineraria*, *P. juliflora*, *Calotropis procera* and *Capparis decidua*. Large proportion of the land is covered by sandy grass, which also acts as soil binder in the sand dune landscape. The plant species like *Acacia Senegal* and *Khair* (*Capparis decidua*) found in the region also have medicinal values.

Impact on land

In the proposed Solar park, the major activity that is expected to cause an impact on land is earthwork excavation, cutting & Filing for the construction. Since the proposed site is desert wasteland it can be inferred that impact on land during construction activities is likely to be marginal and has necessitate appropriate plans for safety measures.

Impact on Noise Levels

The Major Source of noise pollution will be due to material Vehicle movement, excavation, piling and concrete mixing in the project area. Noise levels may be in the range of 80-95 dB (A) and can affect the personnel working at the site and the population residing near the project site. Hence the use of proper personal protective equipment (PPE) such as earmuffs is recommended to mitigate any adverse impact of the noise generated by any equipment or mode. Noise Generating tools such as Generators, pneumatic tools should be avoided during late working hours.

During the operation phase the movement of vehicles is expected to contribute some noise pollution. Creation of green belt & roadside plantation will act as a sound barrier & is recommended as a positive impact to minimise the dispersion of sound to surroundings.

Impact on Water Quality

Dirty Water & Other Solid waste generated from the pooling stations and office areas if not managed will affect the aesthetics of the stations & office and create unhygienic conditions. Adequate number of garbage collection bins should be placed at prominent locations within the proposed pooling station & office. A plan for the timely clearing of the garbage bins and disposal of the waste should be worked out. Display boards to create awareness among the public on solid waste management, should be placed at strategic locations within the proposed pooling station and the office area.

Impact on Climate

Proposed RVUNL Solar Park has negligible & no changes in climatic conditions are anticipated. Central Electricity Authority has published in June 2018 "CO₂ Baseline Database for the Indian Power Sector User Guide Version 13.0. Accordingly, the CDM methodologies which have been approved to date by the CDM Executive Board distinguish a range of different emission factors. In the Indian context,



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the following four are most relevant, and were therefore calculated for the Indian Grid based on the underlying station data:

- (1) The weighted average emission factor describes the average CO₂ emitted per unit of electricity generated in the grid. It is calculated by dividing the absolute CO₂ emissions of all power stations by the total net generation. Net generation from so-called low-cost/must-run sources (hydro and nuclear) is included in the denominator.
- (2) Simple operating margin (OM): The operating margin describes the average CO₂ intensity of the existing stations in the grid which are most likely to reduce their output if a CDM project supplies electricity to the grid (or reduces consumption of grid electricity).
- (3) Build margin (BM): The build margin reflects the average CO₂ intensity of newly built power stations that will be (partially) replaced by a CDM project. In accordance with the Grid Tool, the build margin is calculated in this database as the average emissions intensity of the 20% most recent capacity additions in the grid based on net generation. The build margin covers units commissioned in the last five years.
- (4) Combined margin (CM): The combined margin is a weighted average of the simple operating margin and the build margin. By default, both margins have equal weights (50%). However, CDM project developers may choose to argue for different weights. In particular, for intermittent and non dispatch able generation types such as wind and solar photovoltaic, the Grid Tool allows to weigh the operating margin and build margin at 75% and 25%, respectively.

Accordingly, weighted average emission factor, simple operating margin (OM), build margin (BM) and combined margin (CM) of the Indian Grid for FY 2016-17 (adjusted for cross-border electricity transfers), are respectively 0.82, 0.96, 0.87, 0.92 t CO₂/ MWh. Based on illustration of above document, emission reduction has been calculated & RVUNL Solar Park will effect CO₂ emission reduction of 2428237 tones /year. This is the beneficial environmental impact

Project Information		
1	Type	RVUNL Solar Park
2	Size	1310 MW
3	Capacity Utilisation factor	20%
4	Projected Annual Generation (net)	2358000 MWh
5	Commissioning Year	2024
6	Year of CDM registration	-
7	Grid	Indian
8	CDM methodology	ACM0002 / Version 17.0
B Baseline Emission Factor Calculation:		
1	Calculation method	Combined Margin



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2	Data vintage for projection of emission reductions:	For OM: 2016-17 For BM: 2016-17
3	Data vintage for verification of emission reductions:	Same as for projection (emission factor fixed ex ante)
4	Accounting of imports	Mandatory
5	Weights for combined margin:	Solar generation from solar park will be dispatchable. operating margin: 50% Build margin: 50%
C	Emission Reduction Calculations:	
1	Values in t CO ₂ /MWh:	OM=0.96 BM=0.87 CM=0.92
2	Projected emission reductions, t of CO ₂	2428237
3	Actual emission reductions	Weighted Average: Monitored net generation x monitored weighted average Combined Margin: Monitored net generation x fixed combined margin

Impact On Air Quality

Suspended Particulate Matter (SPM) and dust are the major sources of air quality impacts during the construction phase. However, area has dust storms for few days in summer and dust getting floated in the air with moderate wind. Dust particles settle in few days. With respect to the existing ambient air quality or with respect to compliance of ambient air quality standards, the impact on air quality during the construction phase of the project is temporary in nature and adequate dust suppression measures during construction and thereafter will significantly reduce the impacts. During the operation phase the particulate Matter /dust and emission from the vehicular traffic movement will also have temporary impact on the existing ambient air quality of the project area. However, creation of green belt & roadside plantation will bring a positive impact to the air quality.

6.3 Social Impacts (+ve & -ve, Social)

The broad Social Impacts of setting-up Solar Park could be related to

- (i) Settlement / Demography,
- (ii) Livelihood & Economy
- (iii) Health & Safety.



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The proposed RVUNL Solar park is being setup on Government owned land (no private ownership land is being used), these impacts would be minimum. However, for development & upliftment of the local community RVUNL has been considered this under CSR Activities.

6.4 Mitigation Measures

As setting-up a Solar Park does not impact the environment significantly therefore mitigations measures are minimal. However, creation of green belt, roadside plantation & Enhancing the Road Connectivity will bring positive impact to air quality and minimise the noise level with development of local community.

7 Statutory & Legal Framework.

Establishment of solar park is governed by Government policies to promote RE generation from solar energy. Govt. of Rajasthan has formulated Rajasthan Solar Energy Policy 2014. This policy (vide clause 10) provides for development of Solar Park in the State of Rajasthan by special purpose vehicle formed as a subsidiary of RREC or by Private sector developer with the approval of RREC or by a joint venture company with maximum 50% share of the state to create solar park of 500 MW and above capacity. Private Park developer permitted to have allocation of Govt. land and also to acquire agricultural land in excess of ceiling as per Agricultural holding Act and to create common infra-structure. This policy has extended all benefits of policy to private sector Park also. This policy permits setting up of Solar Power Plants in Rajasthan (i) under Guidelines of MNRE/ National solar Mission, (ii) for sale to Discoms of Rajasthan (iii) sale under RE certification scheme, or (iv) for supply to captive use, sale to third party and other states through open access.

Revision of this policy is under consideration of the State Government. Draft Rajasthan Solar Energy Policy 2019 has been notified and comments of stake holders / Public have been obtained. This policy have retained setting up of power plants as per policy of 2014 , except REC mechanism based power projects and in addition provides for (i) decentralised solar power plants of 0.5 to 3.0 MW capacity, (iii) grid connected roof top solar power plants, (iv) solar power plants with storage facility, (v) solar energy based Electrical vehicle charging stations and (vi) decentralised and off grid solar applications as per MNRE guidelines (including hybrid system) or sale to individuals through own distribution system, power supply to remote villages/ Dhanis through local solar grid and SPV pumping system for irrigation use. It provides for khatedar to set up solar power plant or sublet his holding for setting up of solar power plant without conversion of land.

Solar Park developer (RVUNL) and solar power producers have to abide by the electricity Act and rules and regulations made there under.

A brief discussion is given below.

- (i) Section 9(2) of this act provides for right of the person, who has constructed a captive generating plant and maintains and operates such plant to have the right to open access for the purposes of carrying electricity from his captive generating plant to the destination of his



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use. Section 38(2)(d), 39(2)(D) and 40(c) provides the function respectively of CTU, STU and transmission licensee to provide non-discriminatory open access to its transmission system for use by any licensee or generating company on payment of the transmission charges; or

- (ii) any consumer as and when such open access is provided by the State Commission under sub-section (2) of section 42, on payment of the transmission charges and a surcharge thereon, as may be specified by the appropriate Commission.
- (iii) Section 42(2) provides for the State Commission to introduce open access in such phases and subject to such conditions, (including the cross subsidies, and other operational constraints) as may be specified. These sections provide that cross subsidy surcharge shall be utilised for the purpose of meeting the requirement of current level cross-subsidy and shall not be leviable in case open access is provided to a person who has established a captive generating plant for carrying the electricity to the destination of his own use. The electricity Rules 2005 provides for the prerequisite of a power plant to be classified as captive power plant. Its revision has been proposed.
- (iv) Section 42(4) provides for payment of additional surcharge, as may be specified by the State Commission, to meet the fixed cost of such distribution licensee arising out of his obligation to supply. Thus depending on use of transmission system for open access, transmission charges of injection and drawl end State transmission system and central transmission system, as applicable, is payable. Similarly depending on use of distribution system for open access, wheeling charges (on distribution system of injection and drawl end), as applicable is payable.
- (v) The Regional Load Despatch Centre under section 28 is the apex body to ensure integrated operation of the power system in the concerned region and is interalia responsible for optimum scheduling and despatch of electricity within the region, exercise supervision and control over the inter-State transmission system and for carrying out real time operations for grid control and despatch of electricity within the region in accordance with the Grid Standards and the Grid Code. Under sec. 29, the Regional Load Despatch Centre may give such directions and exercise such supervision and control as may be required for ensuring stability of grid operations and for achieving the maximum economy and efficiency in the operation of the power system in the region under its control and every licensee, generating company, generating station, substation and any other person connected with the operation of the power system shall comply with the direction issued by the Regional Load Despatch Centres. Such directions to intra-state transmission licensee or any other licensee or generating company or sub-station is to be issued through the State Load Despatch Centre (SLDC) to ensure that such directions are duly complied with the licensee or generating company or sub-station.
- (vi) Under section 32, the State load despatch centre is the apex body to ensure integrated operation of the power system in a State and is interalia responsible for carrying out real time operations for grid control and despatch of electricity within the State in accordance with the Grid Standards and the State Grid Code and Every licensee, generating company,



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generating station, substation and any other person connected with the operation of the power system has to comply with the direction issued by the State Load Dispatch Centre.

(vii) Under the provisions of the Electricity Act, Central Govt. has constituted:-

The Central Electricity Authority (CEA) to regulate matters inter alia to specify (i) the technical standards for construction of electrical plants, electric lines and connectivity to the grid; and (ii) the safety requirements for construction, operation and maintenance of electrical plants and electric lines; and (iii) the Grid Standards for operation and maintenance of transmission lines and specify the conditions for installation of meters for transmission and supply of electricity; and the Central Electricity Regulatory Commission (CERC) for regulating inter-state generation, transmission and trading and licensing for latter two.

The regulations, and their amendments notified by these regulating authorities, are applicable to Solar Park Developer (RVUNL) and solar power generator and users (in Rajasthan) are as under:-

(A) *CEA Regulations:*

- (i) CEA (Installation and Operation of Meters) Regulations, 2006.
- (ii) CEA (Measures relating to safety and electric supply) regulations 2010
- (iii) CEA (technical standards for construction of electric Plants and electric lines) regulations 2010
- (iv) CEA (technical standards for Connectivity to Grid) regulations 2007 CEA (safety requirements for Operations, Construction and maintenance of electric plants and electricity lines) Regulations 2011

(B) *CERC Regulations:*

- (i) CERC (Grant of Connectivity, Long-term Access and Medium-term Open Access in inter-State Transmission and related matters) Regulations, 2009
- (ii) CERC (Open Access in inter-State Transmission) Regulations, 2008
- (iii) CERC (Open Access in Inter-State Transmission) Regulations, 2004.
- (iv) CERC (Terms and Conditions for recognition and issuance of Renewable Energy Certificate for Renewable Energy Generation) Regulations, 2010
- (v) CERC (Deviation Settlement Mechanism and related matters) Regulations, 2014.
- (vi) CERC (Indian Electricity Grid Code) Regulations, 2010.

(C) *RERC regulations:*

- (i) RERC (Terms and Conditions for Open Access) Regulations, 2016.
- (ii) RERC (Terms and Conditions for Determination of Tariff for Renewable Energy Sources - Wind and Solar Energy) Regulations, 2014
- (iii) RERC (Rajasthan Electricity Grid Code) Regulations 2008.
- (iv) RERC (Renewable Energy Certificate and Renewable Purchase Obligation



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Compliance Framework) Regulations, 2010.

- (v) RERC (Forecasting, Scheduling, Deviation Settlement and Related Matters of Solar and Wind Generation Sources) Regulations, 2017.

Under sec. 3 Central Govt. has notified revised Tariff Policy on 28th January, 2016. The revised policy vide clause 6.4(2) has specified that the States shall endeavour to procure power from renewable energy sources through competitive bidding to keep the tariff low, except from the waste to energy plants and that Procurement of power by Distribution Licensee from renewable energy sources from projects above the notified capacity, shall be done through competitive bidding process, from the date to be notified by the Central Government. Consequent to this, Reg 7 of CERC's tariff regulations specifies that no annual generic tariff shall be determined for the technologies mentioned in Clause (a) of this Regulation which interilic includes solar power. It also specifies that Financial and Operational norms of the regulations would be the ceiling norms while determining the project specific tariff.

Ministry of Power vide resolution dated 3rd August 2017 has issued Guidelines for Tariff Based Competitive Bidding Process for long term Procurement of Power by distribution licensees from Grid Connected Solar PV Power Projects having size of 5 MW and above, under the provisions of Section 63 of the Electricity Act, 2003. In considerations to State Government direction dated 13.1.2015 under sec. 108 of the Electricity Act, the RERC vide its orders fixed the transmission tariff of RVPN for Solar Power Projects to be commissioned during the period 01.04.2015 to 31.03.2018 or for a capacity of 2,000 MW, whichever is earlier, at a rate equal to 50% of normal transmission tariff (Rs. Per MW) as applicable to conventional power for a period of 25 years.

In respect of captive consumers, RERC tariff regulations as above, provides for injected energy to be adjusted time block-wise against drawl and excess injected energy of each time block to be cumulated till the end of the month and shall be set off against the cumulative drawl of Discom. Remaining energy, at the end of the month will be entitled to 60% of energy charges applicable for large industrial power tariff in respect of 10% of unutilised banked energy after the end of the same month.

List of Clearances /licenses/permissions to be obtained by SPPDs issuing agency.



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Table 16: Permits and Approvals

SL.No	Clearances	Acts	Approving Agency	Applicability to the Project	SPPD entire (for 1310 MW)
1	Permission for removal of tree growth within the terminal/ depot area Felling conversion and removal from stump site	Forest Conservation Act 1980 Rajasthan Tree Preservation Act 1976	Local Divisional Forest Officer	Applicable	✓
2	Permission for Withdrawal of Ground Water	Environment Protection Act 1986	Central Ground Water Board / Water Resource Department, Rajasthan	Applicable	✓
3	Permission for Withdrawal of Surface Water from Reservoirs/ Ponds/ Irrigation canals	Rajasthan State Water Policy,	Irrigation Department, Rajasthan	Applicable (if the contractor is extracting surface water)	✓
4	Crushers, Cement Batching Plant, Hot Mix Plant	Air (Prevention and Control of Pollution) Act. 1981	Rajasthan Pollution Control Board	Applicable	✓
5	Disposal of Construction Waste and liquid effluent from Labor camps	Water (Prevention and Control of Pollution) Act 1974	Rajasthan Pollution Control Board	Applicable	✓
6	Employing Labor	Executing Agency of Building and other construction act, 1996	District Labor Commissioner	Applicable	✓
7	Registration of Workers	Labor welfare Acts	District Labor Commissioner	Applicable	✓



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SL.No	Clearances	Acts	Approving Agency	Applicability to the Project	SPPD entire MW) (for 1310
8 (i)	Approval of installation of 220 kV, 132 kV lines & 33 kV Cables	The section 68 of Electricity Act 2003	Govt. of Rajasthan, Energy department	Applicable	✓
8 (ii)	Approval for Electrical Installations (400 kV lines and pooling substation)	The Electricity Act 2003 and the Central electricity Authority (measures relating to safety and electric supply) regulations 2010	Electrical Inspector / Chief Electrical Inspector to Government (CEIG)	Applicable	✓
8 (iii)	Approval for Electrical Installations (within solar PV power plant)	The Electricity Act 2003 and the Central electricity Authority (measures relating to safety and electric supply) regulations 2010	Electrical Inspector / Chief Electrical Inspector to Government (CEIG)	Applicable	
9	PTCC Permission	The Electricity Act 2003 and the Central electricity Authority (measures relating to safety and electric supply) regulations 2010)	Power and Telecommunication Coordination Committee.	Applicable	✓
10	Registration of solar Power Plant	Rajasthan solar policy	Registration authority- RREC	Applicable	

Any other statutory permission, if required, shall be taken by SPD/SPPD



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8 Project Financials.

8.1 Cost Estimates of Solar Park

Solar Park cost estimate is based on layout of allocated land and infrastructure development as discussed above. The project cost is estimated for solar park capacity of 1310 MW. However, the park capacity will be enhanced further in phase -II. Total Estimated cost for RVUN Solar Park Phase-I 1310 MW will be Rs. 897.24 Crores details described in tables below considering all the aspects discussed above.

- **Land:**

Maximum land has been allotted by the State Government as per the Rajasthan Land Revenue Act, 1956 and Rajasthan Land Revenue (Allotment of Land for Setting-up of Power Plant based on Renewable Energy Sources) (Amendment) Rules, 2014.

The size of land allotment has been considered with tracker-based module mounting system, further size of land per mw will be allotted based on new & updated technology. The Maximum Land allocation for RVUNL Solar Park is of Govt. land at prevailing DLC rates at the time of allotment. Total amount for land & two-year lease rent is 50.044 Crore. Annual lease rent payable as per the rules shall be levied as 7.5% per annum of DLC with a yearly incremental of 5%. Further some small patches of land coming in between the allotted land shall be acquired from private owners if required with the mutual agreement between both the parties.

Annual lease will be payable by the SPPD.

Land Lease Charges			
Description	Qty	UOM	Estimated Cost (Rs in Lakhs)
Land Purchase Charges (Govt Land)	3026.99	Hectares	3640.40
Pvt Land Purchase & Registration Charges	LS	Hectares	1000.00
Sub Total			4640.40
Land Lease Charges for 2 years	5 % per annum of the land premium.		364.00
Total			5004.40

- **Site Development:**

The Site development activities shall include the following:

- 1- Major Land Levelling & dressing work is for development of boundary, road, switchyard, PSS & water infrastructure.
- 2- Construction of Boundary Fencing & Main Gate.



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The levelling of land in Solar Power Plants is generally not preferred to optimise the costs & the natural slope is adjusted through the structures only. Hence only dressing works for only essential development has been considered & major land undulations, levelling & grading have not been considered.

Boundary wall has been considered as a security point of view & healthy operations of projects. The total land size for solar park is 3026.99 Ha & the peripheral fencing of land comes around 195 Kms. Considering the soil of loose type casting of foundation for fencing column post need to be at minimum 1.2 mtr depth from NGL. The fencing will be interconnected with entrance gate at each land packet as the land is divided by no of existing village internal road. However, one main gate will be provided at main PSS & switchyard.

Site Development Charges				
Description	Unit	Qty	Rate (Rs in Lakhs)	Cost (Rs in Lakhs)
Land dressing	LS	LS	1500.00	1500.00
Boundary Fencing & Main Gate	KM	195	39.00	7605.00
Total				9105.00

• Road & Drainage Infrastructure Works:

Existing Main Roads passing through solar park will be widened to 10 meters width (7.5 meters BT and 1.25-meter shoulders either side) and strengthened. New 10mtr wide BT road will also be constructed for RVUNL PSS access & inter connection with existing roads.

Internal Peripheral Secondary has been proposed with 7.5mtr wide WBM road (5.5mtr WBM & 1 mtr shoulder each side) in the entire peripheral boundary of the solar park.

The land selected for the park development is purely desert sand in nature hence water absorption capacity is very high in such type of land. Considering such facts & site scenario Hume Pipe Culverts, Cement Concrete Causeway has been proposed for the development.

Details are tabulated as below:-

Road & Drainage Infrastructure					
S.No	Description	Unit	Qty	Rate (Rs in Lakhs)	Rate (Rs in Lakhs)
1	Construction of New Road 10 m wide including 1.25 m shoulder either side	Kms	16	63.00	1008.00
2	Strengthening of existing roads upto 10 mtr wide approx 35 km	kms	35	29.00	1015.00
3	Construction of New Internal Peripheral Road inside Park area 7.5mtr wide inside the Park plot	Kms	195	18.62	3630.90
Subtotal - Cost of Roads (I)					5,653.90



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4	Cost for Construction of New Hume Pipe Culverts (Considered 900mm dia pipe)	No's	48	5.25	252.00
5	Cost for Construction of New Concrete Causeway (7.5mtr wide)	No's	70	1.48	103.60
Subtotal – Cost of Drainage (II)					355.60
Total Cost Road & Drainage infrastructure(I+II)					6,009.50

• Water Infrastructure:

The major water requirement for O&M purpose is in module cleaning. Ground Water is saline in nature which is neither potable nor useful in module washing. Hence for such requirements of water it has been fulfilled by IGNP Canal which is approx. 20 kms from selected site. Water supply for Solar Park is conceived for Drinking water, washing of the PV modules, green belt development, etc.

The quantum of water required to wash a module is 2.5 litres per Module surface per washing with total 10 washings in a year. For drinking water, sanitations etc, the quantum of water required are considered at the standard norms of 70 litres per person per day.

Raw Water Coming from Panna Lal Pumping Station will be carried out through DI Pipeline of 150mm dia & will be conserved in newly proposed RCC Water tank Reservoir of capacity 100Lakh Litres. Further This raw water will be processed through treatment plant & will be stored in Overhead RCC Water tank of capacity 2.25 lakh liters. Such treated Water will be supplied to SPD's Nearest boundary Point in the allotted plot & further will also be utilised for drinking & Sanitation Purpose in PSS & Admin Building of Solar Park.

Cost Estimates are tabulated below:

Water Infrastructure					
S. No	Description	Unit	Qty	Rate (Rs in Lakhs)	Cost (Rs in Lakhs)
1	Construction of RCC Elevated Water Storage Tank of 5000 gallons (2.25 lakh Litre, 21mtr Height) & its pumping structure	LS	1	437.00	437.00
2	Laying of DI Pipeline 150 mm size from canal/ Panna Lal Pumping Station to SPPD Reservoir	KM	20	26.00	520.00
3	Construction of Reservoir for Capacity 100 Lakh litre for canal closure Purpose	LS	1	1150.00	1,150.00
4	Laying of Pipeline from Reservoir to Overhead water tank & pipeline for water network system to SPD & Local Villagers.	Km	45	18.00	810.00
5	Water Treatment Plant capacity 3Lakh Litre.	LS	1	133.00	133.00
6	ROW for Pipeline Laying	LS	1	200.00	200.00
	Total				3,250.00



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• Street Lighting System

The proposed site will be developed with peripheral lighting around the boundary fencing of approx. 195kms, Primary Road, PSS, Switchyard, Admin Buildings & water storage/treatment plants etc.

Street lighting system has been considered of solar type with battery storage based with 25watts LED bulb fitted over Pole within 30meter from each pole.

Cost Estimate has been tabulated below

Street Lighting System					
S. No	Description	UOM	Qty	Rate (Rs in lakhs)	Cost (Rs in Lakhs)
1	Solar Street Lights along with Roadside (25-Watt LED, 7-meter pole height, 30 meters spacing with battery storage)	No's	8600	0.45	3,870.00
Subtotal –					3,870.00

• Electrical Infrastructure:

The competitive bidding route is being followed for the creation of transmission system. Cost of individual equipment / item provided in letter of awards is observed to be not related with the actual cost estimate. Cost estimate is therefore based on annual plan of RVUNL for FY17-18. Based on this, estimated cost of 400 kV line (with Panther conductor and normal span double circuit & optical fibre terminal) is considered as Rs. 325 Lakhs per circuit km. 400 kV double circuit line between Ramsar Chhota PSS(RVUNL) & GSS(RVUNL) will be strung on standard 400 kV tower with Aluminium alloy AL-59 (Moose equivalent) conductor with reduced span of 320 meters against normal 350 meter. With increased number of towers and Al-59 conductor having higher aluminium content and extra cost of aluminium alloy, estimated cost of 400 kV line with OFGW including provision for GST is considered as Rs. 325.00 lakhs per km.

RVUNL Solar Park has Considered 04 no's of 315 MVA Power transformers costing to Rs. 64 Crores & 1 no's of 100MVA Power transformer costing to Rs 7.5 Crores. 33kV feeder bays from SPD's Control Room will be evacuated at this RVUNL PSS. Cost of substation has been considered based on adjustment in cost of transformer, 3no's of 400 kV bay, cost of shunt reactor and optical fibre terminals. Optical fibre termination is considered for both ends of 400 kV line from RVUNL PSS to RVPNL GSS Splicing (i.e. tee off).

Cost Estimate Details are tabulated below.



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Electrical Infrastructure					
S. No	Description	UOM	Qty	Rate (Rs in Lakhs)	Cost (Rs In Lakhs)
1	Cost of Main Power Transformer (33/400 KV)				-
a)	315 MVA Power transformer	No's	4	1600.00	6,400.00
b)	100 MVA Power Transformer	No's	1	750.00	750.00
2	Cost of Metering Panels & Bidirectional meters	Set	1	4600.00	4,600.00
3	Cost of HT Equipment's (VCB, RMU, Switchgears, HT Panels, Auxiliary T/F, 30No's Incomer 33KV cable of SPD's etc)	Set	10	220.00	2,200.00
4	Cost of HT Cables (HT Panels to Power Trafo & Auxiliary Internal Cables	Km	40	25.00	1,000.00
5	Scada, Server, Relay, Central Protection Monitoring system, Sensors & Alarm including Communication cable	Set	1	1000.00	1,000.00
6	Auxiliary Power Backup System	LS	1	700.00	700.00
7	CCTV & Monitoring System	No's	150	0.30	45.00
8	Switchyard Structure Cost	No's	1	2200.00	2,200.00
9	Registration, Approval, Connection & Inspection Charges	LS	1	500.00	500.00
10	AC Evacuation Transmission Line 400 KV Double Circuit.	Km	15	325.00	4,875.00
11	Interfacing Point at each 30-50MW Block size at 33kV Level including laying of cable upto PSS, protections & switchgears	No's	40	105.00	4200.00
12	3 no's of Bay 400kV with all equipment's (including one spare)	No's	3	2762.50	8,287.50
13	Fees of Interconnection STOA/LTOA.	LS	1310	2.50	3,275.00
14	Mobile Network & Communication enhancement	LS	1	1517.00	1517.00
Subtotal -					41,549.50



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• Admin Building & Other Infrastructure:

The Solar Park is comprised with a RCC Pooling Sub Station building & 4 no's of building which includes Admin/office building and warehouse for storage as per the site conditions. The Park will have setup of security through Watch towers & Portable guard cabins. Cost of such development is tabulated below.

Admin Building & other Infrastructure Setup					
S No	Description	UOM	Qty	Rate (Rs in Lakhs)	Cost (Rs in lakhs)
1	Cost of PSS Building (4000sqft Built Up Area)	Sqft	4000	0.028	112.00
2	Cost of Office Building & Warehouse (1500Sqft Built Up Area *4 no's)	Sqft	6000	0.025	150.00
3	Elevated Watch Tower	No's	50	1.80	90.00
4	Security guard Portable Cabins	No's	30	0.35	10.50
5	Structural Foundation & Canopy work at Interfacing Point	No's	40	10.00	400.00
6	Other Miscellaneous Development Cost	LS	1	643	643.00
Subtotal –					1,405.50

• CSR, Greenery & Other Emergency Expenses.

Considering the Solar Park development Greenery development is also a responsibility to maintain climatic conditions unaffected. Setup of Greenery infrastructure as well nursery has been considered in the development of park cost.

Corporate Social Responsibilities & Community development is also an essential part for any project as well any organisation development. Such factor has also been considered during the cost estimation of solar park. Emergency Expenses like Ambulance, Fire Brigade truck, Internet Connectivity and other safety measures has been considered for healthy development and smooth operation of project.

CSR, Greenery & Other Emergency Expenses.					
S No	Description	UOM	Qty	Rate (Rs in lakhs)	Cost (Rs in Lakhs)
1	Cost of Purchase of Trees & greenery Plantation & nursery work	LS	1	350.00	350.00
2	CSR Activities	LS	1	500.00	500.00
3	Other Safety & Emergency Expenses (Ambulance, fire Brigade, Internet Connectivity Enhancement etc)	LS	1	400.00	400.00
Sub total –					1,250.00



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• Summary Of Estimated Cost

All the cost estimated for development of RVUNL solar park is based on the schedule rate of Rajasthan urban infrastructure development program (RUIDP), Project cost of similar projects, Consultant in house database and discussion with various EPC solar companies. The same has been incorporated considering some market variation.

For O&M Expenditure the estimated cost has been considered as per the CERC Guideline with market price trends.

Project Cost Estimates are Tabulated as below: -

Table 17: Project Cost Estimate for Solar Park

Table 1 : PROJECT COST SUMMARY			
Sr. No.	Particulars		Rs in Lakhs
1	Civil, Electrical and Mechanical Works		
A	Cost of Land		4,640.40
B	Land lease charges during construction		364.04
C	Site Development		9,105.00
D	Roads & Drainage Infrastructure		6,009.50
E	Water Infrastructure		3,250.00
E	Street Lighting System		3,870.00
F	Electrical Infrastructure		41,549.50
G	Admin Building and other establishments		1,405.50
H	CSR & other greenery		1,250.00
	Subtotal (1)		71443.90
2	Preliminary & Pre-operative Expenses	% of Project Hard Cost	-
A	Cumulative PMC & Administration Charges	15.00%	10,716.59
B	Preliminary & Pre-Operative Expenses	1.00%	714.44
C	Contingencies	3.00%	2,143.32
D	Miscellaneous Cost	0.00%	-
E	Financing Charges	0.00%	-
	Subtotal (2)		13,574.35
3	Total Capex (Including Hard & Soft Cost 1+2)		85,018.29
4	Interest During Construction	9.25%	4,706.09
5	Total Project Cost Capex (3+4)		89,724.38



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Sr. No.	Mode of Finance Means of Finance		Rs in Lakhs
1	Loan from Bank/Financial Institution		-
A	Total Loan	71%	63,524.38
	Term Loan from Bank		63,524.38
			-
2	Promoters Contribution	29%	26,200.00
A	MNRE Grant/ CFA @ 20 Lakh/ MW	29%	26,200.00
			-
3	Total (1+2)		89,724.38

Revenue - One Time Upfront Charges

Sr. No.	Calculation of Revenue from Park	Rs in Lakhs
A	One Time Upfront Charges	
1	Project Cost	89,724.38
2	Surplus Fund for Long Term Sustainability	4,000.00
3	MNRE Grant/Subsidy (30% of Capex)	26,200.00
4	Total Project Cost (after deducting CFA) (1+2-3)	67,524.38
B	Total Upfront Cost from Developer after deducting CFA	67,524.38
1	Project Cost (after deducting CFA) per MW	51.545
i)	One Time Upfront Land Cost per MW	3.82
ii)	One Time Upfront Solar Park Development charge per MW	47.73

8.1 Annual O&M Expense

Annual Operation & Maintenance expenses for the RVUNL solar park has been calculated based on market price trend & general practices for civil & utility infrastructure and power evacuation cost as per normative O&M expenses specified in CERC regulations.

The Expenses are detailed in table below.

1	Annual Operation and Maintenance (O&M Exp.)	Rs. Lakhs /p.a.	2,975.00
A	Internal evacuation including Bays, Substation, X-mers, X-mission lines	Rs. Lakhs /p.a.	1,000.00
B	Primary & Secondary Roads including Periodic Maintenance	Rs. Lakhs /p.a.	560.00
C	Water Infrastructure Maintenance	Rs. Lakhs /p.a.	250.00
D	Buildings	Rs. Lakhs /p.a.	250.00
E	Other civil works including fencing	Rs. Lakhs /p.a.	214.00
F	Street lighting	Rs. Lakhs /p.a.	256.00



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G	Establishment, Admin, Manpower	Rs. Lakhs /p.a.	210.00
H	ROE on Surplus Fund for Long Term Sustainability @5.875%	Rs. Lakhs /p.a.	235.00
I	Power Consumption charges		Actual
J	Water Consumption charges		Actual
	Total (A to J)	Rs. Lakhs /p.a.	2,975.00
2	Total Annual Insurance Charges for park	Rs Lakhs/ p.a.	99.64
3	Land Lease charges (7.5% of Land Cost)	Rs Lakhs/ p.a.	273.03
4	Total Annual O&M Expenses (1+2+3)	Rs. Lakhs /p.a.	3347.67
5	Escalation on O & M (2 nd year Onwards)	Percent	5.85%
6	Escalation on Land Lease Charges (2 nd year Onwards)	Percent	5%

7	Summary of Annual Fees		
A	Total Annual O&M Expenses for Park (4)	Rs. Lakhs /p.a.	3347.67
B	Total Annual Fees per MW	Rs. Lakhs per MW	2.555
i)	Annual Fees of Lease Charge on Land @ 7.5%	Rs. Lakhs/ MW	0.21
ii)	Annual Solar Park O&M Expenditure	Rs. Lakhs/ MW	2.34

8.2 Project Summary with Viability & Sensitivity Analysis

1	Financial parameters		
A	Debt equity ratio		71: 29
B	MNRE Grant/ RVUNL Equity	Rs. Lakhs	26,200.00
C	Term Loan - Bank Term Loan	Rs. Lakhs	63,524.38
D	Total Project Cost including Long Term Sustainability Fund	Rs. Lakhs	93,724.38
E	Cost Per MW (Excluding CFA Grant)	Rs. Lakhs/MW	71.545
F	Cost Per MW (After Availing CFA Grant)	Rs. Lakhs/MW	51.545
G	Interest on - Term loan	Percent	9.25%
2	MNRE/ SECI Subsidy		
A	DPR Preparation	Rs. Lakhs	25.00
B	Capital Subsidy/ Grant	% of Project Cost	30.00%
C	Capital Subsidy/ Grant	Rs. Lakhs	26,917.31
D	Capital Subsidy/ Grant (Max.)	Rs. Lakhs/ MW	20.00
E	Capital Subsidy/ Grant (Max.)	Rs. Lakhs	26,200.00
F	Eligible Capital Subsidy/ Grant	Rs. Lakhs	26,200.00
G	Eligible Capital Subsidy/ Grant	Rs lakhs/MW	20.00



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3	Depreciation		
A	Life of Plant Assumed	Years	25
B	Rate of depreciation (As per Companies Act)	Percent	3.80%
C	Residual Value	Percent	5%
D	As Per Income Tax (Avg.)	Percent	13.04%
4	Tax parameters		
A	Income Tax Rate	Percent	34.94%
B	MAT	Percent	17.47%
5	Financial indicators		
A	Project IRR	Percent	12.94%
C	NPV @ 8.93 %	Rs. Lakhs	4,417.15

Based on these assumptions and project cost, the IRR of the solar Park Project for 25 years is 12.94%. Sensitivity analysis of IRR has been worked out with changes in one parameter at a time. The results of sensitivity analysis based on variation on key assumptions in tabulated as below:

Case I

Table: Sensitivity Analysis for variation in cost of project.

Type of Ratio	If Cost of Project Varied By					
		-10%	-5%	0%	5%	10%
	100%	90%	95%	100%	105%	110%
Project Cost	89724.38	81997.89	85861.14	89724.38	93587.63	97540.87
Project IRR	12.94%	9.38%	12.23%	12.94%	12.35%	11.82%
NPV @ 8.93 %	4417.15	2840.20	3987.69	4417.15	4404.65	4392.16

8.3 Tariff Projection for Solar Park

The Projection of tariff for power generation from the RVUNL Solar Park considering all the above-mentioned financial model & assumption is coming in the range of Rs 2.56/kWh to Rs 2.93/kWh.

Falling solar tariffs are driving solar installations in India. Utility-scale solar tariffs have fallen steeply by 82 % from Rs 10.95 per kilowatt hour (kWh) in 2010 to Rs 1.99/kWh in 2020 - largely driven by reduced solar module prices. Technological advancement of solar photovoltaics (PV) and economies of scale have triggered. and sustained. the downward pressure on module prices.

Against the trend in mid-2020, module prices began increasing globally. The price of solar (monocrystalline passivated emitter and rear contact type. commonly termed mono PERC) modules jumped by 38% between August 2020 and March 2022 key factors being the lack of supply of commodities (polysilicon, Glass, copper silver etc) and a surge in freight charges.



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Beyond module price hikes further contributors to the overall short-term inflation of Indian solar tariffs include the surge in demand for electricity and renewables in the post-pandemic world. The hike in goods and services tax (GST) rates on solar cells modules and inverters from 5 to 12%, as well as the rising insurance cost for solar projects.

The average of winning solar tariffs discovered between January 2021 and March 2022 is Rs 2.48/kWh. This is about 25% higher than the record low solar tariff of Rs 1.99/kWh in December 2020.

The main factor contributing to further increases in tariffs is the basic customs duty (BCD) applicable from April 1st 2022 of 25% on imported cells and 40 per cent on imported modules. To avoid this most developers stockpiled imported supplies in Q1 2022 for projects expected to be commissioned this year.

Another major challenge for developers is the Approved list of Models and Manufacturers (ALMM), implemented from 1 April 2022 which mandates the type and source of solar modules to be used for projects under central and state government contracts. The list comprises models and manufacturers of modules ("Solar Photovoltaic, Systems, Devices and Components Goods") certified by the Bureau of Indian Standards (BIS).

ALMM covers only domestic solar modules, precluding Chinese modules - which account for about 70-80% of all solar installations in India so far. The 14 GW of manufacturing capacity (the majority of which is based on out-dated multi-PERC technology) listed in ALMM by April 2022 will not be able to meet total market demand.

The approved list also presents a challenge for developers to source high-efficiency, higher-wattage modules (greater than 600 watts peak, or Wp) in India. The lesser generation capacity of the older technology modules will make it quite difficult for developers to sustain bidding of aggressive tariffs.

In the short-term it is expected that factors such as BCD and ALMM will sustain high domestic module prices, and so have an adverse impact on future utility scale solar tariffs. Even if developers employ stockpiled BCD-free imported modules. JMK Research anticipates solar project auctions will witness a tariff hike of 22% above the record-low tariff in the near term to reach Rs 2.43/kWh (US\$0.0323/kWh).

In the case of a project for which the developer procures domestic modules by Q1 2023 tariffs are likely to be only 14% higher than the record-low solar tariff. For developers depending on imported modules, the tariffs are expected to be 48% more (due to module price inflation, BCD, GST rate hike etc).

9 Social-Economic Impact

9.1 Estimate of Job creation during Solar Park construction and O & M.

Regarding estimation of job creation potential of Solar Power Projects in India, there is a study conducted jointly by two research institutes of international repute namely "Council on Energy, Environment and Water, New Delhi" and "Natural Resources Defence Council, New York" in August 2014 titled "Solar Power Jobs: Exploring the Employment Potential in India's Grid-Connected Solar Market".



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The study report has estimated job potential for Full Time Equivalent (FTE) i.e. equals to one employee working full time in a year (5 working days and 53 weeks), for difference phases of development of Solar Plants of various capacities.

As per the report, estimated job potential for Solar Power Plant larger than 25 MW is as follows:

During Construction & Commissioning period	0.35 FTE/MW for Skilled Manpower	8.08 FTE/ MW for Unskilled Manpower
During Operations & Maintenance (O&M) period	0.33 FTE/MW for Skilled Manpower	0.60 FTE/MW for Unskilled Manpower

Accordingly, for RVUNL Solar Park of 1310MW capacity the estimated job potential has been worked out as mentioned below:

- 460 number of Skilled and 10585 number of Unskilled Manpower during Construction & Commissioning works for both SPD & SPPD which includes construction of Roads, street lighting, building, water infrastructure, transmission lines, substations, installation of solar power plants, greenery development etc.
- 433 number of Skilled and 786 number of Unskilled Manpower during Operations & Maintenance (O&M) of the above referred works.

9.2 GDP Impact in the state

Increased supply of electricity at cheaper tariff would create more industries, thus revolutionise in investment & creating more jobs opportunities with an positive impact on overall GDP of the State and Nation.

9.3 Estimate reduction of CO₂ emission

Considering this project commissioning it will affect at Combined Margin (CM) 2428237 tons of CO₂ emission per year.

Details of land allotted to RVUNL as per Khasra by Collectorate Office, Bikaner			
Village – Ramsar Chhota (Phase -1, Solar Park 1310MW)			
S. No	Khasra No.	Land (in hectares)	Remarks
1	46	21.9289	
2	48	67.3927	
3	64	1.3658	
4	66	2.3016	



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5	68	1.7958	
6	76	44.6165	
7	78	26.8862	
8	81	28.7706	
9	86	47.4619	
10	88	8	
11	91	4.186	
12	93	3.933	
13	95	187.0908	
14	126	75	
15	136	197.1194	
16	137	97.529	
17	139	76.6245	
18	141	137.9469	
19	143	94.0133	
20	145	83.7824	
21	147	39.09	
22	150	41.7711	
23	152	122.4551	
24	154	69.062	
25	156	6.007	
26	158	25.1916	
27	159	125.3638	
28	161	28.9729	
29	166	88.2845	
30	168	6.7405	
31	171	33.6394	
32	173	26.823	
33	175	46.2605	
34	177	4.0342	
35	179	8.2328	
36	181	15.6057	
37	185	16.0356	
38	187	13.2281	
39	189	3.7307	
40	191	4.8562	
41	194	2.2384	
42	196	23.889	
43	198	18.6787	



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44	202	10.3827	
45	204	6.2853	
46	210	96.4288	
47	219	51.3444	
48	54/1	4.2492	
49	54/2	5.9944	
50	58/1	0.3667	
51	58/2	13.6834	
52	84/1	0.4173	
53	84/2	37.2816	
54	74/2	12.7349	
55	91/2	77.3074	
Total Land (In Hectares.)		2294.4122	

Details of land Khasra No in Bandarewala village.

S. No	Khasra No	Land in hectare
1	966/230	4.61
2	232	3.65
3	700/440	1.91
4	440	8.34
5	968/426	4.89
6	659/428	7.81
7	972/428	7.28
8	427	17.41
9	423	5.41
10	424	10.3
11	420	11.58
12	243	3.87
13	416	9.2
14	417	4.68
15	1013/416	1.9
16	470	14.48
17	414	8.26
18	1010/414	2.85
19	251	12.76
20	1015/251	2.31
21	414	8.26
22	486	8.94
23	351	13.03
24	465	1.51



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25	466	5.5
26	1001/464	1.42
27	467	6.04
28	999/468	2.26
29	473	5.27
30	665/472	6.1
31	475	9.68
32	682/471	1.62
33	470	14.48
34	479	8.83
35	481	12.87
36	480	5.95
37	1320/479	4.14
38	480	5.95
39	1232/482	2.21
40	1231/483	5.12
41	483	10.72
42	497	7.29
43	498	11.28
44	499	15.52
45	506	3.15
46	1239/505	6.53
47	1227/512	1.8
48	508	1.32
49	509	2.53
50	1317/496	4.78
51	522	1.13
52	360	12.05
53	1218/521	1.7
54	521	6.44
55	1245/516	3.94
56	670/518	8.82
57	1220/526	4.2
58	523	11.02
59	347	12.39
60	549	12.77
61	552	8.14
62	564	12.68
63	565	1.59



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64	566	7.96
65	567	3.97
66	562	12.14
67	568	2.39
68	1263/573	2.32
69	1262/575	2.92
70	575	2.88
71	580/575	4.36
72	576	4.53
73	1199/576	5.84
74	562	12.14
75	555	12.91
76	556	4
77	544	1.5
78	547	10.93
79	545	9.34
80	546	11.88
81	537	8.86
82	539	14.6
83	542	4.94
84	541	1.72
85	1169/548	2.7
86	1168/547	2.38
87	398	10.46
88	394	4.83
89	393	9.23
90	395	3.16
Total in Ha		732.58

Annexures-

Annexure I – General Land Layout Plan

Annexure II – Infrastructure Details

Annexure III – Land drawing with Google map

Annexure IV – S/S SLD & External Transmission line Drwg. (SLD)

Annexure V– Power Map of state

Annexures are attached separately.